

Packaged, Integrally Geared Centrifugal Air Compressors for Petroleum, Chemical, and Gas Industry Services

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Introduction

Users of this standard should be aware that further or differing requirements may be needed for individual applications. This standard is not intended to inhibit a vendor from offering, or the purchaser from accepting, alternative equipment or engineering solutions for the individual application. This may be particularly appropriate where there is innovative or developing technology. Where an alternative is offered, the vendor should identify any variations from this standard and provide details.

Annex A contains data sheets which purchasers are encouraged to use.

Annex B contains illustrations of typical mounting configurations.

Annex C specifies requirements for lateral analysis.

Annex D contains forms which may be used to indicate vendor drawing and data requirements.

Annex E shows the impact for inlet throttle control vs inlet guide vanes.

Annex F specifies requirements for determining residual unbalance.

Annex G contains an inspector's checklist.

Annex H contains illustrations of nomenclature for integrally geared centrifugal air compressors.

Annex I contains information relative to allowable nozzle loads from purchaser's piping.

This standard requires the purchaser to specify certain details and features. A bullet (•) at the beginning of a paragraph indicates that either a decision by, or further information from, the purchaser is required. Further information should be shown on the data sheets (see example in Annex A) or stated in the quotation request and purchase order.

In this standard, U.S. customary units (USC) are included in brackets for information.

Packaged, Integrally Geared Centrifugal Air Compressors for Petroleum, Chemical, and Gas Industry Services

1 Scope

1.1 This standard covers the minimum requirements for constant-speed, packaged, general purpose, integrally-geared centrifugal air compressors, including their accessories for use in the petroleum, chemical, and gas industry services. This standard is not applicable to machines that develop a pressure rise of less than 0.35 bar (5.0 psi) above atmospheric pressure, which are classed as fans or blowers.

NOTE Special purpose and process applications, including Process Air Services, are covered by API 617.

- **1.2** Equipment covered by this standard is considered non-critical, usually spared and may be either of two classifications, basic or special duty. The purchaser shall specify which of the two classifications applies.

Basic packages are vendors' standard packages of proven design, and include minimal additional requirements.

Special duty packages are typically specified for installations that require higher availability, and include additional features and requirements.

1.3 Additional or overriding requirements applicable only to packages that have been specified as "Special Duty" are noted at the end of each section (see 6.12, 7.10, 8.5, D.2.5, and D.3.5.5).

2 Normative References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) available at the agreed time, applies. Standards referenced in the documents are undated but refer to the specific editions referenced in this section.

ANSI/API Standard 541, Form-Wound Squirrel Cage Induction Motors—500 Horsepower and Larger

ANSI/API Standard 546, Brushless Synchronous Machines—500 kVA and Larger

API Standard 547, General-purpose Form-wound Squirrel Cage Induction Motors 250 Horsepower and Larger

API Standard 611, General-purpose Steam Turbines for Petroleum, Chemical, and Gas Industry Services

API Standard 614, Lubrication, Shaft-sealing, and Control-oil Systems and Auxiliaries for Petroleum, Chemical and Gas Industry Services

ANSI/API Standard 670, Machinery Protection Systems

AGMA 6011 ¹, Specification for High Speed Helical Gear Units

AGMA 9000-C90, Flexible Couplings—Potential Unbalance Classification

AGMA 9002-B04, Bores and Keyways for Flexible Couplings (Inch Series)

ASME B1.1 ², Unified Inch Screw Threads (UN and UNR Thread Form)

¹ American Gear Manufacturers Association, 1001 N. Fairfax Street, Suite 500, Alexandria, Virginia, 22314, www.agma.org.

² ASME International, 2 Park Avenue, New York, New York 10016-5990, www.asme.org.

ASME B1.3, *Screw Thread Gaging Systems for Acceptability: Inch and Metric Screw Threads (UN, UNRm UNJ, M, MJ)*

ASME B1.13M, *Metric Screw Threads: M Profile*

ASME B16.1, *Cast Iron Pipe Flanges and Flanged Fittings Classes 125 and 250*

ASME B16.5, *Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Addenda A*

ASME B16.11, *Forged Fittings, Socket-Welding and Threaded*

ASME B16.42, *Ductile Iron Pipe Flanges and Flanged Fittings Classes 150 and 300*

ASME B16.47, *Large Diameter Steel Flanges NPS 26 Through NPS 60 Addenda A*

ASME B17.1, *Keys and Keyseats*

ASME B31.3, *Process Piping*

ASME, *Boiler and Pressure Vessel Code (BPVC)*

— Section VIII, *Division 1: Rules for Construction of Pressure Vessels*

— Section IX, *Welding and Brazing Qualifications*

ASME PTC-10, *Performance Test Code on Compressors and Exhausters*

ASTM A193/A193M³, *Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications*

ASTM A194/A194M, *Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both*

ASTM A275/A275M, *Standard Test Method for Magnetic Particle Examination of Steel Forgings*

ASTM A395/A395M, *Standard Specification for Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures*

ASTM A515/A515M, *Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service*

ASTM A536, *Standard Specification for Ductile Iron Castings*

ASTM B111/B111M, *Standard Specification for Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock*

ASTM D4304, *Standard Specification for Mineral and Synthetic Lubricating Oil used in Steam or Gas Turbines*

ASTM D5445-05, *Standard Practice for Pictorial Markings for Handling of Good*

ASTM E94, *Standard Guide for Radiographic Examination*

³ ASTM International, 100 Barr Harbor Drive, West Conshohocken, Pennsylvania 19428, www.astm.org.

ASTM E186, *Standard Reference Radiographs for Heavy-Walled (2 to 4 1/2-in. (51 to 114-mm) Steel Castings*

ASTM E446, *Standard Reference Radiographs for Steel Castings up to 2 in. (51 mm) in Thickness*

AWS D1.1/D1.1M⁴, *Structural Welding Code—Steel Errata*

ANSI/B11 Standards B11.19-2010⁵, *Performance Criteria for Safeguarding*

EN 953⁶, *Safety of machinery—Guards—General requirements for the design and construction of fixed and movable guards*

European Union ATEX Directive 94/9/EC⁷

IEC 60079 (all parts)⁸, *Electrical apparatus for explosive atmospheres*

IEEE 841⁹, *Standard for Petroleum and Chemical Industry-Severe Duty Totally Enclosed Fan-Cooled (TEFC) Squirrel Cage Induction Motors-Up to and Including 370 kW (500 hp)*

ISA S12.4¹⁰, *Instrument Purging for Reduction of Hazardous Area*

ISO 7-1:1994¹¹, *Pipe threads where pressure-tight joints are made on the threads*

ISO 261:1998, *ISO General Purpose Metric Screw Threads—General Plan Second Edition*

ISO 1328-1, *Cylindrical Gears-ISO System of Accuracy—Definitions and Allowable Values of Deviations Relevant to Corresponding Flanks of Gear Teeth*

ISO 1940-1, *Mechanical vibration—Balance quality requirements for rotors in a constant (rigid) state—Part 1: Specification and verification of balance tolerances*

ISO 3117:1977, *Tangential Keys and Keyways*

ISO 3448:1992, *Industrial Liquid Lubricants—ISO Viscosity Classification Second Edition*

ISO 5389, *Turbocompressors—Performance Test Code*

ISO 6708, *Pipework Components—Definition and Selection of DN (Nominal Size) Second Edition*

ISO 8068:2006, *Lubricants, industrial oils and related products (class L)—Family T (Turbines)—Specification for lubricating oils for turbines*

ISO 8501, *Preparation of Steel Substrates Before Application of Paints and Related Products—Visual Assessment of Surface Cleanliness*

⁴ American Welding Society, 8669 NW 36 Street, #130, Miami, Florida 33166-6672, www.aws.org.

⁵ B11 Standards Inc., P.O. Box 690905, Houston, Texas 77269

⁶ European Committee for Standardization, Avenue Marnix 17, B-1000 Brussels, Belgium, www.cen.eu.

⁷ European Parliament and the Council, Rue de la Loi, 175 B-1048, Brussels, Belgium

⁸ International Electrotechnical Commission, 3 rue de Varembe, P.O. Box 131, CH-1211 Geneva 20, Switzerland, www.iec.ch.

⁹ International Electrical and Electronics Engineers, 445 Hoes Lane, Piscataway, NJ 08854

¹⁰ International Society for Automation, 67 T. W. Alexander Drive, Research Triangle Park, NC 237709

¹¹ International Organization for Standardization, 1, ch. de la Voie-Creuse, Case postale 56, CH-1211 Geneva 20, Switzerland, www.iso.org.

ISO 8821:1989, *Mechanical Vibration—Balancing—Shaft and Fitment Key Convention*

ISO 9606-1:2012, *Qualification testing of welders—Fusion welding—Part 1: Steels*

ISO 14120, *Safety of machinery. Guards. General requirements for the design and construction of fixed and movable guards*

ISO 15607:2005, *Specification and qualification of welding procedures for metallic materials—General rules*

MSS SP-55¹², *Quality Standard for Steel Castings for Valves, Flanges and Fittings and Other Piping Components—Visual Method for Evaluation of Surface Irregularities*

NEC Article 110¹³, *Requirements for Electrical Installations*

NEMA SM23, *Steam Turbines for Mechanical Drive Service*

NEMA 250, *Enclosures for Electrical Equipment (1000 Volts Maximum)*

NFPA 70¹⁴, *National Electrical Code*

NFPA 496, *Standard for Purged and Pressurized Enclosures for Electrical Equipment*

SSPC SP6¹⁵, *Commercial Blast Cleaning NACE No. 3-2000 (Steel Structures Painting Manual, Chapter 2—Surface Preparation Specifications)*

TEMA¹⁶, *TEMA Book of Standards*, Ninth Edition

3 Terms, Definitions, Acronyms, and Abbreviations

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1.1

alarm point

Preset value of a measured parameter at which an alarm is actuated to warn of a condition that requires corrective action.

3.1.2

anchor bolts

Bolts used to attach the equipment to the support structure (concrete foundation or steel structure).

cf. hold-down bolt (3.1.15).

3.1.3

approve

Written documentation confirming an agreement.

¹² Manufacturers Standardization Society of the Valve and Fittings Industry, Inc., 127 Park Street, NE, Vienna, Virginia 22180-4602, www.mss-hq.com.

¹³ National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1752, Rosslyn, Virginia 22209, www.nema.org.

¹⁴ National Fire Protection Association, 1 Batterymarch Park, Quincy, Massachusetts 02169-7471, www.nfpa.org.

¹⁵ The Society for Protective Coatings, 40 24th Street, 6th Floor, Pittsburgh, Pennsylvania 15222, www.sspc.org.

¹⁶ Tubular Exchanger Manufacturers Association, 25 North Broadway, Tarrytown, New York 10591

3.1.4**axially split joint**

Joint split with the principal face parallel to the shaft centerline.

3.1.5**bearing housing**

All bearing enclosures, including the gear casing.

3.1.6**Birmingham wire gauge****BWG**

Standard series of sizes arbitrarily indicated by numbers, used in specifying and describing the thickness of sheet metal.

3.1.7**certified point**

Point to which the performance tolerances will be applied.

NOTE This is usually the normal operating point and the vendor will normally require that this point is within the preferred selection range.

3.1.8**commercial fastener**

Fastener manufactured to published consensus standards and stocked by manufacturers or distributors.

3.1.9**critical speed**

Shaft rotational speed at which the rotor-bearing-support system is in a state of resonance.

3.1.10**delivered flow****free air delivered****FAD**

Flow rate determined at the discharge of the aftercooler, or at the compressor discharge if an aftercooler is not provided.

3.1.11**design**

Manufacturer's calculated parameter.

NOTE A term that is typically used by the equipment manufacturer to describe various parameters such as design power, design pressure, design temperature, or design speed. It is not intended for the purchaser to use this term.

3.1.12**gear service factor (SF)**

Factor that is applied to the tooth pitting index and the bending stress number, depending on the characteristics of the driver and the driven equipment, to account for differences in potential overload, shock load, or continuous oscillatory torque characteristics, or a combination thereof.

3.1.13**gear wheel****bull gear**

Low-speed rotor of a gear set.

3.1.14**general purpose application**

Application that is usually spared or is in non-critical service.

3.1.15**hold down bolts****mounting bolts**

Bolts holding the equipment to the mounting plate.

3.1.16**hydrodynamic bearings**

Bearings that use the principles of hydrodynamic lubrication.

NOTE The bearing surfaces are oriented so that relative motion forms an oil wedge or wedges to support the load without shaft-to-bearing contact.

3.1.17**informative**

Information only.

cf normative (3.1.27).

NOTE An informative reference or Annex provides advisory or explanatory information. It is intended to assist the understanding or use of the document.

3.1.18**inlet volume flow**

Flow rate expressed in volume flow units at the conditions of pressure, temperature, compressibility and gas composition, including moisture content, at the compressor inlet flange.

NOTE Inlet volume flow is a specific example of Actual Volume Flow. Actual Volume Flow is the volume flow at any particular location such as interstage, impeller inlet, discharge or compressor discharge.

3.1.19**local**

Position of devices on or near the equipment or console.

3.1.20**material certificate of compliance**

Document by which the vendor certifies that the material represented has been produced and tested in accordance with the requirements of the basic material specification shown on the certificate.

3.1.21**maximum allowable temperature****MAT**

Maximum continuous temperature for which the manufacturer has designed the equipment (or any part to which the term is referred) when handling the specified fluid at the specified maximum operating pressure (not at the MAWP).

3.1.22**maximum allowable working pressure****MAWP**

Maximum continuous pressure for which the manufacturer has designed the equipment (or any part to which the term is referred) when handling the specified fluid at the specified maximum operating temperature (not at the MAT).

3.1.23**maximum discharge pressure**

Maximum suction pressure plus the maximum differential pressure the compressor is able to develop, when operating with the furnished impeller(s) and the fluid with the maximum density.

NOTE Refer to 3.1.25 for the definition of maximum suction pressure.

3.1.24**maximum sealing pressure**

Highest pressure the seals are required to seal during any specified static or operating condition.

3.1.25**maximum suction pressure**

Highest suction pressure the compressor will be subject to in service.

3.1.26**mounting plate**

Device used to attach equipment to foundations; this is either a baseplate or soleplate.

3.1.27**normative**

Required.

cf informative (3.1.17)

NOTE A normative reference or annex enumerates a requirement or mandate of the specification.

3.1.28**NPS****nominal pipe size**

Dimensionless value approximately equal to the diameter in inches.

EXAMPLE NPS $\frac{3}{4}$.

3.1.29**NPT****American National Standard Pipe Taper**

Thread form designation for pipe threads.

EXAMPLE $\frac{3}{4}$ -14 NPT.

NOTE It is comprised of a number representing nominal pipe size followed by the number of threads per inch and the letters NPT representing the thread series.

3.1.30**observed**

Inspection or test where the purchaser is notified of the timing of the inspection or test and the inspection or test is performed as scheduled even if the purchaser or his representative is not present.

3.1.31**operating stability**

Operating range from the compressor rated point to surge at constant speed expressed as a percentage.

3.1.32**owner**

Final recipient of the equipment who may delegate another agent as the purchaser of the equipment.

3.1.33**panel**

Enclosure used to mount, display and protect gauges, switches, and other instruments.

3.1.34**pinion**

High-speed rotor(s) in a gear set.

3.1.35**PMI****positive material identification testing**

Physical evaluation or test of a material to confirm that the material is consistent with the selected or specified alloy material designated.

NOTE Typically this is only a consideration for high-alloy materials.

3.1.36**pressure casing**

Composite of all stationary pressure containing parts of the unit, including all nozzles and other attached parts.

3.1.37**pressure rise to surge**

Difference between the discharge pressure at the rated operating point and discharge pressure at the surge point when the unit is operating at rated inlet conditions, and with a constant inlet guide vane position.

3.1.38**purchaser**

Agency that issues the order and specification to the vendor.

NOTE The purchaser is sometimes the owner of the plant in which the equipment is to be installed, but is often the owner's appointed agent.

3.1.39**radially split**

Split with the joint perpendicular to the shaft centerline.

3.1.40**rated point**

Maximum specified flow rate at the specified discharge pressure when operating at the specified inlet conditions and coolant temperature.

3.1.41**rated speed****100 % speed**

Highest rotational speed (revolutions per minute) required to meet any of the specified operating conditions.

3.1.42**relief valve set pressure**

Pressure at which a relief valve starts to lift.

3.1.43**remote**

Location of a device when located away from the equipment or console, typically in a control room.

3.1.44**shutdown set point**

Preset value of a measured parameter at which automatic or manual shutdown of the system or equipment is required.

3.1.45**soleplate**

Plate attached to the foundation, with a mounting surface for equipment or for a baseplate.

3.1.46**special tool**

Tool which is not a commercially available catalog item.

3.1.47**standard volume flow**

Flow rate expressed in volume flow units at standard conditions as follows:

ISO Standard Conditions

Flow:	Cubic meters per hour (m ³ /h) Cubic meters per minute (m ³ /min)
Pressure:	(01.3 kpa (1.013 bar(a))
Temperature:	15 °C
Relative Humidity:	Dry

U.S. Standard Conditions

Flow:	Standard cubic feet per minute (scfm) Million standard cubic feet per day (mmscfd)
Pressure:	14.7 PSI(a)
Temperature:	60 °F
Relative Humidity:	Dry

3.1.48**standby**

Service state in which a piece of equipment is normally idle or idling and is capable of immediate automatic or manual start-up for continuous operation.

3.1.49**surge**

Volume flow capacity below which a centrifugal compressor becomes aerodynamically unstable.

3.1.50**TIR****total indicator reading****total indicated runout**

Difference between the maximum and minimum readings of a dial indicator or similar device, monitoring a face or cylindrical surface during one complete revolution of the monitored surface.

NOTE For a cylindrical surface, the indicated runout implies an eccentricity equal to half the reading. For a flat face, the indicated runout implies an out-of squareness equal to the reading. If the measured surface in question is not cylindrical or flat, the interpretation of the meaning of TIR is more complex, and can represent orality or surface irregularities.

3.1.51

trip speed

Electric motor driver speed (revolutions per minute) corresponding to the synchronous speed of the motor at maximum supply frequency at the motor terminals.

3.1.52

trip speed

Steam turbine driver speed at which the independent emergency overspeed device operates to shut down the driver.

3.1.53

turndown

Percentage of change in capacity (referred to rated capacity) between the rated capacity and the surge point capacity at the rated head when the unit is operating at rated suction temperature and gas composition.

3.1.54

Ultimate load rating

Load that will produce the minimum acceptable oil film thickness without inducing failure during continuous service ,or the load that will not exceed the creep initiation or yield strength of the babbitt or bearing material at the location of maximum temperature on the pad, whichever load is less.

3.1.55

unit responsibility

Obligation for coordinating the documentation, delivery and technical aspects of the equipment and all auxiliary systems included in the scope of the order.

3.1.56

vendor supplier

Manufacturer or manufacturer's agent that supplies the equipment.

3.1.57

verified

Purchaser's review and acceptance of vendor's certification or documentation of successful completion of the inspection or test.

3.1.58

witnessed

Inspection or test where the purchaser is notified of the timing of the inspection or test, and a hold is placed on the inspection or test until the purchaser or the purchaser's representative is in attendance.

3.2 Acronyms and Abbreviations

ABMA	American Bearing Manufacturers Association
AC	alternating current
AENOR	Spanish Association for Standardization and Certification
AF	amplification factor
ANSI	American National Standards Institute
API	American Petroleum Institute

ASME	American Society of Mechanical Engineers
BPVC	<i>Boiler and Pressure Vessel Code</i>
PTC	power test code
ASTM	American Society for Testing and Materials
ATEX	ATmospheres EXplosives
AWS	American Welding Society
B11	B11 Standards Inc.
BFV	butterfly valve
BSI	British Standards Institute
BTH	below the hook
BWG	Birmingham wire gauge
CEN	European Committee for Standardization
cf	(Latin conferre) confer or compare—cross reference
CSA	Canadian Standards Association
DBSE	distance between shaft ends
DC	direct current
DCS	distributed control system
DIN	Deutsches Institut für Normung
DN	nominal diameter
EEA	European Economic Area
EN	European normal standard
ES	extended surface
EU	European Union
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IGV	inlet guide vanes
IP	ingress protection
IPF	instrumented protective functions
ISO	International Organization for Standardization
MAWP	maximum allowable working pressure
MDMT	minimum design metal temperature
MSS	Manufacturers Standardization Society
NACE	National Association of Corrosion Engineers
NDT	nondestructive testing
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
NPS	nominal pipe size
PLC	programmable logic controller
PMI	positive material identification

Ra	arithmetic average roughness
RF	radio frequency
RTD	resistance temperature detector
SI	international system of units
SPL	sound pressure level
SSPC	Society for Protective Coatings
TC	thermocouple
TEMA	Tubular Exchanger Manufacturers Association
UN	United Nations
UNC	unified thread—course
UNF	unified thread—fine
UNRC	unified thread—rolled course
UNS	unified numbering system
USC	U.S. customary units
VDDR	vendor drawing and data requirements

4 General

4.1 Unit Responsibility

4.1.1 The compressor vendor shall assume unit responsibility for all equipment and auxiliary systems included in the scope of the order, and shall ensure that all sub-vendors comply with the requirements of this standard and all reference documents.

4.1.2 The technical aspects to be considered by the compressor vendor include, but are not limited to, such factors as the power requirements, speed, rotation, general arrangement, couplings, dynamics, noise, lubrication, sealing system, material test reports, instrumentation, piping, conformance to specifications and testing of components.

4.2 Nomenclature

Informative guides to integrally-gear air compressor nomenclature can be found in Annex B and Annex H.

5 Requirements

• 5.1 Units of measurement

The purchaser will specify whether data and drawings, supplied to this standard shall use the SI or USC system of measurements.

NOTE Dedicated data sheets for SI units and for USC are provided in Annex A.

5.2 Statutory Requirements

The purchaser and the compressor vendor shall determine the measures to be taken to comply with any governmental codes, regulations, ordinances, directives, or rules that are applicable to the equipment its packaging and any preservatives used.

Equipment installed in the European Economic Area (EEA) shall comply with all applicable European Union Directives.

NOTE The European Economic Area in 2013 includes the countries of the European Union plus Norway, Iceland and Liechtenstein.

• 5.3 Document Requirements

The hierarchy of documents shall be specified by the purchaser.

NOTE Typical documents include company and industry specifications, meeting notes, and modifications to these documents.

6 Basic Design

6.1 General

6.1.1 Equipment Reliability

6.1.1.1 Only equipment that is considered field proven is acceptable.

NOTE Purchasers can use their engineering judgment in determining what equipment is field proven.

- 6.1.1.2 If specified, the vendor shall provide the documentation to demonstrate that all equipment proposed is “field proven”.

6.1.1.3 In the event no such unit is available, the vendor shall submit an explanation of how their proposed equipment can be considered field proven.

NOTE A possible explanation can be that all components comprising the assembled machine satisfy the field proven definition.

- 6.1.1.4 If specified, an installation/reference list shall be submitted for the proposed compressor model noting analogous operating conditions.

6.1.1.5 Purchaser shall specify the period of required uninterrupted continuous operation. Shutting down the equipment to perform planned maintenance or inspection during the specified uninterrupted operation is not acceptable.

NOTE 1 It is realized that there are some services where this objective is easily attainable and others where it is difficult.

NOTE 2 Auxiliary system design and design of the process in which the equipment is installed are very important in meeting this objective.

NOTE 3 Paragraph D.2.3.1k requires the vendor to identify any component or maintenance requirement that would result in the need to shut down the equipment within the uninterrupted operational period.

6.1.1.6 Vendor shall advise in the proposal any component designed for finite life.

NOTE It is recognized that these are design criteria.

6.1.2 Operating Conditions

- 6.1.2.1 The purchaser shall specify all operating conditions as well as the certified point.
- 6.1.2.2 In the specification of operating conditions, either a realistic relative humidity at rated conditions or the maximum operating inlet temperature related dew point shall be specified.

NOTE 1 A relative humidity of 100 % does not occur at the maximum operating inlet temperature, but rather only after the temperature has dropped substantially. Dew point higher than 92 °F (33 °C) seldom occurs anywhere.

NOTE 2 Volume flow increases to account for overstated moisture content can result in oversized equipment and compromised performance.

6.1.2.3 The equipment shall be capable of operating within the entire performance map at all specified operating conditions, as well as accommodating other conditions such as momentary surge, settle-out, trip, and startup.

6.1.3 Sound Pressure Level

- **6.1.3.1** Control of the sound pressure level (SPL) of all equipment furnished shall be a joint effort of the purchaser and the vendor having unit responsibility. The equipment furnished by the vendor shall conform to the maximum allowable sound pressure level specified. In order to determine compliance, the vendor shall provide expected values for maximum sound pressure level data per octave band for the equipment in steady state operation. ISO 3740, ISO 3744 and ISO 3746 or ASME PTC 36, may be consulted for guidance.

6.1.3.2 The vendor shall indicate what special silencing measures, if any, are proposed in order to meet the specified levels.

6.1.3.3 If noise enclosures are supplied, the enclosure design shall ensure full access to the equipment for operational and maintenance purposes. The design of the enclosure shall not obstruct any required cooling of the equipment. The final enclosure design shall be agreed by purchaser and vendor.

6.1.4 Packaged Equipment

The final scope of the package and supply of any loose shipped components shall be agreed between the purchaser and the vendor. Pending agreement, the default shall be that the compressor package shall be supplied fully assembled, and shall include:

- a) integrally geared centrifugal air compressor;
- b) coupling and coupling guard;
- c) baseplate (or structural framework);
- d) intercoolers, moisture separators (if required), drain system, and aftercooler;
- e) lubrication oil system;
- f) instrumentation;
- g) unit control panel;
- h) driver;
- i) inlet air filter;
- j) inlet throttle device (suction throttle valve or inlet guide vanes);
- k) interstage air piping;
- l) discharge blowoff valve;
- m) discharge blowoff silencer;
- n) discharge check valve;

- o) inlet and discharge expansion joints;
- p) other accessories as noted in this standard.

NOTE Inlet piping from air inlet filter to compressor inlet control device, discharge piping between compressor package flange and discharge check valve, piping to blowoff valve, and mounting of accessory components is typically supplied by the purchaser. The aftercooler is often shipped loose. The blowoff valve and discharge check valve are typically shipped loose for field installation by the purchaser.

• 6.1.5 Environmental Conditions

The equipment, including all auxiliaries, shall be suitable for operation under the environmental conditions specified by the purchaser. Specified conditions shall include whether the installation is indoors (heated or unheated) or outdoors (with or without a roof), maximum and minimum temperatures, unusual humidity, and dusty or corrosive conditions.

6.1.6 Cooling Water Systems

Unless otherwise specified, cooling water system or systems shall be designed for the following conditions:

Water Velocity over heat exchange surfaces	1.5 to 2.5 m/s	5 to 8 ft/s
Maximum allowable working pressure (MAWP)	≥700 kPa (7 bar)	≥100 psig
Test Pressure (≥1.5 MAWP)	≥1050 kPa (10.5 bar)	≥150 psig
Maximum pressure drop	≥100 kPa (1 bar)	15 psi
Maximum inlet temperature	30°C	90°F
Maximum outlet temperature	50°C	120°F
Maximum temperature rise	20 °C	30 °F
Minimum temperature rise	10 °C	20 °F
Water side fouling factor	0.18 m ² K /kW	0.001 hr-ft ² - °F/Btu
Corrosion allowance for carbon steel shells	1.5 mm	1/16 in.

The vendor shall notify the purchaser if the criteria for minimum temperature rise, and velocity over heat exchanger surfaces result in conflict. The criterion for velocity over heat exchange surfaces is intended to minimize water-side fouling; the criterion for minimum temperature rise is intended to minimize the use of cooling water. If such a conflict exists, the purchaser shall approve the final selection.

NOTE Based on site coolant conditions and user experience, the purchaser will sometimes specify a different coolant-side fouling factor. For example, for a closed loop glycol system, 0.0005 hrft² °F/Btu is usually adequate; conversely, for poorer quality coolant, 0.002 hrft² °F/Btu (or higher) can be prudent.

6.1.7 Package Arrangement

6.1.7.1 The arrangement of the package (including piping coolers, pumps, and controls) shall provide adequate clearance areas and safe access for operation and maintenance.

6.1.7.2 All equipment shall be designed to permit rapid and economical maintenance. Major parts such as casing components and bearing housings shall be designed and manufactured to ensure accurate alignment on reassembly.

NOTE This can be accomplished by the use of shoulders, cylindrical dowels, or keys.

6.1.7.3 Provisions shall be made for complete venting and draining of liquid-filled systems.

6.1.7.4 All utility and auxiliary customer piping connections are to be located as near as possible to the edge of the skid.

NOTE The intent is to allow clear and easy access to customer connections.

6.1.8 Motors and Electrical Components

- **6.1.8.1** Motors, electrical components, and electrical installations shall be suitable for the area classification (class, group, and division or zone) specified by the purchaser, and shall meet the requirements of the applicable sections of specified standards such as IEC60079 or NFPA 70 (NEC), Articles 500, 501, 502, 504 and 505, as well as local codes. Purchaser will advise on applicability of European Union (EU) ATEX Directive 94/9/EC.

6.1.8.2 Locations for installed equipment can be classified as hazardous electrical areas or they can be unclassified. An unclassified area is considered non-hazardous; therefore, motors, electrical instrumentation, equipment, components, and electrical installations for unclassified areas are not governed by hazardous area electrical codes.

6.1.8.3 If an installation location is classified as hazardous, motors, electrical instrumentation, equipment, components, and electrical installations shall be suitable for the hazardous electrical area classification designation specified.

- **6.1.8.4** All applicable electrical codes shall be specified. Local electrical codes that apply shall be provided by the purchaser upon request.

6.1.9 Performance Criteria

6.1.9.1 The equipment (compressor, driver and ancillary equipment) shall perform on the test stand(s) and on their permanent foundation within the specified acceptance criteria. After installation, the performance of the package shall be the joint responsibility of the purchaser and the vendor who has unit responsibility.

6.1.9.2 The compressor total head curve shall be developed from the differential pressure measurement between the compressor inlet flange and the compressor final-stage discharge flange. The purchaser and vendor shall agree on the pressure drop considerations for the inlet filter, aftercooler, check valves, and associated piping.

6.1.9.3 At rated operating conditions, the overall performance shall provide a minimum of 10 % continuous pressure rise from rated capacity to surge.

6.1.10 Purchaser Connections

All openings or nozzles for purchaser connections shall be DN 12 ($1/2$ NPS) or larger and shall be in accordance with ISO 6708. Sizes DN 32, DN 65, DN 90, DN 125, DN 175 and DN 225 ($1\ 1/4$, $2\ 1/2$, $3\ 1/2$, 5, 7, and 9 NPS) shall not be used.

6.1.11 Bolting and threads

- **6.1.11.1** The threading shall conform to ASME B1.1, ASME B1.13M, or ISO 261 as specified.

NOTE 1 ASME B 1.1 covers general inch series screw threads, ASME B1.13M and ISO 261 covers general metric screw threads.

NOTE 2 For the purposes of this provision, ISO 261 is equivalent to ASME B 1.13M.

Glossary of terms for screw threads can be found in ASME B 18.12-2001.

6.1.11.1.1 If ASME B1.1 threads have been specified, the thread series shall be variable pitch series UNC or UNRC, or constant pitch series 4, 6, or 8-UN or UNR. Diameters shall be selected from Table 2 Column 1 of ASME B 1.1. The threads shall be Class 2 for bolting, studs and nuts. For other threads and nuts they shall be Class 2 or 3.

In order to prevent galling, when ASME B 1.1 Class 3 external threads are used, the tolerance for maximum material conditions shall be modified to prevent zero clearance.

6.1.11.1.2 If ISO 261 has been specified, the thread series shall be coarse or 3, 4, 6 or 8 pitch. Diameters shall be selected from Table 2, Column 1 of ISO 261. The threads shall be Class 6g for bolting and studs, and Class 6H for nuts. For other threads, they shall be Class 6g or 4h for external threads, and Class 6H or 5H for internal threads.

In order to prevent galling, when ISO class h/H position is specified for mating components, the tolerance for maximum material conditions shall be modified to prevent zero clearance.

6.1.11.2 Type J profile threads shall not be used.

6.1.11.3 Thread gaging shall meet requirements of 6.1.11.3.1 through 6.1.11.3.3.

6.1.11.3.1 Inspection or gaging requirements of threads shall be identified in accordance with ASME B 1.1, 1989, *Section 6—Screw Thread Designation*.

6.1.11.3.2 All threaded products shall be visually inspected for gross defects. This visual inspection shall be made without magnification and is intended to detect such gross defects as missing or incomplete threads, defective thread profile, torn or ruptured surfaces and cracks, etc.

6.1.11.3.3 Threads used in joining casing pressure containing components shall be inspected in accordance with 6.1.11.3.3.1 or 6.1.11.3.3.2.

6.1.11.3.3.1 Commercial fasteners shall be manufactured in accordance with the requirements of ASME B 18.18.2M or shall be procured from distributors having quality plans in accordance with ASME B 18.18.2M.

6.1.11.3.3.2 All threads used on non-commercial fasteners shall be manufactured in accordance with ASME B1.3M-1986 thread gaging system 21 using GO, NO-GO gaging.

6.1.11.4 Adequate clearance shall be provided at all bolting locations to permit the use of socket, box or hex key wrenches.

6.1.11.5 Slotted-nut or spanner-type bolting shall not be used unless specifically approved by the purchaser.

6.1.11.6 Manufacturer's marking shall be located on all fasteners 6 mm ($\frac{1}{4}$ in.) and larger (excluding washers and headless set screws). For studs, the marking shall be on the nut end of the exposed stud end.

NOTE A set screw is a headless screw with an internal hex opening on one end.

6.1.12 Mounting Surfaces

Mounting surfaces on equipment shall meet the following criteria:

- a) shall be machined to a finish of $6.3\ \mu\text{m}$ (250 $\mu\text{in.}$) arithmetic average roughness (Ra) or better;
- b) shall be in the same horizontal plane within $25\ \mu\text{m}$ (0.001 in.) to prevent a soft foot;
- c) each mounting surface shall be machined within a flatness of $80\ \mu\text{m}$ per linear meters (0.001 in. per linear foot) of mounting surface;

- d) different mounting planes shall be parallel to each other within 50 μm (0.002 in.);
- e) upper machined or spot faced surface shall be parallel to the mounting surface;
- f) hold-down bolt holes shall be drilled perpendicular to the mounting surface or surfaces;
- g) bolt holes shall be 15 mm ($1/2$ in.) larger in diameter than the hold down bolt;
- h) bolt holes shall be spot faced where necessary to accommodate washers, fasteners and tools.

NOTE Spot face is typically unnecessary if surface is perpendicular to bolting within 1 degree.

6.2 Pressure Casings

6.2.1 The allowable tensile stress used in the design of the casing for any material shall not exceed 25 % of the minimum ultimate tensile strength for that material at the maximum specified operating temperature. The vendor shall state the internationally recognized standard from which the ultimate tensile strength value is obtained. For cast materials the allowable tensile stress shall be multiplied by a casting factor of 0.8 for steel or 0.9 for cast and ductile iron unless additional casting NDE is applied. The thickness of the casing shall be suitable for the maximum working and test pressure and shall include a corrosion allowance of at least 3 mm (0.125 in.). Manufacturing data report forms, third party inspections, and stamping as specified in pressure vessel codes are not required.

6.2.2 The maximum allowable working pressure of each casing shall be at least 1.10 times the maximum discharge pressure of the stage.

6.2.3 For casing joint bolting, the allowable tensile stress determined in 6.2.1 shall be used to determine the total bolting area based on hydrostatic load and gasket preload as applicable. The bolting preload stress shall not exceed 75 % of the bolting material minimum yield. During hydrotest, the bolting preload stress shall not exceed 90 % of the bolting material minimum yield.

6.2.4 Axial clearances between impellers and casings shall be adjustable.

NOTE Due to shape of impellers and volute, changing axial clearance will also change radial clearances.

6.3 Casing Connections

6.3.1 Compressor first-stage inlet and final stage outlet connections shall be furnished with machined flanges and through bolting.

6.3.2 All openings or nozzles for piping connections on pressure casings interfacing with piping provided by the purchaser shall be DN 20 (NPS $3/4$) or larger, and shall be in accordance with ASME B31.3 or ISO 6708. Sizes NPS $1\frac{1}{4}$, $2\frac{1}{2}$, $3\frac{1}{2}$, 5, 7, and 9 (DN 32, DN 65, DN 90, DN 125, DN 175 and DN 225) shall not be used.

6.3.3 Connections welded to the casing shall meet the material requirements of the casing, including impact values, corrosion allowance and temperature-pressure rating, rather than the requirements of the connected piping. All welding of connections shall be completed before the casing is hydrostatically tested (see 8.3.2).

6.3.4 For connections other than main process connections, if flanged or machined and studded openings are impractical, threaded connections for pipe sizes not exceeding DN 40 ($1\frac{1}{2}$ NPS) may be used with purchaser's approval as follows:

- a) on non-weldable materials, such as cast iron;
- b) where essential for maintenance (disassembly and assembly).

6.3.5 Threaded openings for tapered pipe threads shall conform to ASME B1.20.1 or ISO 7-I as specified. If ISO 7-1 has been specified, tapered or straight internal threads shall also be specified. Bosses for pipe threads shall conform to ASME B16.5.

6.3.6 Threaded openings not required to be connected to piping shall be plugged with steel plugs in accordance with ASME B16.11. Thread tape shall not be used.

6.3.7 Flanges

6.3.7.1 Unless otherwise specified, the CLASS system applies and all flanges shall conform to ASME B16.1, B16.5, B16.42 or B16.47 Series B as applicable, except as specified in 6.3.7.1.1, 6.3.7.1.2, and 6.3.7.3. Class 200 and 400 flanges shall not be used.

- **6.3.7.1.1** If specified, ASME B 16.47 Series A steel flanges shall be provided for applicable CLASS system flanges.

6.3.7.1.2 If the CLASS system applies, cast and ductile iron flanges shall be flat faced. Class 125 flanges shall have a minimum thickness equal to Class 250 for sizes NPS 8 and smaller.

6.3.7.2 If the PN system is specified, all flanges shall conform to EN 1092-1 or EN 1092-2 as applicable, except as specified in 6.3.7.2.1 and 6.3.7.2.2.

- **6.3.7.2.1** If the PN system applies, PN 6 and PN 10 shall not be used.

6.3.7.2.2 If the PN system applies, cast and ductile iron flanges shall be flat faced. PN 16 and PN 25 flanges shall have a minimum thickness equal to PN 40 for sizes DN 200 and smaller.

6.3.7.3 Flat-faced flanges with full, raised-face thickness are acceptable on casings of all materials. Flanges in all materials that are thicker or have a larger outside diameter than required by the applicable ASME or EN standard are acceptable. Non-standard (oversized) flanges shall be completely dimensioned on the arrangement drawing. If oversized flanges require studs or bolts of non-standard length, this requirement shall be identified on the arrangement drawing.

- **6.3.8** Machined and studed connections shall conform to the facing and drilling requirements of ASME B16.1, B16.5, B16.42, B16.47, EN 1092-1 or EN 1092-2, as specified. Studs and nuts shall be provided and shall arrive installed, the first 1.5 threads at both ends of each stud shall be removed.

6.3.9 To minimize nozzle loading and facilitate installation of piping, machine flanges shall be parallel to the plane shown on the general arrangement drawing to within 0.5 degrees. Studs or bolt holes shall straddle centerlines parallel to the main axes of the equipment.

6.3.10 All of the purchaser's connections shall be accessible for disassembly without requiring the machine, or any major part of the machine, to be moved.

6.4 External Forces and Moments

6.4.1 The maximum allowable forces and moments that may be imposed on the package by the purchaser's piping shall be stated in the proposal.

NOTE 1 For integrally geared compressors, it is not possible to give a formula to calculate the maximum allowable piping forces and moments on each casing flange. The limiting criteria are the gear contact pattern and the impeller/stator gap. The maximum value of the external forces and moments, which leads to acceptable deformations and therefore acceptable changes of the gear contact pattern and the impeller/casing gap, depends on various parameters. The possible combinations are nearly endless. Each manufacturer has limits based on experience for each volute size and gear case combination for a given specific machine. The values are available from the manufacturer with the quotation. See Annex I for more information.

NOTE 2 It is a common practice on integrally geared compressors to supply expansion joints in order to minimize the piping loads on the machine flanges, and to insure that piping loads are within the allowable limits for the particular unit.

6.4.2 The maximum allowable forces and moments shall be shown on the outline drawing.

6.5 Rotating Elements

6.5.1 Shafts

6.5.1.1 Shafts shall be forged or hot-rolled alloy steel, and machined throughout their entire length.

6.5.1.2 The rotor shaft sensing areas to be observed by the radial probes shall be concentric with the bearing journals and shall meet the requirements of 6.5.1.2.1 through 6.5.1.2.3.

6.5.1.2.1 All sensing areas (both radial vibration and axial position) shall be free from stencil and scribe marks or any other surface discontinuity, such as an oil hole or a keyway, for a minimum of one probe-tip diameter on each side of the probe. These areas shall not be metallized, sleeved, or plated.

6.5.1.2.2 The final surface finish shall be a maximum of $0.8\text{ }\mu\text{m}$ ($32\text{ }\mu\text{in.}$) R_a , preferably obtained by honing or burnishing.

6.5.1.2.3 These areas shall be properly demagnetized to the levels specified in API 670 or otherwise treated so that the combined total electrical and mechanical runout does not exceed:

- a) for radial sensing areas 25 percent of the maximum allowed peak-to-peak vibration amplitude or $6\text{ }\mu\text{m}$ (0.25 mil), whichever is greater;
- b) for areas to be observed by axial-position probes, $12.7\text{ }\mu\text{m}$ (0.5 mil).

6.5.1.3 Chrome plating and other methods of building up the shaft at the journal area shall not be used.

6.5.1.4 All shaft keyways shall have fillet radii conforming to ASME B17.1, ISO 3117, or other applicable national or international standard.

6.5.2 Impellers

6.5.2.1 The impeller material shall be stainless steel, of cast or forged and milled construction.

6.5.2.2 The vendor's proposal shall describe in detail the type of impeller construction and the method of attachment to the shaft.

6.5.3 Gears

6.5.3.1 Gearing shall be designed and manufactured to the tolerances specified in ISO 1328-1, Grade 5 or better.

6.5.3.2 The gear unit shall be rated in accordance with AGMA 6011 using minimum service factors of 1.4 for induction motor driven units and 1.6 for steam-turbine and synchronous motor driven units. The rating shall be based on the driver nameplate rating.

6.5.3.3 Gear wheels and pinion hardness combinations shall be in accordance with the values recommended in AGMA 6011.

6.5.3.4 The tooth portion of pinions shall be integrally forged with their shaft.

6.5.3.5 Gear wheels shall be of forged construction, and shall be assembled on the shaft with an interference fit.

6.6 Seals and Sealing Systems

6.6.1 Air and oil shaft seals shall be provided to achieve the following:

- a) contain compressed air inside the compressor casings,
- b) prevent oil from entering the compressor casings and contaminating the compressed air,
- c) prevent oil from leaking out of the bearing housing into the atmosphere,
- d) prevent contamination of the oil system or compressed air by atmospheric dirt or moisture.

6.6.2 There shall be an atmospheric space between the air and oil seals.

6.6.3 The sealing system shall be furnished complete with piping, filters, instrumentation, and necessary start-up interlocks as applicable. This system, including air consumption, shall be fully described in the proposal.

6.6.4 Seal operation shall be suitable for all specified operating conditions, including suction throttling, startup, shutdown, standby, and momentary surge. The type of standby operation shall be agreed upon by the purchaser and the vendor.

6.7 Dynamics

6.7.1 Critical Speeds

6.7.1.1 For information on critical speeds, refer to Annex C.

6.7.1.2 Resonances of structural support systems that are within the vendor's scope of supply and that affect the rotor vibration amplitude shall not occur within the specified separation margins (see C.2.4) unless the resonances are critically damped. The effective stiffness of the structural support shall be considered in the analysis of the dynamics of the rotor-bearing-support system.

NOTE Resonances of structural support systems can adversely affect the rotor vibration amplitude.

6.7.1.3 The vendor shall determine that the drive-train (turbine, gear, motor, and the like) critical speeds (rotor lateral, system torsional, blading modes, and the like) will not excite any critical speed of the machinery being supplied and that the entire train is suitable for the rated speed and any starting-speed detent (hold-point) requirements of the train. A list of all undesirable speeds from zero to trip shall be submitted for purchaser review and included in the instruction manual for purchaser guidance (see Annex D, Item 26d). These undesirable operating speeds shall comply with the separation margin stated in C.2.10.

6.7.1.4 For the purposes of this standard, resonant conditions of concern are those with an amplification factor (AF) equal to or greater than 2.5.

6.7.2 Lateral Analysis

The vendor's standard critical speed values that have been analytically derived and proven by testing of previously manufactured compressors of the same frame size are acceptable. A report is not required.

6.7.3 Torsional Analysis

6.7.3.1 The vendor's torsional critical speed values that have been analytically derived and proven by successful operation of nominally identical previously manufactured compressor drive trains are acceptable. A report is not required.

6.7.3.2 The undamped torsional natural frequencies of the complete train shall be at least 10 % above or 10 % below any possible excitation frequency unless a report is provided that shows the stresses will be acceptable. This report shall include a summary of the stress analysis, and shall demonstrate that the resonances have no adverse effect on the complete train. This report shall also include a clear statement of the assumptions made in the analysis regarding the magnitude of excitation and the degree of damping.

6.7.3.3 Torsional criticals at two or more times running speeds shall preferably be avoided or in systems in which corresponding excitation frequencies occur, shall have no adverse effect. In addition to multiples of running speeds, torsional excitations that are not a function of running speeds or that are non-synchronous in nature shall be considered in the torsional analysis when applicable, and shall have no adverse effect.

6.7.3.4 The vendor shall perform a transient torsional vibration analysis for synchronous-motor-driven units. The acceptance criteria for this analysis shall be agreed upon by the purchaser and the vendor.

6.7.4 Vibration and Balancing

6.7.4.1 Manufacturer's standard balancing procedure shall be used.

6.7.4.2 When spare rotating elements are supplied, they shall be dynamically balanced to the same tolerances as the main rotating elements.

6.7.4.3 During the shop test of the machine as assembled with balanced rotors and operating at its rated speed, the peak-to-peak amplitude of unfiltered vibration, including runout, in any plane, measured on the pinion shaft adjacent and relative to each radial bearing shall not exceed the following value or 40 μm (1.5 mils), whichever is less—See Equation (1):

In SI units:

$$A = 25.4 \times (12000/N)^{(1/2)} \quad (1)$$

In USC units,

$$A = (12000/N)^{(1/2)}$$

where

A is the amplitude of unfiltered vibration, in μms (mil) true peak to peak;

N is the rated speed, in revolutions per minute.

NOTE These limits are not to be confused with the limits specified in Annex C.3 for shop verification of unbalanced response.

6.7.4.4 Electrical and mechanical runout shall be determined and recorded by rolling the rotor in V blocks at the journal centerline while measuring runout with a non-contacting vibration probe, and with a dial indicator at the centerline of the probe location and one probe-tip diameter to either side.

6.7.4.5 Accurate records of electrical and mechanical runout for the full 360° at each probe location shall be included in the mechanical test report.

6.8 Bearings and Bearing Housings

6.8.1 Bearings—General

6.8.1.1 Unless otherwise specified, hydrodynamic radial and thrust bearings shall be provided.

NOTE Some small, low-power units are being offered with anti-friction bearings on the bull gear.

6.8.1.2 Bearings shall be designed to prevent incorrect positioning.

6.8.1.3 As a design criteria, bearing metal temperatures shall not exceed 100 °C (212 °F) at specified operating conditions with a maximum inlet oil temperature of 50 °C (120 °F). Acceptance value during testing as well as design criteria in the case of an air-cooled machine shall be agreed prior to order.

6.8.2 Radial Bearings

6.8.2.1 Radial bearings shall be designed for ease of assembly, precision bored and of the sleeve or pad type with babbitted replaceable liners, pads, or shells. These bearings shall be equipped with anti-rotation pins and shall be positively secured in the axial direction.

6.8.2.2 The bearing design shall suppress hydrodynamic instabilities and provide sufficient damping over the entire range of allowable bearing clearances to limit rotor vibration to the maximum specified amplitudes (see 6.7.4.3) while the equipment is operating loaded or unloaded at the rated operating speed.

6.8.3 Thrust Bearings

6.8.3.1 Thrust loads from impellers and gears and couplings shall be absorbed by individual thrust bearings on pinions, or transmitted to the gear wheel thrust bearing by means of thrust rider rings fixed to the pinions and gear wheel. All specified operating conditions and start up conditions shall be evaluated for resulting thrust loads.

6.8.3.2 Thrust bearings shall be selected using manufacturer's standard criteria. In sizing thrust bearings, consideration shall be given to the following for each specific application:

- a) shaft speed;
- b) temperature of the bearing babbitt;
- c) deflection of the bearing pad;
- d) minimum oil-film thickness;
- e) feed rate, viscosity, and supply temperature of the oil;
- f) design configuration of the bearing;
- g) babbit alloy;
- h) turbulence of the oil-film.

6.8.3.3 Thrust forces from metallic flexible-element couplings shall be calculated on the basis of the maximum allowable deflection permitted by the coupling manufacturer.

6.8.3.4 If two or more rotor thrust forces are to be carried by one thrust bearing, the resultant of the forces shall be used provided the directions of the forces make them numerically additive; otherwise, the largest of the forces shall be used.

6.8.3.5 Thrust bearings shall be babbitted and arranged for continuous pressurized lubrication. Integral thrust collars are preferred. If replaceable collars are furnished, they shall be positively locked to the shaft to prevent fretting.

NOTE Replaceable collars facilitate assembly and maintenance.

6.8.3.6 The faces of the thrust collar or rider rings shall have a surface finish of $0.4\text{ }\mu\text{m}$ ($16\text{ }\mu\text{in.}$) Ra, or better. After mounting, the axial total indicator runout of either face shall not exceed $13\text{ }\mu\text{m}$ (0.0005 in.).

6.8.4 Bearing Housings

6.8.4.1 Bearing Housings shall be arranged so that bearings can be replaced without disturbing equipment driver or mounting, and unless otherwise agreed shall not require removal of thrust collars or coupling hub(s).

NOTE 1 Bearing replacement can require removal of gear housing cover.

NOTE 2 Removal of coupling and possible adjustment of driver is necessary for removal of anti-friction bearings.

6.8.4.2 Bearing housings shall be arranged to minimize foaming. The drain system shall be adequate to maintain the oil and foam level below shaft end seals. The bearings shall be designed not to exceed $30\text{ }^{\circ}\text{C}$ ($50\text{ }^{\circ}\text{F}$) oil temperature rise and a housing oil outlet temperature of $80\text{ }^{\circ}\text{C}$ ($180\text{ }^{\circ}\text{F}$).

NOTE This is a design criteria. Housing exit temperature is not measured in actual machines.

6.8.4.3 Bearing housings shall be equipped with replaceable labyrinth-type end seals and deflectors or non-contacting carbon ring seals for high speed pinions where the shaft passes through the housing. Lip-type seals may be used with purchaser approval. The design of the seals and deflectors shall effectively retain oil in the housing and prevent entry of foreign material into the housing. This shall be achieved without the requirement for external service such as an air purge or grease.

6.9 Lubrication

6.9.1 Unless otherwise specified, bearings and bearing housings shall be designed for oil lubrication using a mineral oil in accordance with ASTM D4304 or ISO 8068 Type AR.

NOTE Some users and manufacturers prefer to apply synthetic lubricants to this equipment.

6.9.2 Unless otherwise specified, a pressurized oil system shall be supplied in accordance with API 614, except as noted in this standard.

NOTE Many basic systems are purchased with single filters, without oil heaters, without baffles in reservoir, and without indication of suction loss level in the level glass.

- **6.9.3** Lube oil shall be supplied at the required pressure or pressures, as applicable, to the following:

- a) bearings of the integrally geared compressor,
- b) spray nozzles for the gear teeth,
- c) bearings of the driver if specified or required.

6.9.4 The oil reservoir shall be manufacturer's standard design and of material with an oil and coolant-compatible corrosion resistant internal coating.

NOTE In some designs, the oil reservoir is integral with the gear casing or integral with the base frame.

- **6.9.5** If specified, the oil reservoir shall be fabricated from austenitic stainless steel.

6.9.6 If a reservoir oil heater is provided, it may be controlled by an integral heater thermostat unless it is a special duty package, in which case the requirements in 6.12.15 applies.

NOTE API 614 default is controlled by a separate temperature sensing element, but standard units often use an integral thermostat.

6.10 Materials

6.10.1 General

6.10.1.1 Materials of construction shall be the manufacturer's standard for the site environmental and operating conditions specified, except as required or prohibited by this standard.

6.10.1.2 The material specification of all major components shall be clearly stated in the vendor's proposal. Materials shall be identified by reference to applicable international standards, including the material grade. If no such designation is available, the vendor's material specification giving physical properties, chemical composition, and test requirements shall be included in the proposal.

6.10.1.3 External parts that are subject to rotary or sliding motions (such as control linkage joints and adjustment mechanisms) shall be of corrosion-resistant materials suitable for the site environment and of sufficient hardness to resist wear.

6.10.1.4 Minor parts such as nuts, springs, washers, gaskets, and keys shall have corrosion resistance at least equal to that of specified parts in the same environment.

- **6.10.1.5** The purchaser shall specify any agents (including trace quantities) present in the site environment, including constituents that may cause corrosion. If possible, the concentration of each agent shall be stated.

6.10.1.6 The vendor shall select materials to avoid conditions that may result in electrolytic corrosion. Where such conditions cannot be avoided, the purchaser and the vendor shall agree on the material selection and any other precautions necessary.

NOTE If dissimilar materials with significantly different electrical potentials are placed in contact in the presence of an electrolytic solution, galvanic couples that can result in serious corrosion of the less noble material can be created. The *NACE Corrosion Engineer's Reference Book* is one resource for selection of suitable materials in these situations.

6.10.1.7 If the manufacturer of studs and nuts specifies anti-seizure compounds, modified torque values shall be included in the manuals. Molybdenum disulphide lubricants shall not be used. Close tolerance mating parts, such as shaft sleeves, that are made from galling materials and that cannot be disassembled by hydraulic or thermal expansion techniques shall not rely on anti-seizure compound. These items shall have a suitable metallic coating to prevent galling.

6.10.1.8 Bolting for pressure joints shall be high-temperature alloy steel in accordance with ASTM A193 Grade B7. Carbon steel ASTM A194, Grade 2H nuts shall be used.

6.10.1.9 Low-carbon steels can be notch sensitive and susceptible to brittle fracture at ambient or lower temperatures. Therefore, for pressure containing parts made of steel, only fully killed, normalized steels made to fine-

grain practice are acceptable. Steel made to a coarse austenitic grain size practice (such as ASTM A515) shall not be used for pressure containing parts.

6.10.2 Castings

6.10.2.1 Castings shall be free from porosity, hot tears, shrink holes, blow holes, cracks, scale, blisters, and similar injurious defects. Surfaces of castings shall be cleaned by sandblasting, shot-blasting, chemical cleaning, or other standard methods. Mold-parting fins and the remains of gates and risers shall be chipped, filed or ground flush.

6.10.2.2 Pressure containing ferrous castings shall not be repaired except as specified in 6.10.2.2.1 through 6.10.2.2.3.

6.10.2.2.1 Weldable grades of steel castings shall be repaired by welding, using a qualified welding procedure based on the requirements of Section VIII, Division 1, and Section IX of the ASME *BPVC*, or other internationally recognized standard as approved by the purchaser. After major weld repairs, and before hydrotest, the complete repaired casting shall be given a post-weld heat treatment to ensure stress relief and continuity of mechanical properties of both weld and parent metal, and dimensional stability during subsequent machining operations.

6.10.2.2.2 Defects in cast gray iron or ductile iron that are within allowed repair limits of ASTM A395 shall either be repaired by plugging or the part replaced. Larger defects require replacement. The holes drilled for plugs shall be carefully examined, using liquid penetrant, to ensure that all defective material has been removed. Repair by welding is not allowed.

6.10.2.2.3 All repairs that are not covered by the agreed material specification shall be subject to the purchaser's approval.

6.10.2.3 Fully enclosed cored voids, which become fully closed by methods such as plugging, welding, or assembly, shall not be used.

6.10.2.4 All Ductile (Nodular) iron castings shall be produced in accordance with ASTM A395, or other internationally recognized standard as approved by the purchaser. Standard products using gray cast iron ASTM A536 Grades such as 60-40-18 or 80-55-06 may be proposed for purchaser approval.

NOTE The industry has a long history of reliable units in such grades of ASTM A536.

6.10.3 Welding

6.10.3.1 Welding of piping, pressure-containing parts, rotating parts and other highly stressed parts, weld repairs and any dissimilar-metal welds shall be performed and inspected by operators and procedures qualified in accordance with Section VIII, Division 1, and Section IX of the ASME *BPVC* or another purchaser approved standard such as ISO 9606 and ISO 15607 for welding procedures and welding qualification.

6.10.3.2 The vendor shall be responsible for the review of all repairs and repair welds to ensure that they are properly heat treated and nondestructively examined for soundness and compliance with the applicable qualified procedures. Repair welds shall be nondestructively tested by the same method used to detect the original flaw.

- **6.10.3.3** If specified, documentation of major defects shall be submitted to the purchaser prior to any repairs being conducted at the manufacturer's shop and shall include the following:

- a) extent of the repair,
- b) location indicated on sketch, weld map, or drawing;
- c) size;

- d) welding procedure specification;
- e) detailed photographs of the defect prior to any preparatory work and after preparation but prior to the actual repair. If the location of the defect cannot be clearly defined by photographic means, the defect repair preparation shall be indicated on a sketch, weld map or drawing of the affected component.

6.10.3.4 Repairs performed at the manufacturer's shop shall be considered major if any of the following conditions apply:

- a) castings leak during hydrostatic testing;
- b) depth of the repair cavity prepared for welding exceeds 50 % of the wall thickness or 25 mm (1 in.), whichever is smaller;
- c) surface area of all repairs to the part exceeds 10 % of the surface area of the part;
- d) repair cavity is longer than 150 mm (6 in.) in any direction;
- e) repair is to any rotating components.

6.10.3.5 Welding not covered by 6.10.3.1, such as welding on baseplates, non-pressure ducting, lagging, and control panels, shall be performed by welders qualified in accordance with an appropriate internationally recognized structural welding standard such as AWS D1.1 or Section IX of the ASME BPVC or other purchaser approved welding standard.

6.10.3.6 Connections welded to pressure casings shall be installed as specified in 6.10.3.6.1 and 6.10.3.6.2.

6.10.3.6.1 Post-weld heat treatment, if required, shall be carried out after all welds, including piping welds, have been completed.

6.10.3.6.2 All welds shall be heat treated in accordance with Section VIII, Division 1, Sections UW-10 and UW-40, of the ASME BPVC.

• **6.10.4 Low Temperature**

The purchaser shall specify the minimum design metal temperature (MDMT) to be applied to the equipment. If the MDMT is below the minimum allowable temperature for the materials of the equipment, the vendor and the purchaser shall agree and implement measures to assure that the equipment will not be operated with pressure casing at a metal temperature below the MDMT, to avoid brittle failures.

6.11 Nameplates and Rotation Arrows

6.11.1 A nameplate shall be securely attached at a readily visible location on the equipment and on any major piece of auxiliary equipment.

6.11.2 Rotation arrows shall be cast in or attached to each major item of rotating equipment at a readily visible location.

6.11.3 Nameplates and rotation arrows (if attached) shall be of austenitic stainless steel or nickel-copper alloy (UNS N04400). Attachment pins shall be of the same material. Welding to attach the nameplate to the casing is not permitted.

6.11.4 As a minimum, the following data shall be clearly stamped or engraved on the compressor nameplate:

- a) vendor's name,
- b) serial number,
- c) size, model and type,
- d) rated capacity,
- e) rated discharge pressure,
- f) purchaser's item number
- g) maximum allowable working pressure,
- h) purchase order number,
- i) compressor weight,
- j) year built,
- k) if required, Certification Markings such as ATEX or CE.

Units shall be consistent with those used on the data sheets.

6.12 Additional Basic Design Requirements for Special Duty Packages Only

6.12.1 Jackscrews, guide rods, cylindrical casing-alignment dowels, or other appropriate devices, or a combination thereof, shall be provided to facilitate disassembly and reassembly.

6.12.1.1 Guide rods shall be of sufficient length to prevent damage to the internals or casing studs by the casing during disassembly and reassembly.

6.12.1.2 Lifting lugs or eyebolts shall be provided to lift only the top half of the gear casing.

6.12.1.3 If jackscrews are used as a means of parting contacting faces, one of the faces shall be relieved (counterbored or recessed) to prevent a leaking joint or improper fit caused by marring of the face.

6.12.2 Sufficient pressure and temperature taps shall be provided to monitor or troubleshoot performance of each stage and intercoolers.

6.12.3 Gearing shall be manufactured to the gear tooth quality tolerances specified in ISO 1328-1, Grade 4.

6.12.4 Unless the vendor is able to demonstrate that functionally identical equipment has operated satisfactorily under the conditions stated in 6.7.2, a damped unbalanced response analysis to assure acceptable amplitudes of vibration at any speed from zero to trip shall be conducted as per C.2.10.

- **6.12.5** If specified, any damped unbalanced response analysis required by 6.12.4 shall be confirmed by test stand data in accordance with Annex C.

6.12.6 Unless the vendor is able to demonstrate that functionally identical equipment has operated satisfactorily under the conditions stated in Item 6.7.3.1, a torsional vibration analysis of the complete coupled train shall be

conducted as required in C.7. In this case, the vendor shall also be responsible for directing the modifications necessary to meet the requirements of 6.7.3.2 through 6.7.3.4.

6.12.7 Excitations of undamped torsional natural frequencies can come from many sources, which shall be considered in the analysis. Sources of excitation to be considered include, but are not limited to, the following:

- a) gear characteristics such as unbalance and pitch line runout, and cumulative pitch error;
- b) startup conditions such as speed detents and other torsional oscillations;
- c) torsional transients such as start-ups of synchronous electric motors and transients due to generator phase-to-phase fault or phase-to-ground fault;
- d) torsional excitation resulting from drivers;
- e) one and two times electrical line frequency;
- f) one and two times operating speeds;
- g) harmonic frequencies from variable frequency fixed speed drives.

6.12.8 Major parts of the rotating elements, such as the shaft and impellers, shall be dynamically balanced. When a bare shaft with a single keyway is dynamically balanced, the keyway shall be filled with a fully crowned half-key, in accordance with ISO 8821. A shaft with keyways 180 degrees apart, but not in the same transverse plane, shall also be filled. The initial balance correction to the bare shaft shall be recorded.

6.12.9 The rotating elements shall be multiplane dynamically balanced during assembly. This shall be accomplished after the addition of each major component. Balancing correction shall be applied only to the elements added. Balancing of impellers by welding is prohibited. Minor correction of other components may be required during the final trim balancing of the completely assembled element. In the sequential balancing process, any half-keys used in the balancing of the bare shaft (see 6.12.6) shall continue to be used until they are replaced with the final key and mating element. On rotors with single keyways, the keyway shall be filled with a fully-crowned half-key. The weight of all half-keys used during final balancing of the assembled element shall be recorded on the residual unbalance work sheet (see Annex F). The maximum allowable residual unbalance per plane (journal) shall be calculated as follows:

In SI units:

$$U_{max} = 6350W/N \text{ for } N < 25,000 \text{ rpm} \quad (2)$$

$$U_{max} = W/3.937 \text{ for } N > 25,000 \text{ rpm} \quad (3)$$

In USC units:

$$U_{max} = 4W/N \text{ for } N < 25,000 \text{ rpm}$$

$$U_{max} = W/6250 \text{ for } N > 25,000 \text{ rpm}$$

where

U_{max} is the residual unbalance, in gram-mm (ounce-in.).

W is the journal static weight load, in kg (lbs.).

N is the rated speed, in revolutions per minute (rpm).

NOTE Balance tolerance above 25,000 rpm is based on an eccentricity of 0.25 μm (10 $\mu\text{in.}$) for each journal static weight load. Unbalance readings are measured at each journal-bearing position with no compensation to actual balance planes.

- **6.12.10** If specified, after the final balancing of each assembled rotating element has been completed, a residual unbalance check shall be performed and recorded in accordance with the residual unbalance work sheet (see Annex F).

6.12.11 Thrust bearings shall be selected such that at start-up or emergency trip conditions, the load does not exceed 75 % , and with any steady-state operating condition the load does not exceed 50 % of the bearing manufacturer's ultimate load rating. The ultimate load rating is the load that will produce the minimum acceptable oil-film thickness without inducing failure during continuous service or the load that will not exceed the creep-initiation or yield strength of the babbitt at the location of maximum temperature on the pad, whichever load is less.

6.12.12 Unless otherwise specified, thrust bearings and radial bearings shall be fitted with bearing-metal temperature sensors.

NOTE Some small pinion bearings do not have space for temperature sensors on the bearing.

- **6.12.13** If specified, installation of bearing-metal temperature sensors shall be in accordance with API 670.
- **6.12.14** If specified, oil cooler tubes shall have a 16 mm ($5/8$ in.) minimum outside diameter, and be made of inhibited admiralty (ASTM B111) with an average wall thickness of 18 BWG.

6.12.15 An austenitic stainless steel oil reservoir shall be supplied non-integral with the equipment casing.

6.12.16 If an electric heater is provided, it shall comply with API 614 requirements including control features.

7 Accessories

7.1 Drivers

7.1.1 General

7.1.1.1 The driver shall be of the type specified, shall be sized to meet the maximum specified operating conditions, including gear and coupling losses, and shall be in accordance with applicable specifications, as stated in the inquiry and order. The driver shall operate under the utility and site conditions specified in the inquiry.

7.1.1.2 The driver, in combination with the controls provided, shall be sized to accept any specified process variations such as changes in the pressure, temperature, relative humidity of the air, cooling water temperature, and plant start-up conditions.

- **7.1.1.3** The driver shall be capable of starting under the conditions specified and the starting method shall be agreed. The driver's starting-torque capabilities shall exceed the speed-torque requirements of the driven equipment. Purchaser shall state if starting conditions are to include minimum ambient temperature and minimum cooling water temperature.

7.1.1.4 The driver nameplate rating (exclusive of the service factor) shall be at least 110 % of the power required at the rated point.

- **7.1.1.5** If specified, the driver nameplate rating (exclusive of the service factor) shall be at least 110 % of the greatest power required for all of the specified operating conditions.

7.1.1.6 The supporting feet of compressor drivers shall be provided with vertical jackscrews.

7.1.2 Electric Motors

- **7.1.2.1** Electric motor drives shall conform to the guidelines in 7.1.2.1.1 through 7.1.2.1.4, or other standard as specified by the purchaser.

7.1.2.1.1 Low voltage induction motors shall be in accordance with IEEE 841.

7.1.2.1.2 General purpose medium voltage induction motors shall be in accordance with API 547.

7.1.2.1.3 Special purpose medium and high voltage induction motors shall be in accordance with API 541.

7.1.2.1.4 Synchronous motors shall be in accordance with API 546.

NOTE API 541, 546, and 547 are applicable to NEMA MG1 or IEC, ISO as specified. Currently, there is no equivalent IEC or ISO standard to IEEE 841, and IEEE 841 is NEMA MG1 based.

- **7.1.2.2** The motor's starting torque shall meet the requirements of the driven equipment at a reduced voltage of 80 % of the normal voltage or another value as may be specified. The motor shall accelerate to full speed within 20 seconds or some other period of time agreed upon by the purchaser and the vendor.

7.1.3 Steam Turbines

7.1.3.1 Steam turbine drivers shall conform to API 611. Steam turbine drivers shall be sized to deliver continuously the power required by 7.1.1.4 or 7.1.1.5 with the specified normal steam conditions.

7.1.3.2 The steam turbine shall be equipped with a NEMA Class D constant speed governor as specified in NEMA SM 23. Unless otherwise specified, an electronic governor shall be furnished.

7.2 Couplings and Guards

7.2.1 Couplings

7.2.1.1 Couplings between drivers and driven equipment shall be supplied by the manufacturer of the driven equipment and shall meet the requirements of 7.2.1.2 through 7.2.1.8.

7.2.1.2 Couplings shall be all-metal flexible element, spacer-type couplings manufactured to meet AGMA 9000 Class 9, and shall comply with the following:

- a) flexible elements shall be of corrosion resistant material;
- b) couplings shall be designed to retain the spacer if a flexible element ruptures;
- c) coupling hubs shall be steel;
- d) the distance between the driven and driver shaft ends (distance between shaft ends or DBSE) shall be at least 125 mm (5 in.) and shall permit removal of the coupling, bearings, seal and rotor, as applicable, without disturbing the driver, or driver coupling hub;

NOTE The DBSE dimension usually corresponds to the nominal coupling spacer length.

- e) provision shall be made for the attachment of alignment equipment without the need to remove the spacer or dismantle the coupling in any way;

NOTE 1 One way of achieving this is to provide at least 25 mm (1 in.) of bare shaft between the coupling hub and the bearing housing where alignment brackets can be located.

NOTE 2 This can be impractical on smaller units.

- **7.2.1.3** If specified, couplings shall be balanced to ISO 1940-1 Grade G6.3.

7.2.1.4 The coupling spacer shall be of sufficient length to allow maintenance of the compressor, including shaft alignment, without requiring the compressor or driver to be removed.

7.2.1.5 Flexible couplings shall be keyed to the shaft. Keys and keyways, and their tolerances shall conform to AGMA 9002, Commercial Class.

7.2.1.6 Flexible couplings with cylindrical bores shall be mounted with an interference fit. Cylindrical shafts shall comply with (AGMA 9002) and the coupling hubs shall be bored to the following tolerances (ISO 286-2):

a) for shafts of 50 mm (2 in.) diameter and smaller—Grade N7;

b) for shafts larger than 50 mm (2 in.) diameter—Grade N8.

7.2.1.7 If the coupling hubs require removal for maintenance, they shall be furnished with tapped puller holes of at least 10 mm (0.375 in.) diameter.

7.2.1.8 The couplings and coupling-to-shaft juncture shall be designed and manufactured to be capable of transmitting power at least equal to the power rating of the motor including service factor.

7.2.2 Coupling Guards

7.2.2.1 Each coupling shall have a coupling guard that is removable without disturbing the coupled elements and shall meet the requirements of 7.2.2.2 through 7.2.2.4.

7.2.2.2 Coupling guards shall enclose the coupling and the shafts to prevent personnel from contacting moving parts during operation of equipment train. Allowable access dimensions shall comply with specified standards such as ANSI B11.19, ISO 14120 or other applicable nationally recognized standards.

7.2.2.3 Guards shall be constructed with sufficient rigidity to withstand a 900 N (200 lb) static point load (or force) in any direction without the guard contacting moving parts.

7.2.2.4 Guards shall be fabricated from solid sheet or plate with no openings. Guards fabricated from expanded metal or perforated sheets may be used if the size of the openings does not exceed 10 mm (0.375 in.). Guards shall be constructed of steel, brass, aluminum or non-metallic (polymer) materials. Guards of woven wire shall not be used.

- **7.2.2.5** If specified, spark-resistant guards of agreed material shall be supplied.

NOTE Many users consider pure aluminum and aluminum alloys with less than 5 % iron and a maximum content of 2 % magnesium or 0.2 % copper to be spark resistant. However, local regulations sometimes invoke standards, such as EN 13463-1, which prohibit aluminum or non-metallic materials within potentially explosive atmospheres without an "ignition hazard assessment" (risk analysis) and report are processed in accordance with the standard. Nickel-copper alloys (UNSN0440X or UNSN0550X) and copper-based alloys (e.g. brass, bronze, aluminum bronze, beryllium bronze) are generally considered to be spark resistant. Nickel based alloys including alloy 600 (UNSN06600) and alloy 625 (UNSN06625) are considered spark resistant.

7.3 Baseplate/Support Structure

7.3.1 Unless otherwise specified, the compressor and all other machine components shall be supported on a rigid steel frame. The frame may have full-length structural members in contact with the foundation or it may have support feet. The term baseplate shall refer to either design.

NOTE 1 Some units are now designed with the unit's base integrally cast with the gearbox, and with the driver either flange-mounted or foot-mounted or on tubular rails.

NOTE 2 See Annex B for examples of baseplate/support structural types.

7.3.2 A baseplate shall be a single fabricated steel unit, unless the purchaser and the vendor agree that it may be fabricated in multiple sections. Multiple-section baseplates shall have machined and doweled mating surfaces which shall be bolted together to ensure accurate field reassembly.

NOTE A common arrangement especially for larger units has the air and oil coolers skid separate from the main baseplate for compressor and driver. Coolers skid can be bolted to main baseplate.

7.3.3 The baseplate shall have major load-bearing members under the mounting surfaces of the major components.

7.3.4 The baseplate/structure shall be provided with lifting attachments meeting the requirements of 7.3.4.1 through 7.3.4.3.

7.3.4.1 Attachments shall be provided for at least a four-point lift.

7.3.4.2 Lifting attachments for the module shall be designed using a maximum allowable stress of one-third of the specified minimum yield strength of the material.

NOTE Design of lifting attachments can be in accordance with standards such as ASME BTH-1, *Design of Below-the-Hook Lifting Devices*.

7.3.4.3 Lugs or trunnions that are attached by welding shall be continuous welds, and shall be 100 % NDE tested in accordance with the applicable code.

NOTE Lifting features for individual components are not covered by the above.

7.3.5 Lifting the baseplate complete with all equipment mounted shall not permanently distort, or otherwise damage the baseplate or the equipment mounted on it.

7.3.6 The bottom of the baseplate between structural members shall be open. If the baseplate is installed on a concrete foundation, accessibility shall be provided for grouting under all fixed, load-carrying structural members.

NOTE Some designs have gas coolers integral with the baseplate and have load carrying members which cannot be fixed due to a needed allowance for thermal expansion.

7.3.7 Mounting surfaces shall be provided for the integrally geared compressor and all drive train components. The mounting surfaces shall be at least 25 mm (1 in.) larger than the foot of the mounted equipment to allow leveling of the baseplate without removal of the equipment. The surfaces shall meet the following criteria:

- a) they shall be machined after the baseplate is fabricated;
- b) they shall be machined to a finish of 6.3 μm (250 $\mu\text{in.}$) Ra or better;
- c) they shall have corresponding pads in the same horizontal plane within 50 μm (0.002 in.) when measured in an unrestrained condition on a flat machined surface at the place of manufacture;

- d) they shall have each mounting surface machined within a flatness of 160 μm per linear meter (0.002 in. per linear ft) of mounting surface;
- e) mounting planes for different equipment shall have different mounting planes parallel to each other within 400 μm per m (0.005 in. per ft). This requirement shall be met by supporting on a flat machined surface and clamping the baseplate at the foundation bolt holes only.

7.3.8 The baseplate shall be drilled only for drivers that are shop fitted. The baseplates shall be supplied with leveling screws. Baseplates that are to be grouted shall have 50-mm (2-in.) radius outside corners (in the plan view). Mounting surfaces that are not to be grouted shall be coated with a rust preventive immediately after machining.

NOTE UngROUTed installation is common for this equipment and because of thermal growth, some base frame designs need one end of the support structure left free to slide.

7.3.9 Unless otherwise specified, anchor bolts will be furnished by the purchaser.

NOTE Typically anchor bolts are field installed and are needed in the field much earlier than the equipment delivery. However, it is common practice in some locals to require the vendor to supply the anchor bolts.

7.3.10 Driver support mounting surfaces shall be machined to allow the installation of vendor supplied austenitic stainless steel, precut, full bearing shim packs, 3 mm–6 mm (0.125 in.–0.250 in.) thick with no more than five shims in the pack between the driver and each mounting surface. Laminated shims are not acceptable. Shims shall be slotted so they can be installed and removed without removing the fasteners.

7.3.11 If the supported driver weighs more than 225 kg (500 lb), the driver mounting plates shall be furnished with axial and lateral jackscrews the same size as or larger than the vertical jackscrews. The lugs holding these jackscrews shall be attached to the mounting plates so that the lugs do not interfere with the installation or removal of the equipment, jackscrews, or shims. If the equipment is too heavy to use jackscrews, other means shall be provided.

NOTE The integral gearbox is the fixed point and adjustments are made on the driver.

7.3.12 The underside mounting surfaces of the baseplate shall be in one plane to permit use of a single-level foundation.

7.3.13 Drip pan(s) shall be placed beneath oil-carrying components. If drip pans cannot be installed, the vendor shall provide a drip lip or drain rim around the perimeter of the baseplate to contain oil leaks. One inch (25 mm) minimum drain fittings shall be provided.

7.4 Controls and Instrumentation

7.4.1 General

7.4.1.1 Instrumentation and installation shall conform to the requirements of API 614 or purchaser supplied specifications, or both, except as noted below:

7.4.1.2 Controls and instrumentation, and equipment and wiring shall be designed for outdoor installation.

- **7.4.1.3** Controls and instrumentation, equipment and wiring shall have either a minimum ingress protection level of IP 65 as detailed in IEC 60529, or a NEMA 4X minimum rating per NEMA Standard Publication 250, as specified.

7.4.1.4 The manufacturer's standard microprocessor based control and instrumentation system shall be provided.

- **7.4.1.5** The microprocessor shall be capable of communicating with the purchaser's distributed control system (DCS). The purchaser shall advise the communication protocol to be used.

7.4.1.6 All conduit/cable trays, armored cable and supports shall be designed and installed so that it can be easily removed without damage, and shall be located so that it does not hamper removal of bearings, seals, or equipment internals.

7.4.1.7 Neither piping without breakout points, cable trays nor rigid conduit shall be routed over the cases of axially (horizontally) split rotating machinery. They shall not be routed over or in front of removable heads on vessels and exchangers, or where the piping impairs the functionality of inspection openings or panel doors.

7.4.2 Control Systems

- **7.4.2.1** The purchaser will specify which of the following compressor capacity control modes shall be furnished by the vendor:

a) capacity modulation (inlet throttle device or variable inlet guide vanes) used when constant discharge pressure to surge is required and when the system air demand is relatively constant;

NOTE Typical performance curve impact for inlet throttle butterfly valve versus variable inlet guide vanes is shown in Annex E.

b) Automatic dual control-capacity modulation plus intermittent (load-unload) mode control for smaller air demand;

c) An automatic start and automatic stop control.

7.4.2.2 If more than one mode is specified, a means to change to any mode shall be supplied. If multiple compressors are to be operated in parallel, the control system proposed shall include all the necessary controls to permit the operation of all compressors on the same control mode or individual units on separate control modes.

7.4.2.3 A compressor surge recognition and protection system shall be furnished.

NOTE 1 Surge recognition (surge detection) continually monitors the discharge pressure for a significant drop over a short period of time, indicating that the compressor has surged. The system then protects from repeated surge by opening the blowoff valve.

NOTE 2 Typically an on/off blowoff valve is provided, and is controlled by monitoring motor amps or fluctuation in discharge pressure.

7.4.2.4 An automatic driver-overload control system shall be included to permit continuous operation at minimum ambient air and water temperatures without exceeding the nameplate rating (excluding service factor, if any).

7.4.2.5 Manual override at the control panel shall be provided to allow manual operation of the inlet throttle device and discharge blowoff valve. The system shall provide bumpless transfer from manual to automatic for smooth mode transfer. The surge protection system shall remain in effect even when the manual override is active.

7.4.2.6 To reduce driver load during startup of a motor-driven compressor, automatic unloading of the compressor by closing the inlet throttle device and opening the discharge blowoff valve shall be provided by the vendor. (An auxiliary source of control air or nitrogen may be required for initial startup.)

7.4.2.6.1 The control system shall provide a "soft" shutdown (or unloaded condition) in which the inlet valve is closed and the blowoff valve is opened prior to terminating the power source to the driver except for an emergency stop. This feature allows for less severe surging when stopping the unit.

7.4.2.6.2 The control system shall also provide warning to the operator that a hot-start condition exists for the motor driver because the unit was shut down and an adequate cool-down time period has not occurred for restart of the driver.

7.4.3 Instrument and Control Panels

7.4.3.1 A control panel from which startup and shutdown can be accomplished shall be provided and shall include the following:

- a) components for control systems as defined in 7.4.2.1, exclusive of the inlet throttle device or variable inlet guide vanes and discharge blowoff valve;
- b) a control mode selector (see 7.4.2.2);
- c) manual override and adjustment of control valves (see 7.4.2.5);
- d) digital-readout pressure measurements;
- e) digital-readout temperature measurements;
- f) a display for annunciation (see 7.4.5.2);
- g) control devices for alarms and shutdowns;
- h) a common alarm indication along with a method to acknowledge and reset the common alarm;
- i) the capability for starting and stopping the package from the control panel or remote;
- j) vibration measurement and readout instruments (see 7.4.4.6);
- k) self-diagnostics to check that the microprocessor and all instruments are functioning properly;
- l) logging of the compressors cumulative operating time;
- m) logging of the total number of compressor starts;
- n) on/off switch for panel power;
- o) on/auto/standby switch for auxiliary oil pump;
- p) auxiliary pump running indicator;
- q) lubrication oil heater status indicator.

7.4.3.2 The panel shall be fully enclosed.

7.4.3.3 Unless otherwise specified, the panel shall be mounted on the package baseplate.

NOTE Panels are sometimes installed off-skid or in a remote location (control building).

7.4.3.4 The panel shall include the following:

- a) a display designed to be visible in all conditions including darkness or direct sunlight;

NOTE Weather hoods with lighting are typically used for control panel displays for outdoor installations without a roof.

- b) shielding of the devices in the panel for protection from 5 watts radio-frequency (RF) interference at 1 m (3 ft) using commercial frequency bandwidths;

NOTE Some components incorporate suitable electrical noise immunity, which means they do not need additional shielding.

c) cooling for devices within the panel if the temperature inside the panel exceeds the electronic hardware temperature rating;

NOTE This is typically of concern for ambient conditions above 40 °C (100 °F).

d) a thermostatically controlled interior panel heater for units if required by the ambient condition;

e) driver, instrumentation, and control power separated if in the same cabinet;

f) a method of disconnecting power for devices inside the control panel.

- **7.4.3.5** If specified, internal light or lights and electrical receptacle shall be provided.

NOTE 1 Lighting can facilitate safely carrying out maintenance activities.

NOTE 2 For panels installed in hazardous areas which have electrical receptacles, external warning labels are mandated by electrical codes.

7.4.3.6 If required to meet the area classification, purging shall be provided in accordance with ISA S12.4 or with NFPA 496, Type X, Y, or Z; as required to meet the area classification.

7.4.4 Instrumentation

7.4.4.1 Except as noted in 7.4.4.2 through 7.4.4.5, the following instrumentation shall be in accordance with API 614:

- a) switches,
- b) transmitters,
- c) temperature indicators (gauges),
- d) thermowells,
- e) thermocouples and RTDs (for in-line instruments only),
- f) liquid level instruments,
- g) pressure indicators,
- h) flow indicators,
- i) solenoid valves,
- j) pressure-limiting valves and safety relief valves,
- k) control valves and regulators.

Refer to API 614 for bulleted paragraph options that may need to be selected regarding these instruments.

NOTE Certain instruments such as thermowells, level instruments and pressure indicators standardly use stainless steel which might not be acceptable in some services.

7.4.4.2 Signals may be generated from transmitters or transducers.

7.4.4.3 Thermowells

7.4.4.3.1 Temperature sensing elements that are located in pressurized or flooded lines shall have DN 12 (NPS 1/2) minimum thermowells made of austenitic stainless steel.

- **7.4.4.3.2** If specified, thermowells shall be at least DN 19 (NPS 3/4).

7.4.4.4 Pressure Indication

Pressure indications shall be on the local panel display screen.

7.4.4.5 Sight flow indicators

Sight flow indicators shall be furnished in the atmospheric oil-drain return lines that are external to the equipment.

NOTE Sight flow indicators are not feasible from individual compressor bearing drains, and sometimes are not feasible from the gear casing drain if the reservoir is integral or close-coupled to the casing.

7.4.4.6 Vibration and Position Detectors

7.4.4.6.1 Each bearing adjacent to an impeller shall be provided with a vibration-monitoring system consisting of the following:

- a) two non-contacting shaft vibration sensing probes—radially oriented and nominally 90° orientation;

NOTE 1 It is not always feasible to have two probes on some small pinion shafts.

- b) oscillator-demodulators;

- c) a readout instrument.

NOTE 2 The vibration monitoring system supplied as standard is often substantially different from an API 670 standard system and might not interface with other user systems.

7.4.4.6.2 Provision shall be made for mounting of two vibration accelerometers generally in the area of diagonal corners of the gearbox. Manufacturer's standard mounting shall be furnished.

7.4.5 Alarms and Shutdowns

7.4.5.1 General

- **7.4.5.1.1** Transmitters/transducers, sensors, control devices, and annunciation function shall be furnished as specified by purchaser and shall be mounted by the vendor.
- **7.4.5.1.2** The purchaser shall specify the alarms and shutdowns required. Minimum required alarms and recommended shutdowns are listed in Table 1.

7.4.5.1.3 Discrete, direct mounted, field switches for alarm and instrumented protective functions (IPFs) are not allowed. A signal transmitter/transducer and/or trip amplifier shall always be used.

7.4.5.1.4 Unless otherwise specified, the alarm/shutdown system shall comply with the requirements of 7.4.5.1.4.1 through 7.4.5.1.4.6.

Table 1—Equipment Monitoring

Condition	Alarm	Recommended Shutdown
High vibration of compressor	X	X
High last-stage air temperature (inlet)	X	X
Low lube-oil pressure ^f	X	X
High oil-supply temperature	X	
High oil filter differential pressure	X	
Low sealing-system pressure ^a	X	
Operation of the standby oil pump	X	
Low-lube level in reservoir ^b	X	
Reservoir oil temperature ^f	X	
High inlet-air filter differential pressure	X	
High vibration of driver ^c	X	
Panel purged	X	^d
Surge recognition	X	
Permissive start contact ^e	X	
^a If applicable. ^b With oil heater cutout. ^c If specified. ^d If required. ^e Separate pilot-light indication. ^f Start permissive.		

NOTE It is generally understood and accepted that with some systems, particularly those based on conventional direct acting instruments, complete compliance with the requirements of 7.4.5.1.4.1 through 7.4.5.1.4.6 is sometimes not achievable.

7.4.5.1.4.1 For every shutdown parameter an alarm shall be provided with the alarm point set at a lesser deviation from the normal condition than the associated shutdown point.

- **7.4.5.1.4.2** Any alarm parameter, reaching the alarm point, shall initiate an audible warning or flashing light or both as specified. It shall be possible to determine which parameter initiated the alarm.
- **7.4.5.1.4.3** Any shutdown parameter, reaching the shutdown point, shall cause the equipment to shutdown and shall initiate an audible warning or a flashing light or both, as specified, which shall be distinguishable from those associated with an alarm. It shall be possible to determine which parameter initiated the shutdown.

7.4.5.1.4.4 If any component of the alarm/shutdown system malfunctions, an alarm shall be initiated, and shall be distinguishable from alarms resulting from malfunction of the equipment. To accomplish this, redundant sensors may be required.

7.4.5.1.4.5 If any malfunction of a component of the shutdown system results in the system being unable to recognize a shutdown condition, the equipment shall automatically shutdown and an alarm shall be initiated. This alarm shall be distinguishable from shutdowns resulting from malfunction of the equipment (fail-safe system).

- **7.4.5.1.4.6** If a non-fail-safe system is specified, a failure that results in the system being unable to recognize an alarm or shutdown condition shall also result in all other shutdown and alarms remaining functional.

7.4.5.2 Annunciator

The vendor shall furnish first-out annunciation either as a separate device or as a function contained within the control system (e.g. a section of the PLC or microprocessor used for control of the compressor). The sequence of operation shall be as specified in 7.4.5.2.1 through 7.4.5.2.4.

7.4.5.2.1 The alarm condition shall be individually acknowledged by activating an alarm-silencing function via the keypad or display system.

7.4.5.2.2 When the alarm is acknowledged, the flashing display or alarm shall change to steady display of alarm. The annunciator shall be capable of indicating a new alarm (with a flashing display) if another function reaches an alarm condition, even if the previous alarm condition has been acknowledged but still exists.

7.4.5.2.3 Alarm and shutdown set points shall have default values set by the vendor. These values shall be field configurable with a user-defined password or key.

7.4.5.2.4 Connections shall be provided for a common remote alarm and a common remote shutdown indication.

NOTE Typically this would be in the form of a relay dry (unpowered) contact.

7.4.5.3 Alarm and Shutdown Devices

- **7.4.5.3.1** If specified, with the exception of the final shutdown device (motor contacts), alarm and shutdown instruments shall be arranged to permit testing of the control circuit, including if possible the actuating element, without interfering with normal operation of the equipment. To minimize risk, alarm and shutdown devices shall be bypassed one at a time.

7.4.5.3.2 The vendor shall provide a clearly visible light on the panel to indicate when shutdown circuits are in a test bypass mode.

7.4.5.3.3 The vendor shall furnish with the proposal a complete description of the alarm and shutdown facilities to be provided.

7.4.6 Electrical Systems

7.4.6.1 Electrical Systems shall meet requirements of API 614, except as modified below.

7.4.6.2 Electrical starting and supervisory controls may be either AC or DC.

7.4.6.3 To guard against accidental contact, enclosures shall be provided for all terminal strips, relays, switches and other energized parts.

7.4.6.3.1 Electrical power wiring shall be segregated from instrument and control signal wiring both externally and, as far as possible, inside enclosures.

7.4.6.3.2 Inside enclosures which may be required to be opened with the equipment in operation, for example, for alarm testing or adjustment, shall be provided with secondary shields or covers for all terminal strips, and other exposed parts carrying electrical potential in excess of 50 volts.

7.4.6.3.3 Maintenance access space shall be provided around or adjacent to electrical equipment or in accordance with the appropriate code such as the National Electrical Code, Article 110, or other internationally recognized standard as approved by the purchaser.

7.4.6.4 Electrical terminals shall meet the requirements of 7.4.6.4.1 through 7.4.6.4.3.

7.4.6.4.1 No terminal blocks shall be located in wire-ways.

7.4.6.4.2 The terminals shall be straight-through compression or spring clamp type with shrouded screws (dead front) and center tapping for test purposes.

7.4.6.4.3 Terminal block connections shall be single level (not tiered).

7.4.6.5 The panel shall contain two, bare-soft copper grounding connections. One shall be used for a signal ground and the other for an equipment ground bus. The instrument case shall not be grounded through the steel of the panel.

- **7.4.6.6** Control, instrumentation, and power wiring, that are not within a fully enclosed panel or other enclosure, shall be in the form of armored cable or shall be run in metal conduit as specified, and shall meet the following:
 - a) cables shall be supported on cable trays;
 - b) conduit shall be properly supported to avoid damage caused by vibration;
 - c) conduit shall be isolated and shielded to prevent interference between different services;
 - d) conduits may terminate (and in the case of the leads to temperature elements, shall terminate) with a length of flexible metal conduit, long enough to facilitate maintenance without removal of the conduit.

7.4.6.7 Internal vibration probe or thermocouple leads exposed to lube-oil turbulence shall be sufficiently anchored to prevent fatigue failures due to excessive movement.

7.5 Piping

7.5.1 General

7.5.1.1 Piping shall meet requirements of API 614 except as specifically modified below:

- **7.5.1.2** If specified, a manifolded cooling water piping system shall terminate with flanged single-supply and single-return connections at the edge of the package. It is not necessary to provide flanged connections for tubing systems, however, terminal connections shall be supplied.

7.5.1.3 The minimum requirement for piping material shall be as specified by API 614 except as allowed below, including requirements in 7.5.2.

7.5.1.4 Special pipe fittings in air, water or atmospheric oil service may be acceptable with purchaser approval.

NOTE Such fittings facilitate maintenance and allow for misalignment of close-coupled systems.

7.5.1.5 Flanges mating with iron compressor flanges shall be flat faced.

7.5.1.6 Butterfly valves are acceptable for water balance valves DN 80 (NPS 3) and larger and for inlet air throttling valves. They shall not be used for other services unless approved by the purchaser.

7.5.1.7 Gaskets and packing for flanges, valves, and other components shall not contain asbestos.

7.5.2 Oil Piping

7.5.2.1 Oil piping, tubing, and fittings downstream of filters (excluding slip-on flanges), shall be stainless steel (see API 614). Oil drain lines and piping upstream of filters may be manufacturer's standard.

7.5.2.2 Oil drains shall be sized to run no more than half-full and shall be arranged or sloped to ensure good drainage using manufacturer's proven practices.

7.5.2.3 Pipe joints downstream of the oil filter (filter to supply points) shall be butt-welded. Threaded connections shall be used for instrument connections and where tubing is used.

7.5.3 Instrument Piping

Instrument piping shall meet requirements of API 614, except bleeder valves are required between instruments and their isolating valves. Combinations of isolating and bleeder valves may be used.

7.5.4 Air Piping

7.5.4.1 Air Piping should be designed with adequate supports to prevent undue loads on compressor flanges, including transient loads such as blowoff.

7.5.4.2 The inlet air piping from the air filter shall be of corrosion-resistant material to avoid ingestion of rust into the compressor.

7.6 Intercoolers and Aftercoolers

7.6.1 Intercoolers and aftercoolers shall meet requirements of API 614, except as specifically noted in 7.6.2 through 7.6.8.

7.6.2 The vendor shall provide an inter-cooler between each compression stage. Unless otherwise specified, an aftercooler shall be provided after the final compression stage.

7.6.3 Unless otherwise specified, the coolers shall have continuous-bleed notched gate valves to permit removal of liquid.

NOTE As compared to drain traps, notched valves reduce delivered air by a small amount.

7.6.4 Unless otherwise specified, intercoolers and aftercooler shall be of the water-cooled shell and tube type with water on the tube side. A removable-bundle design is required. Tubes shall not have an outside diameter of less than 15 mm ($5/8$ in.), and the tube wall shall not have a thickness of less than 18 BWG, 1.25 mm (0.049 in.). Each cooler shall be sized to accommodate the total cooling load of the associated stage.

NOTE 1 Default is water-cooled shell and tube type, but other exchanger types and cooling media are commonly applied.

NOTE 2 Due to physical limitations, smaller units are commonly supplied with 10 mm ($3/8$ in.) tubes and thinner walls.

7.6.5 Double-pipe coolers and finned double-pipe designs may be furnished only if specifically approved by the purchaser.

7.6.6 Unless otherwise specified, cooler shells shall be of carbon steel and painted on the inside with a suitable coating for corrosion protection; tube sheet shall be carbon steel, painted on each side with a suitable coating for corrosion protection; and tubes shall be of manufacturer's standard copper alloy.

NOTE 1 Typical tube materials are 90-10 copper-nickel or hard drawn CI 220 copper.

NOTE 2 Some plant locations warrant consideration of alternative materials to combat atmospheric corrosion.

- **7.6.7** ES (extended surface) aluminum plate fin design shall be used. Purchaser shall specify if the plate fins require a corrosion resistant coating. Bundle-to-internal-shell seals shall be made of silicon rubber or stainless steel. U-bend tubes are not permitted.

7.6.8 The vendor shall include in the proposal complete details of any proposed plate and frame or air-cooled exchangers.

7.7 Inlet Air Filter

7.7.1 The vendor shall furnish a dry-type multistage, high-efficiency air intake filter suitable for mounting outdoors. Unless otherwise specified, the filter shall be shipped loose for field installation by purchaser. This filter shall be provided with the following:

- a) differential pressure alarm instrumentation and indication;
- b) filter shall be designed such that the first-stage (prefilter) elements may be changed while the unit is operating;
- c) weather hood(s) or louvers;
- d) clean pressure drop across the filter elements shall not exceed 5.0 millibar (2 in.) water gauge at the specified design conditions;
- e) removal of a minimum of 99.5 % of particulate sized 2 micron or larger over the inlet capacity range;
- f) element(s) designed to withstand pressure reversal from compressor surge;
- g) carbon steel components shall be coated to resist internal and external corrosion;
- h) internal fasteners and hardware downstream of the final filter element shall be stainless steel.

7.7.2 Many configurations and arrangements are available. Thus, the purchaser will need to specify any required specific features.

NOTE 1 The filter can be elevated some distance above the compressor for certain plant locations subject to unusual conditions such as sand storms.

NOTE 2 Inlet piping between filter and the compressor is typically supplied by the purchaser.

7.8 Discharge Blowoff Silencer

7.8.1 The vendor shall furnish a flanged discharge blowoff silencer. The silencer is typically shipped loose for field installation by the purchaser.

7.8.2 Silencer construction shall be suitable for service in an unprotected location. The silencer preferably should be located immediately downstream of the discharge blowoff valve and oriented as specified.

7.9 Special Tools

7.9.1 If special tools and fixtures are required to disassemble, assemble, or maintain the unit, they shall be included in the quotation and furnished as part of the initial supply of the machine. For multiple-unit installations, the requirements for quantities of special tools and fixtures shall be agreed upon by the purchaser and the vendor. These or similar special tools shall be used during shop assembly and post-test disassembly of the equipment.

7.9.2 If special tools are provided, each tool shall be labeled using metal stamps or have a permanently attached stainless steel tag to indicate its intended use. Tools which do not exceed one meter in length, width or height, and that weigh less than 40 kg shall be packaged in one or more rugged metal boxes and shall be marked "special tools for (tag/item number), box x of x." Larger tools do not need to be boxed, but shall be stamped or permanently stainless steel tagged to indicate both its intended use and the tag/item number of the equipment for which they are intended.

7.10 Additional Accessories Requirements for "Special Duty" Packages Only

- **7.10.1** If specified, the driver nameplate rating shall be no less than the power required (including losses from shaft-driven oil pump, coupling, and gear) if the compressor is operated unthrottled (inlet throttle device wide open) at the specified low-ambient operating conditions. The purchaser will specify the inlet air temperature and the inlet cooling water temperature to be used by the vendor in calculating the maximum unthrottled power.

NOTE The specified inlet temperature is not necessarily the minimum ambient temperature.

- **7.10.2** If specified, the vendor shall commercially blast, in accordance with ISO 8501, Grade Sa2 or SSPC SP6, all grout contact surfaces of the baseplate, and coat those surfaces with a primer compatible with epoxy grouting.

7.10.3 The extent of communication between the vendor's microprocessor and the purchaser's distributed control system (DCS) shall be agreed between the purchaser and the vendor.

NOTE Communication can be read only or read and write, or a combination of the two for different functions.

- **7.10.4** If specified, all alarm and shut-down sensors shall be directly connected to the central DCS or shut-down system, or both. This purchaser supplied system shall provide the necessary first-out annunciation.

7.10.5 If direct connection to a central system is specified in 7.10.4, the vendor shall furnish pre-wired interconnection box(es) for routing signals to the central system.

- **7.10.6** If specified, facilities shall be provided for local indication of alarm conditions. Requirements shall be agreed between the vendor and the purchaser.
- **7.10.7** If specified, a surge avoidance system shall be provided.

NOTE 1 Surge avoidance (anti surge control) system monitors the discharge pressure and the inlet flow and ensures through modulated opening of a blowoff valve that the compressors stays out of surge at all operating conditions.

NOTE 2 Typically this requires additional instrumentation for measuring flow, pressure and temperature, a modulating type anti-surge (blowoff) valve, and additional control logic.

- **7.10.8** If specified, the compressor control system shall have the capability of continuously recording data at multiple intervals and save them for a period of time from just prior to an alarm or trip as an aid for troubleshooting compressor operational problems.
- **7.10.9** If specified, a tapped and plugged hole shall be provided for mounting a probe to sense axial position of the gear wheel. Manufacturer shall advise if their thrust bearing arrangement makes it more advantageous to utilize axial position probes on the pinions instead of the gear wheel.

7.10.10 Gear casing shall have machined surfaces for mounting the purchaser's accelerometers in accordance with API 670. This requirement is for purchaser's field diagnostics of gear condition. Manufacturer shall locate mounting to best serve this function.

- **7.10.11** If specified, vibration and axial position transducers shall be supplied, installed, and calibrated in accordance with API 670 and provisions for phase reference (phase angle probes) shall also be made on all pinions in accordance with API 670.
- **7.10.12** If specified, vibration and axial position monitors shall be supplied, installed, and calibrated in accordance with API 670.
- **7.10.13** If specified, a bearing-temperature monitor shall also be supplied and calibrated in accordance with API 670.

NOTE Bearing distress is picked up as vibration in pinions—so this is not always specified.

7.10.14 The control system shall maintain a chronological record of the shutdowns. The panel shall have the capability of storing operational parameters related to the chronological shutdowns in a nonvolatile memory (with battery-backup if needed). The vendor and the purchaser shall determine the required parameters to be stored.

- **7.10.15** If specified, each alarm sensing device and each shutdown sensing device shall be furnished as separate devices.
- **7.10.16** If specified, a pilot light shall be provided on the incoming side of each supply to indicate that the circuit is energized. The pilot lights shall be installed on the control panel.
- **7.10.17** Piping wall thickness shall conform to the minimum requirement of API 614. Where space does not permit the use of NPS $\frac{1}{2}$, $\frac{3}{4}$, or 1 pipe, seamless tubing may be furnished in accordance with API 614. Stainless steel fittings shall be furnished with stainless steel tubing. The make and model of fittings shall be subject to purchaser's approval.
- **7.10.18** If specified, oil piping on external return lines and upstream of oil filters shall be stainless steel (excluding slip-on flanges).

7.10.19 All oil piping components such as flanges, valves, control valve bodies or heads, and relief valves shall be made of steel as a minimum.

- **7.10.20** If specified, intercooler and aftercooler tube sheet shall be of brass, and tubes shall be of inhibited admiralty.

7.10.21 Intercoolers and aftercooler shall be in accordance with TEMA Class C and shall be constructed with removable channel covers.

- **7.10.22** If specified, intercoolers and aftercoolers shall incorporate a horizontal air flow pattern through the tube bundle.
- **7.10.23** If specified, demister pads shall be included in the intercooler design to effect a 98 % removal efficiency of 5 micron or greater droplet size.

8 Inspection, Testing, and Preparation for Shipment

8.1 General

8.1.1 Inspection, testing, and preparation for shipment shall be in accordance with API 614, except as noted below:

- **8.1.2** If specified, the purchaser's representative, the vendor's representative, or both, shall indicate compliance in accordance with the inspector's checklist (see Annex G) by initialing, dating, and submitting the completed checklist to the purchaser prior to shipment.

8.2 Inspection

8.2.1 General

8.2.1.1 The vendor shall keep the following data available for at least 20 years:

- a) material certificates of compliance for shafts, pinions, gear wheels, and impellers;
- b) documentation to verify that the requirements of this specification have been met, for the required level of service;
- c) results of documented tests and inspections, including fully identified records of all heat treatment and nondestructive examinations.

8.2.1.2 Unless specifically agreed, pressure-containing parts shall not be painted until the specified inspection and testing of the parts is complete. Auxiliaries and components that arrive at the packager already tested and painted may be included in subsequent assembly testing without having to meet this requirement.

8.2.2 Material Inspection

8.2.2.1 Material inspection including major drive train components shall meet requirements of API 614, except as noted in 8.2.2.2 and 8.2.2.3.

8.2.2.2 Castings shall also be visually inspected per MSS SP-55 or purchaser approved standard.

8.2.2.3 Regardless of the generalized limits presented in this section, it shall be the vendor's responsibility to review the design limits of all materials and welds in the event that more stringent requirements are specified. Defects that exceed the limits imposed in the applicable material standard or API 614, shall be removed to meet the quality standards cited, as determined by additional magnetic particle or liquid penetrant inspection as applicable before repair welding.

8.2.3 Mechanical Inspection Prior to Run Test

8.2.3.1 Each component (including cast-in passages of these components) and all piping and appurtenances shall be inspected to ensure they have been cleaned and are free of foreign materials, corrosion products, and mill scale.

8.2.3.2 The gear contact pattern shall be checked in a static test with all pinions in place (Soft Dye method) or after the combined mechanical and performance test run (Hard Dye method). Unmodified profile leads shall show a minimum contact of 60 % of tooth contact along the axis, 30 % radially-with no edge loading. For crowned gear teeth, 50 % centered contact is acceptable.

8.3 Testing

8.3.1 General

8.3.1.1 The equipment shall be tested in accordance with 8.3.2 through 8.3.4.

8.3.1.2 The oil to be used in operation, in accordance with 6.9.1, should be used in the mechanical and performance tests. If different oil is used, oil viscosity shall be adjusted via oil temperature for comparability.

8.3.2 Hydrostatic Tests

8.3.2.1 Components designed and fabricated to an internationally recognized pressure design code or standard shall be pressure tested in accordance with that code or standard. Compressor casings, interstage piping and other pressure containing components not designed to a specific code or standard shall be tested hydrostatically with liquid

at a minimum of one and one-half times the maximum allowable working pressure of the component but not less than 1.5 bar (20 psi).

8.3.2.2 The chloride content of liquids used to test austenitic stainless steel materials shall not exceed 50 parts per million. To prevent deposition of chlorides on austenitic stainless steel as a result of evaporative drying, all residual liquid shall be removed from tested parts at the conclusion of the test.

8.3.2.3 Tests shall be maintained for a sufficient period of time to permit complete examination of parts under pressure. The hydrostatic test shall be considered satisfactory when neither leaks nor seepage occurs through the pressure containing parts or joints is observed for a minimum of 30 minutes. Complex systems may require a longer testing period to be agreed upon by the purchaser and the vendor. Seepage past internal closures required for testing of segmented cases and operation of a test pump to maintain pressure are acceptable.

8.3.2.4 Gaskets used during hydrotest of an assembled casing shall be of the same design as supplied with the casing.

8.3.2.5 Following hydrostatic testing, all equipment subassemblies shall be cleaned and dried to prevent corrosion.

8.3.3 IMPELLER OVERSPEED TEST

8.3.3.1 An overspeed test to 115 % of rated speed or 110 % of the driver maximum continuous speed (whichever is greater) shall be performed for a minimum duration of one minute. Impellers shall be examined for dimensional changes and cracking in high-stress areas, unless the conditions of 8.3.3.2 are met. Impeller dimensions identified by the manufacturer as critical (such as bore and outside diameter) shall be measured before and after the overspeed test. Any permanent deformation of the bore or other critical dimensions outside drawing tolerances shall be resolved to the satisfaction of the vendor and the purchaser.

- **8.3.3.2** With purchaser approval, no inspection/dimensional check is required of the impellers provided the following minimum criteria are met:

- a) the design impeller stress at max continuous speed does not exceed 60 % of published material yield strength at the highest stress point of the impeller;
- b) vibration signatures comparison for rated speed before and after the impeller overspeed test exhibit negligible change;
- c) forgings or castings used for the impeller are radiographic examined;
- d) impellers are of a design of proven success employing this approach.

NOTE Some manufacturers carry out the overspeed test in the casing during the mechanical run test.

- **8.3.3.3** If specified, after the overspeed test, each impeller shall be examined by liquid penetrant methods.

8.3.4 Combined Mechanical and Performance Tests

8.3.4.1 The combined mechanical and performance test of the package, in accordance with vendor's standard test procedure, shall be conducted at rated operating speed for a continuous two-hour minimum period. The purchaser and the vendor shall agree upon equipment and accessories to be included in the scope of the test and the test class.

8.3.4.2 Aerodynamic performance test shall be in accordance with either ASME PTC-10 or ISO 5389 as agreed between purchaser and vendor.

8.3.4.3 If control panel is provided in the order scope, it shall be used during the test. All control and safety functions shall be verified.

8.3.4.4 If parallel compressors are furnished with software to control sequencing of multiple compressors, the vendor shall analytically simulate satisfactory operation of the lead/lag control system.

8.3.4.5 All oil pressures, viscosities, and temperatures shall be within the range of operating values recommended in the vendor's operating instructions for the specific unit being tested. Performance data shall be obtained only after bearing and lube-oil temperatures have stabilized.

8.3.4.6 During the running test, peak-to-peak vibration levels shall be recorded for each stage at operating speed.

8.3.4.7 Performance shall be calculated using the test raw data, reduced to the specified site-rated conditions, including expected inlet air filter losses, cooling water temperatures and flows, tube side fouling factors, and all mechanical, blowdown, and condensate losses in accordance with the vendor's standard procedure.

8.3.4.8 The requirements of 8.3.4.8.1 through 8.3.4.8.4 shall be met before the combined mechanical and performance test of the package is performed.

8.3.4.8.1 All joints and connections shall be checked for tightness and any leaks shall be corrected.

8.3.4.8.2 Test stand oil filtration shall not exceed 10 microns nominal. Oil-system components downstream of the filters shall meet the cleanliness requirements of API 614 before any test is started.

8.3.4.8.3 If the job lube system is not used for the package test, a functional test of the job lube system shall be performed, including verification of calibration and operation of all valves and instrumentation.

8.3.4.8.4 Total indicated runout measurements (combined electrical and mechanical) of the pinion probe areas and calibration records for flow, pressure, temperature, and vibration-measuring devices utilized during the test shall be available to the purchaser's representative for review.

8.3.4.9 The requirements of 8.3.4.9.1 through 8.3.4.9.5 shall be met during the combined mechanical and performance test.

8.3.4.9.1 With the compressor operating at its certified discharge pressure, the delivered flow reduced to certified conditions specified on the data sheets shall have zero negative tolerance (that is, -0 % tolerance on the certified point flow).

8.3.4.9.2 The required power referred to the gear wheel shaft, at the certified point, including mechanical and convection losses, shall not exceed 104 % of the value quoted for the certified point.

8.3.4.9.3 Overall pressure rise shall meet the criteria of 6.1.9.3.

8.3.4.9.4 Compressor vibration levels shall be recorded at every performance data point, and except at the surge condition shall meet the criteria of 6.7.4.3, and if applicable 8.5.10.

8.3.4.9.5 The performance test shall verify the expected turndown flow at the specified rated discharge pressure.

8.3.4.10 If replacement or modification of bearings or seals or dismantling of the case to replace or modify other parts is required to correct mechanical or performance deficiencies, the initial test will not be acceptable, and the final shop tests shall be run after these replacements or corrections are made.

8.3.5 Final Inspection

The purchaser's representative may perform a final inspection prior to shipment, including verification of tagging, dimensional inspection, review of scope of supply, and documentation review.

8.3.6 Test Data

Immediately upon completion of each mechanical and performance test, copies of the data logged and the as-tested performance data shall be submitted to the purchaser.

8.4 Preparation for Shipment

8.4.1 Equipment shall be prepared for the type of shipment specified, including blocking of the rotor when necessary. Blocked rotors shall be identified by means of corrosion-resistant tags attached with stainless-steel wire. The preparation shall make the equipment suitable for six months of outdoor storage from the time of shipment, with no disassembly required before operation except for inspection of bearings and seals. If storage for a longer period is contemplated, the purchaser will consult with the vendor regarding the recommended procedures to be followed.

8.4.2 Except for machined surfaces, all exterior surfaces that may corrode during shipment, storage, or in service shall be given at least one coat of the manufacturer's standard paint. The paint shall not contain lead or chromates.

NOTE Austenitic stainless steels are typically not painted except for service in high-chloride environments such as chlorine plants, offshore, and sea coast facilities.

8.4.3 All exterior machined surfaces except for corrosion-resistant material shall be coated with a rust preventive which can be removed by solvent flushing. The type of rust preventive used shall be indicated on a tag attached to the equipment. Included in these procedures shall be recommended methods of removing preservatives in the field prior to startup. Vendor shall advise Purchaser what effect, if any, preservations have on the life of any components, particularly elastomeric materials such as gaskets and seals.

- **8.4.4** If specified, all equipment supplied by vendor including, but not limited to, the lube oil system components, instruments, and monitoring devices, shall be permanently tagged with equipment numbers issued by the Purchaser. Tags shall be 28 gauge minimum stainless steel. RTD or TC assemblies, pressure gauges, etc. may, if necessary, have their tags attached with stainless steel wire, 20 gauge minimum.
- **8.4.5** Auxiliary piping connections furnished on the purchased equipment shall be impression stamped or permanently tagged to agree with the vendor's connection table or general arrangement drawing. Service and connection designations shall be indicated.

8.4.6 Lifting points and lifting lugs shall be clearly identified on the equipment or the equipment package both in English and the language for the country of destination or via international pictograms. The recommended lifting arrangement shall be identified on the boxed equipment.

8.4.7 The package shall be identified with item and serial number. Material shipped separately shall be identified with securely affixed, corrosion-resistant metal tags indicating the item and serial number of the equipment for which it is intended. In addition, crated equipment shall be shipped with duplicate packing lists, one inside and one on the outside of the shipping container.

8.4.8 If spare rotating elements are purchased, they shall be prepared and crated for unheated indoor storage for a period of at least three years.

8.4.9 One copy of the vendor's standard installation instructions shall be packed and shipped with the equipment.

8.4.10 The vendor shall provide the purchaser with the instructions necessary to preserve the integrity of the storage preparation after the equipment arrives at the job site and before startup.

8.4.11 Small or fragile parts shall be removed, tagged, protected, packaged and fastened to the unit skid or baseplate or with other loose shipped components. All contract probes and instruments shall preferably be shipped installed in the equipment.

8.4.12 Shipping units shall be marked with industry standard cautionary symbols indicating center of gravity, sling or lifting points, top heavy packages, fragile and liquid contents, moisture sensitive contents etc. per ASTM D5445-05, *Standard Practice for Pictorial Markings for Handling of Goods*.

8.4.13 Package markings shall include:

- a) purchaser's order number and tag number,
- b) shipping unit piece number,
- c) gross weight.

8.5 Additional Inspection, Testing, and Preparation for Shipment Requirements for "Special Duty" Packages Only

- **8.5.1** If specified, the vendor shall keep final assembly maintenance and running clearances for at least 20 years.

8.5.2 Impellers that are welded or machined from other than investment castings, forgings, or bar stock, shall be 100 % radiographed and inspected. The radiographs, if compared with the standard reference radiographs within ASTM E446 for steel castings up to 50 mm (2 in.) in thickness or standard reference radiographs for heavy walled 50 mm–100 mm (2 in.–4 in.) steel castings within ASTM E186, shall show a casting quality equal to or better than Severity Level 2 for Categories A, B, and C (Types 1–4). Defects per categories D, E, and F are unacceptable. The methods of radiographic examination shall be in accordance with ASTM E94.

8.5.3 Inspection of the impeller is required following overspeed testing per 8.3.3.3.

8.5.4 All gear wheel and pinion teeth shall be 100 % magnetic particle inspected in accordance with ASTM A275. Cracks are not acceptable. Linear indications due to metallic inclusions larger than 1.5 mm (0.06 in.) located in the tooth flanks or roots shall be reported to the purchaser for disposition. Linear indications are defined as indications whose length is at least three times the width. Acceptance or rejection shall be decided on a case-by-case basis and shall be agreed upon by the purchaser and the vendor.

8.5.5 The vendor shall verify that dimensions of all rotating components and stationary gas path components fall within the drawing tolerances. Dimensional nonconformance shall be reported to the purchaser within five days after approval of the non-conformance by the vendor's engineering department.

8.5.6 At least one month before any mechanical or performance test, the vendor shall submit a full detailed test plan for the Purchaser's review and approval.

- **8.5.7** If specified, the combined test shall be for a continuous 4-hour period.
- **8.5.8** If specified, a minimum of five test points shall be recorded, including surge, certified point, and maximum capacity.
- **8.5.9** If specified, an unthrottled test curve shall be produced.

- **8.5.10** If specified, while the equipment is operating at rated speed, sweeps shall be made for vibration amplitudes at frequencies other than synchronous. As a minimum, these sweeps shall cover a frequency from 0.25 times to 8 times the rated speed of the shaft being observed. If the amplitude of any discrete, nonsynchronous vibration exceeds 20 % of the allowable vibration as defined in 6.7.4.3, the purchaser and the vendor shall agree on requirements for any additional testing and on the equipment's suitability for shipment.

8.5.11 During the combined test, the difference between inlet- and drain-oil temperatures shall not exceed 30 °C (50 °F).

- **8.5.12** If specified, the requirements of 8.5.12.1 through 8.5.12.3 shall be met after the combined mechanical and performance test is completed.

8.5.12.1 The bearings, seals, and gearing shall be inspected.

- **8.5.12.2** If due to the design of the integrally geared compressor, inspection of the bearings and seals requires disassembly of any pinion rotor, the purchaser shall specify either:

a) to inspect the bearings one time and retest in accordance with 8.3.4, or

b) to forego inspection of the bearings and seals based upon analysis of test data.

8.5.12.3 The gear contact pattern shall be checked using the hard-bluing method with all pinions in place. Unmodified profile leads shall show a minimum contact of 70 % of tooth contact along the axis, 30 % radially, with no edge loading.

8.5.13 If a damped unbalanced response analysis is required by 6.12.4, shop verification test of the unbalanced response analysis shall be performed.

8.5.14 Spare rotating elements with specified service conditions different from the contract rotating elements shall be given a combined mechanical and performance test.

- **8.5.15 Optional Tests**

The purchaser shall specify whether either of the shop tests specified in 8.5.15.1 or 8.5.15.2 shall be performed.

- **8.5.15.1 Guide Vane Test**

If specified, the package shall be tested at the number of guide vane settings specified by the purchaser. Each setting shall include surge, rated, and maximum capacity.

- **8.5.15.2 Spare Rotor Test**

If specified, spare rotating elements with duplicate performance to the contract rotating elements shall be given a mechanical test only in accordance with the requirements of this standard.

9 Vendor's Data

9.1 General

9.1.1 The purchaser may specify the content of proposals, meeting frequency and vendor data content/format identified in Annex D. Annex D provides a general outline of information that potentially may be requested by the purchaser.

9.1.2 If specified, the information described in Annex D shall be provided.

Annex A (informative)

Typical Datasheets

A.1 Introduction

This annex contains typical datasheets that may be used by the purchaser and the vendor. Native formats of these datasheets are available and intended for reproduction and free transmittal between all involved parties. The datasheets are presented in USC units and SI units. A representation of the datasheets is enclosed in this annex; however, MS Excel format datasheets have been developed and are available, for purchase from API publications distributors, with this standard. The MS Excel electronic datasheets may have additional functionality over printed hard copies.

IMPORTANT NOTE Electrical Area Classification Note that the selections available allow for both Europe and US designations

TEXT COLOR CODE: GREEN = TYP. BY PURCHASER RED = TYP. BY VENDOR BLUE = BY PURCHASER OR VENDOR

PROJECT UNITS (5.1)

Customary

CLIENT: _____ PROJECT TITLE: _____ JOB NUMBER: _____ EQUIPMENT NUMBER: _____ EQUIPMENT SERVICE: _____ SERIAL NUMBER: _____ REQ / SPEC NO. : _____ / _____ PURCH ORDER NO. _____	Cells highlighted yellow as thus are applicable to "Special Duty" only When you have completed the DS highlight the whole page and set format / cells / pattern to "none" (Delete this note on completion) HIERARCHY OF DOCUMENTS (if not specifically given in Inquiry/Purchase Order): <table style="width: 100%;"> <tr> <td style="width: 50%;">1 _____</td> <td style="width: 50%;">6 _____</td> </tr> <tr> <td>2 _____</td> <td>7 _____</td> </tr> <tr> <td>3 _____</td> <td>8 _____</td> </tr> <tr> <td>4 _____</td> <td>9 _____</td> </tr> <tr> <td>5 _____</td> <td>10 _____</td> </tr> </table> COMMENTS: _____ _____ _____ _____	1 _____	6 _____	2 _____	7 _____	3 _____	8 _____	4 _____	9 _____	5 _____	10 _____
1 _____	6 _____										
2 _____	7 _____										
3 _____	8 _____										
4 _____	9 _____										
5 _____	10 _____										

DATA SHEETS					
ITEM No.	ATT	ITEM No.	ATT	ITEM No.	ATT
COMPRESSOR					
GEAR					
MOTOR					
OTHER					

APPLICABLE OVERLAY STANDARD(S) _____

Rev	Date	Description	By	CHK	APPV		
INTEGRALLY GEARED AIR COMPRESSOR DATA SHEET			DATA SHEET No. _____				
			Sheet 1 of _____				

Figure A.1—Typical Datasheets for Centrifugal Air Compressors

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET									
SI UNITS (bar)									
1	APPLICABLE TO: ☐ PROPOSAL ☐ PURCHASE ☐ AS BUILT								
2	FOR: _____ UNIT: _____								
3	SITE: _____ NO. REQUIRED: _____								
4	SERVICE: _____ DRIVER ITEM NO.: _____								
5	☐ CONTINUOUS ☐ INTERMITTENT ☐ STANDBY (3.1.48) SPARED BY: _____ (months)								
6	NOTE: INFORMATION TO BE COMPLETED BY: ☐ PURCHASER ☐ MANUFACTURER ☒ PURCHASER OR MANUFACTURER								
7	GENERAL								
8	COMPRESSOR MFR: _____ MODEL (SIZE AND TYPE): _____ SERIAL NO.: _____								
9	DRIVER MFR: _____ DRIVER TYPE: _____ RATED (BKW): _____ RPM: _____								
10	DRIVE SYSTEM: ☐ DIRECT COUPLED ☐ OTHER _____ DUTY (1.2) ☐ BASIC ☒ SPECIAL DUTY								
11	OPERATING CONDITIONS (6.1.2)					SCOPE OF SUPPLY			
12	(6.1.2.1)					PACKAGE INCLUDES: (6.1.4)			
13	(ALL DATA ON PER UNIT BASIS)					☒ INTEGRALLY GEARED AIR COMPRESSOR ☒ COUPLING AND COUPLING GUARD ☒ BASEPLATE OR STRUCTURAL FRAMEWORK ☒ INTERCOOLERS ☒ AFTERCOOLER ☐ COOLERS MOUNTED ON SEPARATE SKID ☐ AFTERCOOLER SHIPPED LOOSE - MOUNT OFFSKID ☒ DRAIN SYSTEM ☐ MOISTURE SEPARATORS ☒ LUBRICATION SYSTEM ☒ INSTRUMENTATION ☒ UNIT CONTROL PANEL ☐ DRIVER ☒ INLET AIR FILTER ☐ SHIP LOOSE ☒ DISCHARGE BLOWOFF VALVE ☐ SHIP LOOSE ☒ DISCHARGE BLOWOFF SILENCER ☐ SHIP LOOSE ☒ DISCHARGE CHECK VALVE ☐ SHIP LOOSE ☒ INLET EXPANSION JOINTS ☐ SHIP LOOSE ☒ DISCHARGE EXPANSION JOINTS ☐ SHIP LOOSE ☐ MANIFOLD C.W. PIPING TO SINGLE INLET & OUTLET (7.5.1.2)			
14		RATED (3.1.40)	LOW AMB * (7.10.1)	MIN AMB	OTHER				
15	☐ DELIVERED FLOW, M ³ /H (1.013 barA & 0°C DRY)								
16	☐ WEIGHT FLOW, (kg/h) (WET) (DRY)								
17	☐ INLET COOLING WATER TEMP., (°C)								
18									
19	INLET CONDITIONS:								
20	☐ PRESSURE (barA)								
21	☐ TEMPERATURE (°C)								
22	☐ DEWPOINT (°C) (6.1.2.2)								
23	☐ MOLECULAR WEIGHT (MW)								
24	☐ INLET VOLUME, (m ³ /h) (WET / DRY)								
25	☐ INLET CW TEMP FOR MAX UNTHROTTLED POWER (°C)								
26	DISCHARGE CONDITIONS:								
27	☐ PRESSURE (barA)								
28	☐ TEMPERATURE (°C)								
29									
30	PERFORMANCE:								
31	☐ MAX (BkW) REQUIRED (ALL LOSSES INCL)					PACKAGE BASEPLATE / SUPPORT STRUCTURE (7.3)			
32	☐ (BkW/ 100 m ³ /h) AIR DELIVERED					☐ FULL LENGTH STRUCTURAL MEMBERS ON FOUNDATION ☐ SUPPORT FEET ONLY IN CONTACT WITH FOUNDATION ☐ ONE END LEFT UNGROUTED FOR THERMAL GROWTH ☐ UNIT MAY BE FABRICATED IN MULTIPLE SECTIONS (7.3.2)			
33	☐ INPUT SPEED (rpm)								
34	☐ ESTIMATED SURGE, (m ³ /h) (@ ABOVE SPEED)								
35	☐ MAX (DIRTY) ΔP ACROSS INLET FILTER, (bar)								
36	ΔP INCLUDED IN CALCULATION: ☐ YES ☐ NO					INTER- AND AFTER-COOLERS			
37	☒ AFTERCOOLER OUTLET TEMP., (°C)					COOLERS ☐ PURCHASER APPROVES DOUBLE PIPE (7.6.5) ☐ PURCHASER APPROVES FINNED DOUBLE PIPE (7.6.5) ☐ ALUMINUM PLATE-FINS REQUIRE COATING (7.6.7) ☒ COOLERS REQUIRE HORIZONTAL AIR FLOW (7.10.25) ☒ COOLERS REQUIRE DEMISTER PAD (7.10.26) ☐			
38	☐ PERFORMANCE CURVE NO.								
39	☐ % RISE TO SURGE (6.1.12.2)								
40	☐ % TURNDOWN								
41	☒ CERTIFIED OPERATING POINT (3.7 & 6.1.2.1)								
42	* UNTHROTTLED PERFORMANCE FOR DRIVER SIZING								
43									
44									
45	REMARKS:								
46									
47									
48									
49									
50									
51									
52									
DATA SHEET NO.		REV.			SHEET 2 of				

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET SI UNITS (m ³ /h)									
ITEM SPECIFIC INSTALLATION INFORMATION					SPECIFICATIONS				
LOCATION (6.1.5) 1 ☐ INDOOR ☐ HEATED ☐ UNDER ROOF 2 ☐ OUTDOOR ☐ UNHEATED ☐ PARTIAL SIDES 3 ☐ GRADE ☐ MEZZANINE ☐ _____ 4 ☐ OTHER _____					NOISE SPECIFICATIONS (6.1.3.1) ☐ MAXIMUM ALLOWABLE SOUND POWER LEVEL, dBA _____ (@ 1 m) ☐ EXPECTED OVERALL SOUND POWER LEVEL, dBA (6.1.3.1) _____ (@ 1 m) ☐ ACOUSTIC HOUSING (6.1.3.3): ☐ YES ☐ NO ☐ PROPOSED SILENCING MEASURES: (6.1.3.2) _____				
AREA ELECTRICAL CLASSIFICATION: (6.1.8.1) 6 ☐ ZONE or DIV _____ GROUP _____ TEMP CODE _____ 7 ☐ APPLICABLE ELECTRICAL CODES (6.1.8.4) _____ 8 ☐ LOCAL ELEC CODES (6.1.8.4): _____ ☐ ATEX/DIR 94/MEC					MISCELLANEOUS SPECIFICATIONS: ☐ NON-ASME WELDING F NOT AWS D1.1: (6.10.3.5) _____ BOLT THREADING TO CONFORM TO (6.1.11.1): ☐ ASME B1.1 ☐ ASME B1.13M ☐ ISO 281 ☐ _____				
PAINTING (6.4.2): 10 ☐ MANUFACTURER'S STD _____ 11 ☐ OTHER _____					☐ UTILITY CONSUMPTION (9.2.3.1)				
PREPARATION FOR GROUT SURFACES: (7.10.2) 13 BASEPLATE GROUT: (7.10.3) ☐ EPOXY ☐ CEMENT ☐ NONE 14 ☐ MFR STD 15 ☐ OTHER _____					STEAM: OIL HEATER: _____ (kg/h) OTHER _____ (kg/h) ELECTRIC: _____ (kW) LOCKED ROTOR _____ AMPS FULL LOAD _____ AMPS MAIN LO PUMP _____ AUX LO PUMP _____ OIL HEATER _____ (kW) SPACE HEATER _____ (kW) CONTROL SYSTEM LOAD: _____ (kW)				
SHIPMENT: (8.4.1) 19 ☐ DOMESTIC ☐ EXPORT ☐ EXPORT BOXING REQD 20 ☐ OUTDOOR STORAGE OVER _____ MONTHS (8.4.1)					COOLING WATER: LO COOLER INTER-COOLER INTER-COOLER INTER-COOLER AFTER-COOLER OTHER QUANTITY, (L/min) _____ OUTLET TEMP, (°C) _____ PRESS DROP, (bar) _____ TOTAL CW, (L/min) _____				
SPACE REQUIREMENTS, (mm) 23 ☐ COMPLETE UNIT: L _____ W _____ H _____ 24 COOLER SKID (IF SEPARATE) L _____ W _____ H _____ 25 INLET FILTER: L _____ W _____ H _____ 26 AFTERCOOLER (IF SEPARATE) L _____ W _____ H _____ 27 OTHER: L _____ W _____ H _____					AIR / NITROGEN PURGE: Air N ₂ INLET PRESS (barG) QUANTITY (m ³ /h) SEAL SYSTEM ☐ _____ CONTROL PANEL ☐ _____ LO RESERVOIR ☐ _____ INSTR HOUSINGS ☐ _____ CONTROL SYSTEM ☐ _____ TOTAL AIR PURGE, (m ³ /h) _____ TOTAL N ₂ PURGE, (m ³ /h) _____				
WEIGHT: (kg) 31 SKIDDED PACKAGE _____ INTEG GEAR / COMPR _____ DRIVER _____ 32 GEAR UPPER CASE _____ GEAR-WHEEL ROTOR _____ 33 1st ROTOR w/ PINION _____ 2nd ROTOR w/ PINION _____ 34 1st INTERCOOLER BUNDLE _____ 35 AFTERCOOLER _____ BUNDLE _____ 36 BLOWOFF SILENCER _____ CONTROL PANEL _____ 37 MAX FOR MAINTENANCE _____ 39 (IDENTIFY COMPONENT) _____ 40 TOTAL SHIPPING WEIGHT _____					PIPING REQUIREMENTS: ☐ RECOMMENDED STRAIGHT RUN OF PIPE DIAMETER BEFORE SUCTION: _____				
ALLOWABLE PIPING FORCES AND MOMENTS (8.4)									
AXIAL VERT TRANS		COMPRESSOR INLET (at Customer connection)		COMPRESSOR DISCHARGE (if a Customer connection)		PACKAGE OUTLET (at Customer connection)			
		FORCE, (kg)	MOMENT, (N-m)	FORCE, (kg)	MOMENT, (N-m)	FORCE, (kg)	MOMENT, (N-m)		
DATA SHEET NO. _____ 0 _____ REV. _____ 0 _____ SHEET 3 of _____									

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET									
SI UNITS (bar)									
CONSTRUCTION FEATURES									
☐ COMPRESSOR SPEEDS: RATED INPUT: _____ (rpm) TRIP _____ (rpm) BULLGEAR CRITICALS: 1st _____ (rpm) 2nd _____ (rpm) PINION ROTOR CRITICALS: 1st PINION SHAFT 1st _____ (rpm) 2nd _____ (rpm) 2nd PINION SHAFT 1st _____ (rpm) 2nd _____ (rpm) 3rd PINION SHAFT 1st _____ (rpm) 2nd _____ (rpm) 4th PINION SHAFT 1st _____ (rpm) 2nd _____ (rpm) OTHER UNDESIRABLE SPEEDS: (6.7.1.3) _____					☐ INTEGRAL GEAR HOUSING: MATERIAL _____ SPLIT _____ ☐ GEAR WHEEL: (6.5.3), (6.12.2) RATED POWER BASED ON TOOTH SURFACE DURABILITY: _____ (kW) RATED POWER BASED ON TOOTH BENDING: _____ (kW) ☐ MIN AGMA SERVICE FACTOR (6.5.3.2) _____ ☐ ACTUAL S.F. _____ GEAR RIM MATERIAL _____ HARDNESS: _____ GEAR FACE WIDTH: _____ (mm) GEAR CENTER MATL: _____ MECHANICAL EFFICIENCY: _____ % ISO 1328-1 GRADE: _____ PITCH DIA _____ (mm) PITCH LINE VELOCITY _____ (m/s) ROTATION, VIEWED FROM MOTOR SHAFT END: ☐ CW ☐ CCW ☐ PINIONS: (6.5.3), (6.12.3) 1st _____ 2nd _____ 3rd _____ 4th _____ HARDNESS: (BHN) (R _c) _____ ☐ GEAR WHEEL SHAFT: MATL _____ HARDNESS: _____ (BHN) (R _c) BRG SPAN _____ (mm) WEIGHT (w/GEAR) _____ (kg) DIA @ GEAR _____ (mm) DIA @ COUPLING _____ (mm) SHAFT LABYS: TYPE _____ MATL _____ RADIAL BRG TYPE: _____ LENGTH _____ (mm) ALLOW BRG LOAD _____ (bar) ACTUAL LOAD _____ (bar) ☐ GEAR WHEEL THRUST BEARINGS: (6.8.3) LOCATION _____ TYPE _____ MFR _____ AREA _____ (mm ²) THRUST COLLAR (6.8.3.5) ☐ INTEGRAL ☐ REPLACEABLE ALLOW LOAD _____ (bar) ACTUAL LOAD _____ (bar) GAS LOAD _____ (kg) COUPLING LOAD _____ (kg)				
☐ IMPELLERS: (6.5.2) NO. OF IMPELLERS: _____ MATERIAL _____ TYPE CONSTRUCTION: (6.5.2.2) _____ METHOD OF ATTACH: (6.5.2.2) _____					STAGE SPEED IMPELLER DIAMETER TIP SPEED 1st STAGE _____ (rpm) _____ (mm) _____ (m/s) 2nd STAGE _____ (rpm) _____ (mm) _____ (m/s) 3rd STAGE _____ (rpm) _____ (mm) _____ (m/s) 4th STAGE _____ (rpm) _____ (mm) _____ (m/s)				
☐ COMPRESSOR CASING: MODEL _____ CASING SPLIT _____ STG 1 STG 2 STG 3 STG 4 MATERIAL _____ MAWP: (barG) _____ HYDRO TEST: (barG) _____ MAX OPT TEMP: (°C) _____ MIN OPT TEMP: (°C) _____ MINIMUM ALLOWABLE TEMPERATURE FOR MATERIALS: _____ (°C) ☐ MINIMUM DESIGN METAL TEMP (6.10.4) _____ (°C) ☐ MAX ALLOW STRESS FOR MATL (6.2.1) _____ (MPa) ☐ CODE CASTING FACTOR (6.2.1) _____ WELDED CONNECTIONS TO CASINGS—NON-DESTRUCTIVE TESTING REQD: ☒ 100% RADIOGRAPH ☒ MAG PARTICLE ☒ LIQ PENETRANT					BEARINGS FITTED WITH TEMP SENSORS (6.12.12, 6.12.13) ☐ PINION RADIAL BEARING ☐ GEAR WHEEL RADIAL BEARING ☐ THRUST BEARING ☐ BRG TEMP SENSOR PER API 670 (6.12.13) ☐ PROVISION FOR AXIAL POSITION PROBE (7.10.9) ☒ MAIN CONNECTIONS: (6.3) ☐ PN SYSTEM (6.3.7.2)				
☐ COMPRESSOR BEARINGS & BEARING HOUSINGS: BEARING HSG MATERIAL: _____ PINION RADIAL BEARINGS: (6.8.2) STG 1 STG 2 STG 3 STG 4 BRG TYPE _____ ALLOW LOAD: (bar) _____ ACTUAL LOAD: (bar) _____ BRG SPAN: (mm) _____ PINION THRUST BEARINGS: (6.8.3) STG 1 STG 2 STG 3 STG 4 BEARING TYPE: _____ ALLOW LOAD: (bar) _____ ACTUAL LOAD: (bar) _____					SIZE FLANGE RATING FACING POSITION COMPRESSOR INLET COMPRESSOR DISCHARGE PACKAGE OUTLET ATMOSPHERIC BLOWOFF AIR FILTER OUTLET ☒ FLANGE / STUD CONN. CONFORM TO (6.3.7, 6.3.8): ☐ EN 1092-1 ☐ EN 1092-2 ☐ ASME B16.1 ☐ ASME B16.5 ☐ ASME B16.42 ☐ ASME B16.47 ☐ PROVIDE ASME B16.47 SERIES A FLANGES ABOVE NPS 24 (6.3.7.1.1) ☐ OTHER CONNECTIONS: LUBE OIL INLET LUBE OIL OUTLET COOLING WATER INLET COOLING WATER OUTLET PRESSURE GAUGE TEMPERATURE GAUGE CONDENSATE DRAINS NO. SIZE TYPE ☒ SS PIPING UPSTREAM OF FILTERS & EXTERNAL OIL DRAIN LINES (7.10.19) ☒ THREADED CONN. CONFORM TO (6.3.5): ☐ ☐ ASME B1.20.1 TAPER ☐ ISO 7-1 TAPER ☐ ISO 7-1 STRAIGHT				
DATA SHEET NO. 0					REV. 0				
					SHEET 4 of				

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET		REV
SI UNITS (bar)		
1	VIBRATION DETECTORS: (7.4.4.6), (7.10.9-7.10.12)	SHOP INSPECTIONS & TESTS: (8.1 and API 614)
2	☒ TYPE ☐ MODEL	ADVANCE NOTIFICATION REQD. _____ DAYS
3	☐ MFR: _____	
4	☐ NO. AT EACH PINION BEARING: _____ TOTAL NO. _____	
5	☐ NO. AT EACH DRIVER BEARING: _____ TOTAL NO. _____	
6	☐ X&Y RADIAL PROBES CAN BE MOUNTED ADJACENT TO IMPELLERS FOR (7.4.4.6.1a)	VISUAL INSPECTION OF CASTINGS (8.2.2.2)
7	☐ 1st STG ☐ 2nd STG ☐ 3rd STG ☐ 4th STG	HYDROSTATIC TEST (8.3.2)
8	OSCILLATOR-DEMODULATORS (7.4.4.6.1b):	PRE-RUN CHECKS
9	☐ MFR ☐ MODEL	CLEANLINESS CHECK OF COMPONENTS (8.2.3.1)
10	MONITOR SUPPLIED BY (7.4.4.6.1c):	GEAR CONTACT CHECK (8.2.3.2)
11	☐ MFR ☐ MODEL	☐ RESIDUAL UNBALANCE CHECK (8.12.9)
12	☐ LOCATION ☐ ENCLOSURE	IMPELLER OVERSPEED TEST (8.3.3)
13	☐ READOUT SCALE RANGE: _____ ALARM ☐ SET @ _____ (mm/sec ²)	POST OVERSPEED DIMENSIONAL CHECK OF IMP. (8.3.3.1)
14	☐ SHUTDOWN: ☐ SET @ _____ (mm/sec ²) ☐ TIME DELAY _____ SEC	POST OVERSPEED DIMENSIONAL CHECK NOT REQUIRED (8.3.3.2)
15	☐ PER API 670 (7.10.11 & 7.10.13) ☐ TRANSDUCERS ☐ MONITOR	POST OVERSPEED LIQUID PENETRANT TEST OF IMP. (8.3.3.3)
16	BEARING-TEMPERATURE MONITOR:	COMBINED TEST (8.3.4)
17	☐ REQ'D ☐ SUPPLIED BY: _____	PERF. TEST PER (8.3.4.2) ☐ ASME PTC 10 TEST (8.3.5.1) ☐ ISO 5388
18	☐ MFR ☐ MODEL	☐ CONTINUOUS 4-HOUR TEST (8.5.7)
19	AXIAL POSITION MOVEMENT DETECTOR: (7.10.9 & 7.10.11)	☐ MINIMUM 5-POINT TEST (8.5.8)
20	☐ PROVISION ONLY for GEAR WHEEL	☐ UNTHROTTLED TEST CURVE REQUIRED (8.5.9)
21	☐ TYPE ☐ MODEL	☐ VIBRATION SWEEPS TAKEN (8.5.10)
22	☐ READOUT SCALE RANGE: _____ ALARM ☐ SET @ _____ (mm/sec ²)	COMBINED TEST INCLUDES: (8.3.4.1)
23	☐ SHUTDOWN: ☐ SET @ _____ (mm/sec ²) ☐ TIME DELAY _____ (sec)	☐ AIR FILTER ☐ AFTERCOOLER
24	DYNAMICS: (6.7), (6.12.4 - 6.12.7)	☐ CONTROL PANEL ☐ _____
25	☐ CRITICAL LATERAL SPEEDS ARE PROVEN BY PRIOR UNITS (6.7.2)	☐ SOUND-LEVEL TEST FOR REFERENCE
26	☐ DAMPED UNBALANCED RESPONSE ANALYSIS REQD (6.12.4)	
27	☐ CONFIRM DAMPED UNBALANCED RESPONSE ANALYSIS ON TEST STAND (8.12.5)	DAMPED UNBALANCE RESPONSE TEST (8.5.13)
28	☐ TORSIONAL VIBRATION ANALYSIS OF TRAIN REQD (6.12.6)	OPTIONAL SPECIAL DUTY TESTS (8.5.14)
29	☐ RESIDUAL UNBALANCE CHECK REQD (6.12.10)	☐ GUIDE VANE TEST (8.5.15.1)
30	REMARKS	☐ AT _____ NON-100% POSITIONS (8.5.15.1)
31	COUPLINGS: (7.2.1)	☐ SPARE ROTOR MECH/PERF. TEST (8.5.14)
32	☒ TYPE ☐ DISC PACK ☐ DIAPHRAGM ☐ OTHER	☐ SPARE ROTOR MECH ONLY (8.5.15.2)
33	☒ DISK MATL: ☐ COATED W/ _____	☐ OPTIONAL SP DUTY POST TEST REQUIREMENTS (8.5.12)
34	☒ MAKE ☐ MODEL	☐ BRG / SEAL INSP REQUIRES DISASSEMBLY OF ROTOR (8.5.12.2):
35	☐ BALANCE TO GRADE G6.3 (7.2.1.3)	☐ INSPECT BEARINGS & RETEST (8.5.12.2a)
36	☐ SPACER LENGTH _____ (mm) ☐ LIMITED END-FLOAT REQD	☐ FORGO INSP BASED ON DATA (8.5.12.2b)
37	☐ CPLG RATING _____ (kW/100 rpm) @ 1.0 S.F. ACTUAL S.F. _____	FINAL INSPECTION PRIOR TO SHIPMENT (8.3.5)
38	☐ SHAFT JCT RATING: @ DRIVER _____ (kW) @ INPUT SHAFT _____ (kW)	
39	☐ MOUNTING ARRANGEMENT @ INPUT SHAFT: _____	
40	☐ MFR MAX BORE _____ (mm) PROPOSED BORE _____ (mm)	MISCELLANEOUS INSPECTION ITEMS:
41	☒ TAPPED PULLER HOLES REQUIRED FOR COUPLING HUBS (7.2.1.7)	☐ PRIOR DOCUMENTATION ON MAJOR REPAIRS REQD (8.10.3.3)
42	☐ DRIVER HALF-CPLG MTD BY: ☐ DRIVER MFR ☐ COMPR VENDOR	☐ SUBMITTAL OF INSPECTOR'S CHECKLIST (8.1.2)
43	☐ SPARK RESISTENT GUARD REQD (7.2.2.5) ☐ MATERIAL _____	SIGNED BY REP. FOR: ☐ PURCHASER ☐ VENDOR
44	MISCELLANEOUS:	☐ PERMANENTLY TAG ALL COMPONENTS WITH PURCHASER NOS (8.4.4)
45	☐ JOB SPECIFIC IOM I MANUALS ARE REQUIRED (9.4.3)	☐ RETAIN FINAL ASSEMBLY CLEARANCES FOR 20 YEARS (8.5.1)
46	☐ THERMAL RELIEF VALVES PROVIDED BY VENDOR	
47	☐ FOR WATER-COOLED EXCHANGERS	
48	☐ FOR _____	
49	☐ A COORDINATION MEETING WILL BE HELD (9.1.3)	
50	☐ PURCHASER WILL PREPARE COORDINATION MEETING AGENDA (9.1.3)	
DATA SHEET NO. <u>0</u>		REV. <u>0</u> SHEET <u>5</u> of <u> </u>

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET			
SI UNITS (bar)			
LUBE OIL SYSTEM (6.9)			
USE THIS SHEET ONLY IF AN API 614 OIL SYSTEM IS NOT TO BE APPLIED			
BASIC SYSTEM REQ'MENTS--NORMAL OIL FLOW		LUBRICANT: ☐ SYNTHETIC ☐ MINERAL	
LUBE OIL TO (6.9.3) (l/min) (barG) (SSU @ 37.8°C)		☐ DESCRIPTION	
☐ COMP/R/GEAR		☐ MIN ALLOW OIL TEMP FOR STARTING (°C) (SSU)	
☐ DRIVER			
☐ EXT GEAR			
☐ OIL SYSTEM PRESSURES:		☐ SYSTEM COMPONENT SUPPLIERS:	
SUPPLY (barG) PUMP RV SETTING (barG)		MFR MODEL	
SYS DESIGN (barG) HYDROTEST (barG)		MAIN PUMP	
		STANDBY PUMP	
STAINLESS STEEL PIPING (7.10.18)		ELECTRIC MOTOR(S)	
☐ EXTERNAL DRAIN LINES ☐ RETURN LINES & UPSTREAM OF FILTERS		STEAM TURBINE(S)	
OIL COOLER (API614 pt3):		OIL COOLER(S)	
☐ OPERATING PRESS. (barG) SHELL SIDE TUBE SIDE		OIL FILTERS	
☐ MAX ALLOW WORK PRESS. (barG)		ACCUMULATOR(S)	
☐ MAX ALLOW TEMP. (°C)		SUCTION STRAINERS	
☐ FOULING FACTOR		CHECK VALVES	
☐ SURFACE AREA (m²) DUTY (kJ/hr)		TRANSFER VALVE(S)	
☐ REMOVABLE BUNDLE TO BE FURNISHED		PUMP COUPLING	
☐ ASME CODE STAMPED ☐ DESIGNED TO TEMA		PUMP RELIEF VALVES	
☐ TUBES (API614, PT3 or 6.12.14 O.D. (mm) LENGTH (mm)		ELECTRIC HEATER	
NO. BWG ☐ AVG ☐ MIN			
MATERIALS (6.12.12):		PUMPS:	
CHANNELSHEADS SHELL		MAIN STANDBY	
TUBES TUBE SHEETS		☐ HORIZONTAL	
CHANNEL COVERS TUBE SUPPORTS		☐ VERTICAL	
OIL FILTERS (API614 pt3):		☐ SUBMERGED	
☐ MICRON RATING ☐ NOMINAL ☐ ABSOLUTE		☐ MOTOR DRIVEN	
☐ ΔP: (bar) CLEAN DIRTY COLLAPSE		☐ TURBINE DRIVEN	
☐ ELEMENT MAKE MODEL		☐ SHAFT DRIVEN	
☐ NO. ELEMENTS ☐ MEDIA		☐ CENTRIFUGAL	
☐ CORE MATL ☐ HSG MATL		☐ ROTARY	
☐ HSG MAWP (barG) ☐ MAX ALLOW TEMP (°C)		☐ FLANGE CONNECTED	
OIL HEATER (API614 pt3):		☐ (m³/h) (RATED)	
☐ STEAM HEATER REQD ☐ ELECTRIC HEATER REQD		☐ @ (barG) DISCHARGE	
☐ RATING (kJ/hr)		☐ (BkW) @ MAX SSU	
☐ WATT DENSITY (W/m²)		☐ OIL DRIVER (kW)	
OIL RESERVOIR (API614 pt3):		☐ CASING MATERIAL	
☐ RETENTION TIME min ☐ CAPACITY (l)		☐ SPEED	
☐ FREE SURFACE AREA (cm²) ☐ INTERNAL BAFFLES		☐ COUPLING	
RESERVOIR CONFIGURATION (6.9.4 & 6.12.15)		☐ AS 4024.1 GUARD	
☐ SEPARATE VESSEL ☐ INTEGRAL W/ GEAR CASING		☐ MECHANICAL SEAL	
MATERIAL (6.9.5 or 6.12.15) ☐ AUS. ST. STL ☐ C. STL W/ COATING		STANDBY PUMP CONTROL RESET:	
		☐ MANUAL ☐ AUTOMATIC ☐ JOA SELECTOR SWITCH	
NOTES:			
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Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET				REV
SI UNITS (bar)				
CONTROL PANELS AND INSTRUMENTATION (7.4)				
LOCAL CONTROL PANEL				
☐ ELECTRICAL AREA CLASSIFICATION: (6.1.8.1) ZONE / DIVISION _____ GR _____ TEMP CODE _____				
☐ TROPICALIZATION REQUIRED				
PANEL ENCLOSURE REQUIREMENT: (7.4.3.2)				
☒ MINIMUM INGRESS PROTECTION LEVEL (7.4.1.3) _____				
☒ ENCLOSURE MATERIAL: _____				
☒ NEMA TYPE 7 REQUIRED				
PURGE REQUIREMENT: (7.4.3.5)				
☒ NONE ☐ INSTRUMENT AIR ☐ NITROGEN				
☒ CONNECTIONS ONLY for FUTURE PURGE				
☒ TYPE X--REDUCES THE CLASS FROM DIV 1 TO NONHAZARDOUS				
☒ TYPE Y--REDUCES THE CLASSIFICATION FROM DIV 1 TO DIV 2				
☒ TYPE Z--REDUCES THE CLASS FROM DIV 2 TO NONHAZARDOUS				
PANEL FEATURES: (7.4.3.4)				
☐ VIBRATION ISOLATORS ☒ STRIP HEATER (7.4.3.4d)				
☒ INTERNAL COOLING (7.4.3.4c)				
☐ WEATHERHOOD / SHADE (7.4.3.4a) ☐ PILOT LIGHT (7.10.16)				
☐ INTERNAL LIGHTING AND ELECTRICAL RECEPTACLE (7.4.3.5)				
☐ OTHER _____				
WIRING OUTSIDE PANEL TO BE (7.4.6.6):				
☐ ARMORED & RUN IN CABLE TRAY ☐ RUN IN CONDUIT				
MISCELLANEOUS INSTRUMENTATION:				
☐ DN 19 (NPS 3/4) THERMOWELLS REQUIRED (7.4.4.3.2)				
SYSTEM CONTROL MODE (7.4.2.1)				
☐ CAPACITY MODULATION (CONSTANT DISCHARGE PRESSURE) (7.4.2.1 a)				
INLET THROTTLE DEVICE: (BY VENDOR)				
☐ DAMPER ☐ BUTTERFLY VALVE				
☐ GLOBE VALVE ☐ VARIABLE INLET GUIDE VANES				
☐ AUTOMATIC DUAL CONTROL (7.4.2.1 b.)				
☐ _____ (barG) TO _____ (barG) DISCH PRESS				
☐ AUTO START AND STOP (7.4.2.1 c.)				
START _____ (barG) STOP _____ (barG)				
☐ OTHER (DESCRIBE): _____				
☐ UNIT OPERATES IN PARALLEL (7.4.2.2 & 9.3.4.4)				
WITH _____				
MICROPROCESSOR				
☐ COMMUNICATION PROTOCOL (7.4.1.5):				
☐ EXTENT OF COMMUNICATION (7.10.3):				
☐ SYSTEM TO RECORD / SAVE DATA BEFORE ALARM (7.10.8)				
☐ SENSORS DIRECT CONNECTED TO DCS (7.10.4)				
☐ FACILITIES FOR LOCAL INDICATION OF ALARM CONDITION (7.10.6)				
☐ A SURGE AVOIDANCE SYSTEM SHALL BE PROVIDED (7.10.7)				
ALARM & SHUTDOWN DEVICES:				
☐ ALARM BEACON ☐ AUDIBLE ALARM WARNING (7.4.5.1.4.2 & 7.4.5.2.2)				
☐ ALARM & SHUTDOWN DEVICES ARRANGED TO PERMIT TESTING				
DURING OPERATION (7.4.5.3.1)				
☐ SEPARATE SENSOR FOR ALARM & SHUTDOWN (7.10.15)				
☐ AUDIBLE S.D. WARNING ☐ S.D. FLASHING BEACON (7.4.5.1.4.3)				
☐ NON-FAIL-SAFE SYSTEM REQUIRED (7.4.5.1.4.6)				
INSTRUMENT SUPPLIERS:				
☐ (REFER TO PURCHASER'S APPROVED SUPPLIER LIST):				
29	PRESSURE INDICATOR:	MFR	SIZE & TYPE	
30	TEMPERATURE INDICATOR:	MFR	SIZE & TYPE	
31	LEVEL INDICATORS:	MFR	SIZE & TYPE	
32	DIFF PRESSURE INDICATORS:	MFR	SIZE & TYPE	
33	PRESSURE TRANSMITTERS:	MFR	SIZE & TYPE	
34	TEMPERATURE TRANSMITTERS:	MFR	SIZE & TYPE	
35	LEVEL TRANSMITTERS:	MFR	SIZE & TYPE	
36	CONTROL VALVES:	MFR	SIZE & TYPE	
37	PRESSURE RELIEF VALVES:	MFR	SIZE & TYPE	
38	THERMAL RELIEF VALVES:	MFR	SIZE & TYPE	
39	TEMPERATURE CONTROL VALVES:	MFR	SIZE & TYPE	
40	SIGHT FLOW INDICATORS:	MFR	SIZE & TYPE	
41	PURGE FLOW INDICATORS:	MFR	SIZE & TYPE	
42	SOLENOID VALVES:	MFR	SIZE & TYPE	
43	TUBE FITTINGS	MFR	SIZE & TYPE	
44		MFR	SIZE & TYPE	
45		MFR	SIZE & TYPE	
46		MFR	SIZE & TYPE	
NOTES:				
48				
49				
50				
51				
52				
53				
DATA SHEET NO. 0				REV. 0
				SHEET 7 of

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET																		
SI UNITS (bar)																		
INSTRUMENTATION SCOPE OF SUPPLY (7.4.5.1.2)																		
	ELEMENT (SENSOR) 7.4.5.1.1									READ OUT								
	PROVIDED BY			TYPE	LOCATION			INSTALLED BY			PROVIDED BY			LOCATION		ALARM (7.4.5.1.1)	SHUTDOWN (7.4.5.1.1)	REPEAT SIGNAL (3)
	VENDOR	PURCHASER	OTHER	DIRECT READOUT (1)	TRANSDUCER (1) (2)	TRANSMITTER (1) (2)	VENDOR PACKAGE	LOCAL PANEL	PURCHASER PIPING	VENDOR	PURCHASER	OTHER	VENDOR	PURCHASER	OTHER			
1 PRESSURE:																		
2 COMPRESSOR SUCTION _____ STAGE																		
3 COMPRESSOR DISCHARGE _____ STAGE																		
4 LUBE OIL DISCHARGE																		
5 LUBE OIL FILTER ΔP																		
6 LUBE OIL SUPPLY																		
7 AIR FILTER ΔP																		
8																		
9 TEMPERATURE:																		
10 COMPRESSOR SUCTION _____ STAGE																		
11 COMPRESSOR DISCHARGE _____ STAGE																		
12 OIL COOLER INLET & OUTLET																		
13 COMPRESSOR PINION JOURNAL BRG																		
14 BULL GEAR JOURNAL BRG																		
15 BULL GEAR THRUST BRG																		
16 DRIVER JOURNAL BRG																		
17 DRIVER THRUST BRG																		
18 RESERVOIR OIL																		
19																		
20 LEVEL:																		
21 LUBE OIL RESERVOIR																		
22 SEPARATOR																		
23																		
24 VIBRATION:																		
25 RADIAL VIBRATION EACH STAGE																		
26 RADIAL VIBRATION BULL GEAR SHAFT																		
27 AXIAL POSITION BULL GEAR SHAFT																		
28 RADIAL VIBRATION ON DRIVER																		
29 AXIAL POSITION ON DRIVER SHAFT																		
30 ACCELEROMETER ON GEAR BOX																		
31																		
32 FLOW:																		
33 OIL RETURN																		
34 PURGE GAS																		
35 AIR (BY PURCHASER)																		
36																		
37 MISCELLANEOUS:																		
38 STANDBY L.O. PUMP RUNNING																		
39 PANEL PURGE FAILURE																		
40 ANNUNCIATOR PURGE FAILURE																		
41 SURGE RECOGNITION																		
42 OIL HEATER ON																		
43 COMMON REMOTE ALARM INDICATION																		
44 COMMON REMOTE SHUTDOWN INDICATION																		
45																		
46																		
47 NOTES: 1) TRANSMITTERS SUPPLIED BY VENDOR SHALL INCLUDE SENSING ELEMENT																		
48																		
49																		
DATA SHEET NO. 0																		
REV. 0																		
SHEET 8 of																		

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

Duplicate this sheet for however many stages are required. Or use form provided with API 614.

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET

SI UNITS (bar)

(INTER-) (AFTER-) COOLER(S) (7.6)

1 SERVICE OF UNIT: _____ ITEM NO. _____ REV _____

2 SIZE _____ TYPE: _____ ☐ HORIZ ☐ VERT CONNECTED IN ☐ PARALLEL ☐ SERIES

3 SURF/UNIT: (GROSS/EFF) _____ (m²) SHELLS/UNIT: _____ SURFACE/SHELL: (GROSS/EFF) _____ (m²)

PERFORMANCE OF ONE UNIT

	SHELL SIDE	TUBE SIDE
7 ☐ FLUID NAME		
8 ☐ FLUID QUANTITY, TOTAL (kg/h)		
9 VAPOR-IN/OUT		
10 LIQUID-IN/OUT		
11 ☐ TEMPERATURE-IN/OUT (°C)		
12 ☐ SPECIFIC GRAVITY		
13 ☐ VISCOSITY, LIQUID (mPa-s)		
14 ☐ SPECIFIC HEAT, (kJ/kg °C)		
15 ☐ THERMAL CONDUCTIVITY, (kJ/m h °C)		
16 ☐ LATENT HEAT, (kJ/kg °C)		
17 ☐ INLET PRESSURE, (barG)		
18 ☐ VELOCITY, (m/s)		
19 ☐ PRESSURE DROP-ALLOW/CALC, (bar)		
20 ☐ FOULING RESISTANCE-MINIMUM (8.1.6) (m² %/kW)		
21 ☐ HEAT EXCHANGED _____ (kJ/hr) MTD CORRECTED _____ (°C)		
22 ☐ TRANSFER RATE, (kJ/hr m² °C) SERVICE CLEAN _____		

23 ☐ CONSTRUCTION OF ONE SHELL

	SHELL SIDE	TUBE SIDE
24 DESIGN CODE:		
25 DESIGN/TEST PRESSURE, (barG)		
26 DESIGN TEMPERATURE, (°C)		
27 NO. PASSES PER SHELL		
28 CORROSION ALLOWANCE, (mm)		
29 NOZZLES:		
30 SIZE & INLET		
31 RATING OUTLET		
31 RATING VENT-DRAIN		

SKETCH: BUNDLE NOZZLE ORIENTATIONS

32 TUBE NO. _____ O.D. _____ (mm) THK (MIN) (AVG) _____ (mm) LENGTH _____ (m) PITCH _____ (mm) ☐ 30 ☐ 60 ☐ 90 ☐ 45

33 TUBE TYPE _____ TUBE MATERIAL (7.6.6 & 7.10.20) _____

34 SHELL MATL _____ I.D. _____ (mm) O.D. _____ (mm) SHELL COVER MATL _____ (INTEG)(REMOV)

35 CHANNEL OR BONNET MATL _____ CHANNEL COVER MATL _____

36 TUBESHEET-STATIONARY MATL (7.6.6 & 7.10.20) _____ TUBESHEET-FLOATING MATL _____

37 FLOATING HEAD COVER MATL _____ IMPINGEMENT PROTECTION _____

38 BAFFLES-CROSS MATL _____ TYPE _____ % CUT (DIA) (AREA) _____ SPACING: C/C _____ INLET _____ (mm)

39 BAFFLES-LONG MATL _____ SEAL TYPE _____

40 SUPPORTS-TUBE _____ U-BEND _____ TYPE _____

41 BYPASS SEAL ARRANGEMENT _____ TUBE-TUBESHEET JOINT _____

42 GASKETS-SHELL SIDE _____ -TUBE SIDE _____

43 -FLOATING HEAD _____

44 ASME SECTION VIII CODE REQUIREMENTS: ☐ DESIGN & TEST ☐ STAMP ☐ NOT APPLICABLE TEMA CLASS _____

45 WEIGHT/SHELL _____ (kg) FILLED WITH WATER _____ (kg) BUNDLE _____ (kg)

46 ☐ CONDENSATE SEPARATION AND COLLECTION FACILITIES (see API 614)

47 ☐ APPLY HORIZONTAL AIRFLOW PATTERN (7.10.22) ☐ DEMISTER PADS FOR 98% REMOVAL OF 5 µm DROPLET (7.10.23)

48 REMARKS: _____

49 _____

50 _____

51 _____

52 _____

53 _____

54 _____

55 _____

DATA SHEET NO. 0 REV. 0 SHEET 9 of _____

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

This form may be deleted if another motor datasheet is used.

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET

SI UNITS (bar)

REV. _____

ELECTRIC MOTOR DRIVER (7.1.2)

1 MFR _____ MODEL _____ SERIAL NO. _____ FRAME _____

2 DRIVEN EQUIPMENT TYPE _____ DRIVEN EQUIPMENT ITEM NO. _____ MOTOR ITEM NO. _____

3 APPLICABLE SPECIFICATION (7.1.2.1) ☐ IEEE 841 ☐ API 547 ☐ API 541 ☐ API 546 ☐ OTHER _____

☐ OPERATING CONDITIONS

4 **SITE DATA:**

5 ELECTRICAL SUPPLY: VOLT _____ (V) PHASE _____ HERTZ _____

6 ELECTRICAL AREA CLASSIFICATION: ☐ NON-HAZARDOUS

7 ☐ ZONE / DIVISION _____ GROUP _____

8 ATMOSPHERIC MIXTURE: _____

9 IGNITION TEMPERATURE: _____ (°C) TEMP CODE: _____

10 ALTITUDE: ☐ LESS THAN (1000 m) ☐ _____ (m)

11 AMBIENT TEMPERATURE MINIMUM _____ (°C) MAXIMUM: _____ (°C)

12 UNUSUAL CONDITIONS: _____

13 **DRIVE SYSTEM:**

☐ DIRECT CONNECTED ☐ EXTERNAL GEAR

☐ OTHER _____

STARTING: (7.1.2.7.1.2.2)

☐ FULL VOLTAGE ☐ REDUCED VOLTAGE (7.1.2.2) _____ %

☐ LOADED ☐ UNLOADED

☐ VOLTAGE DIP _____ %

☐ LOAD INERTIA _____ (kg-m²)

☒ ACCELERATION TO FULL SPEED (7.1.2.2) _____ (sec)

☐ PERFORMANCE

14 NO LOAD CURRENT: (A) _____

15 FULL LOAD TORQUE: (N-m) _____

16 STARTS PER HOUR: _____ HOT _____ COLD

17 ACCELERATION TIME: _____ (sec)

LOAD	CURRENT (A)	EFFICIENCY	POWER FACTOR
FULL			
75%			
50%			
LOCKED ROTOR			

☐ CONSTRUCTION FEATURES

19 ☐ NAMEPLATE (kW) _____ (rpm) _____ S.F. _____

20 ☐ SIZE MOTOR FOR 110% POWER AT RATED POINT (7.1.1.4)

21 ☐ SIZE MOTOR FOR 110% POWER AT ALL SPECIFIED CONDITIONS (7.1.1.5)

22 ☐ SIZE MOTOR POWER FOR UNTHROTTLED AT LOW AMBIENT (7.10.1)

23 NEMA TORQUE DESIGN: ☐ A ☐ B ☐ C ☐ D

24 ☐ NEMA LOCKED ROTOR kVA CODE LETTER: _____

25 EFFICIENCY: ☐ STANDARD ☐ HIGH ☐ PREMIUM

26 NOISE DESIGN: ☐ STANDARD ☐ LOW NOISE

27 ☐ MAX SOUND PRESSURE LEVEL (dBA) (@ 1 m) _____

28 ☐ EXPECTED SPL (dBA) (@ 1 m) _____

29 ENCLOSURE & COOLING

30 ☐ TEFC / IC411 ☐ TENV / IC400 ☐ EXPLOSION PROOF

31 ☐ WPII / IPW24 ☐ TEWAC / IC81W ☐ TEAAC / IC418

32 MOUNTING: ☒ HORIZONTAL ☐ FOOT MOUNTED

34 MAIN TERMINAL BOX MOUNTING LOCATION: ☐ F-1 ☒ F-2

37 FAN: ☐ REVERSIBLE ☐ UNI-DIRECTIONAL

38 ☐ NON-SPARKING METAL MATERIAL

39 BEARING TYPE: ☐ BALL ☐ ROLLER ☐ SLEEVE

40 BRG LUBRICATION: ☒ GREASE ☐ RING OIL

41 ☐ PRESSURE LUBE (Ø 9.3c)

42 GREASE FITTING: ☒ PLUGGED ☐ ALEMITE ☐ OTHER _____

43 BRG SHIELDING: ☐ SINGLE ☒ DOUBLE ☐ SEALED FOR LIFE

44 **TESTING**

45 IEEE TESTING: ☐ OBSVD ☐ WIT ☐ SUBMIT CERT'D RESULTS

46 ☐ SPECIAL TESTING _____

47 _____

48 _____

49 _____

50 _____

44 **MISCELLANEOUS**

45 PAINTING: ☐ IEEE 841 STD ☐ OTHER _____

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DATA SHEET NO. 0 REV. 0 SHEET 10 of _____

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET	
SI UNITS (bar)	
INLET AIR FILTER / SILENCER: (7.7) ☐ MFR _____ ☐ DESCRIPTION _____ ☐ PIPING CONNECTION _____ ☐ CLEAN ΔP AS QUOTED _____ (bar) ☒ CORROSION PROTECTION _____ ☐ FILTER WILL BE REMOTE MOUNTED BY PURCHASER AT A DISTANCE _____ (m) FROM COMPRESSOR ☐ FILTER WILL BE ELEVATED _____ (m) ABOVE GRADE MODEL _____	DISCHARGE BLOWOFF SILENCER: (7.8) ☐ MFR _____ ☐ DESCRIPTION _____ ☐ FLANGE CONNECTION _____ ☒ MOUNTING ☒ HORIZONTAL ☐ VERTICAL ☒ SUPPORTED BY ☒ PIPING ☐ OTHER _____ ☐ SPL (dBA) _____ (@ 1 m) FROM DISCHARGE OF SILENCER REMARK: _____ MODEL _____
ADDITIONAL NOTES:	
DATA SHEET NO. 0 REV. 0 SHEET 11 of	

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

PACKAGED, INTEGRALLY GEARED CENTRIFUGAL AIR COMPRESSORS (API 672 - 5TH ED) DATA SHEET	
SI UNITS (b.e.r.)	
REV	
1	CENTRIFUGAL AIR COMPRESSOR PERFORMANCE CURVES When this requisition is issued for purchase, the supplier's proposed curves for the selected compressor will be inserted here as a substitute for this sheet. Add pages as needed for stages. The compressor performance and characteristics as given on this performance curve will be a part of the supplier's contractual obligation within the tolerances agreed upon.
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DATA SHEET NO. 0 REV. 0 SHEET 12 of	

Figure A.1—Typical Datasheets for Centrifugal Air Compressors (Continued)

Annex B (informative)

Illustrations of Typical Package Mounting Configurations

B.1 General

B.1.1 Figures B.1 through B.6 show typical package mounting configurations.

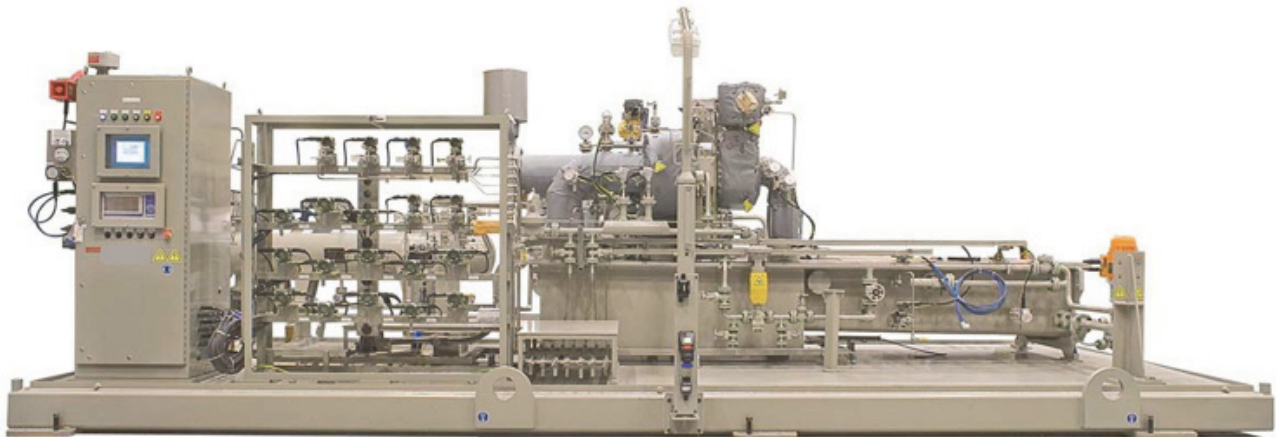


Figure B.1—Package with Full Drain-rim Base

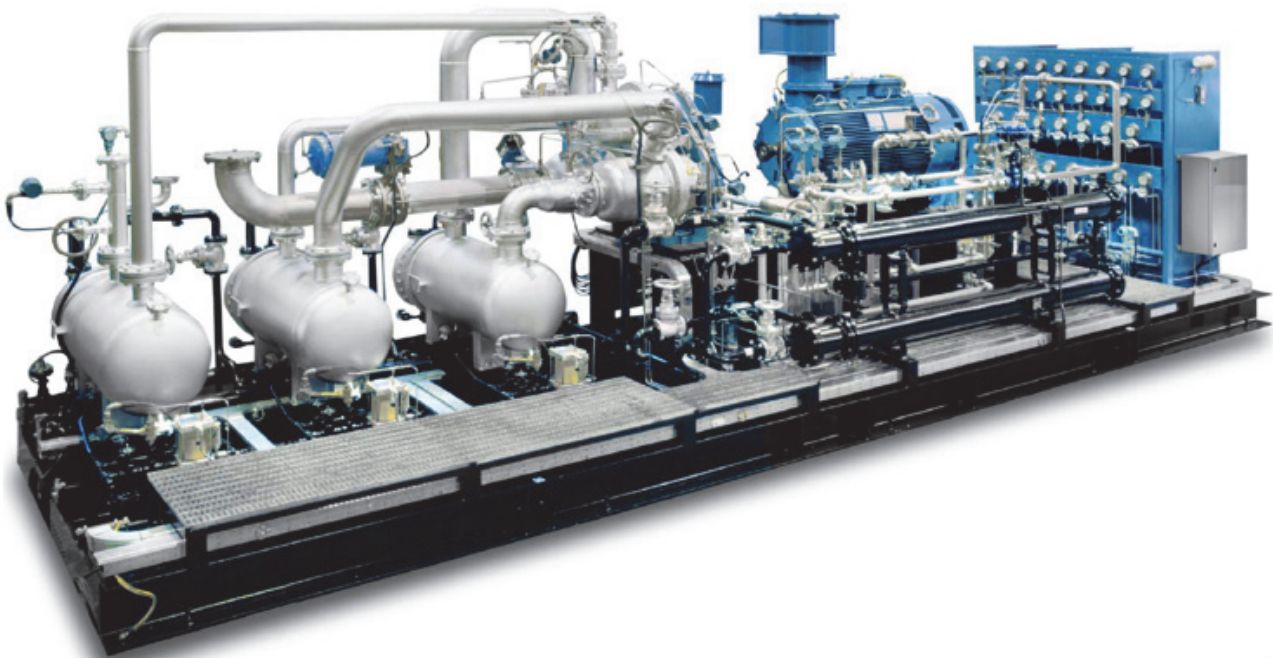


Figure B.2—Package with Open-channel Structural Base



Figure B.3—Package with Foot Mounted Base



Figure B.4— Package with Base Integral with Casing



Figure B.5—Package with Split Base (Separate Exchanger Skid)



Figure B.6—Package with Three-point Spherical Bearings Supports

Annex C **(normative)**

Dynamics

C.1 General

NOTE Refer to API RP 684 for more information on rotor dynamics.

C.1.1 In the design of rotor-bearing systems, consideration shall be given to all potential sources of periodic forcing phenomena (excitation) that shall include, but are not limited to, the following:

- a) Unbalance in the rotor system.
- b) Fluid destabilizing forces from bearings, seals, and aerodynamics.
- c) Internal rubs.
- d) Blade, vane, nozzle, and diffuser passing frequencies.
- e) Gear tooth meshing and side bands.
- f) Coupling misalignment.
- g) Loose rotor system components.
- h) Internal friction within the rotor assembly.
- i) Synchronous excitation from complimentary geared elements.
- j) Control loop dynamics such as those involving active magnetic bearings and variable frequency drives.
- k) Electrical line frequency.

NOTE 1 The frequency of a potential source of excitation can be less than, equal to, or greater than the rotational speed of the rotor.

NOTE 2 When the frequency of a periodic forcing phenomenon (excitation) applied to a rotor bearing-support system coincides with a natural frequency of that system, the system will be in a state of resonance. A rotor bearing support system in resonance can have the magnitude of its normal vibration amplified. The magnitude of amplification and, in the case of critical speeds, the rate of change of the phase angle with respect to speed, are related to the amount of damping in the system.

C.1.2 Resonances of structural support systems that are within the vendor's scope of supply, and that affect the rotor vibration amplitude shall not occur within the specified operating speed range or the specified separation margins (SM) (see C.2.10). The dynamic characteristics of the structural support shall be considered in the analysis of the rotor-bearing-support system (see C.2.4d).

NOTE Resonances of structural support systems adversely affect the rotor vibration amplitude.

C.1.3 The vendor with unit responsibility for the complete drive train shall communicate the existence of any undesirable running speeds in the range from zero to trip speed. This is illustrated by the use of a Campbell (forced frequency) diagram for individual machines or for the complete train, or both, when such has been specified. These diagrams shall be submitted to the purchaser for review and included in the instruction manual.

NOTE Examples of undesirable speeds are those associated with rotor lateral critical speeds with amplification factors (AFs) greater than or equal to 2.5, train torsionals, and vane and blading modes.

C.2 Lateral Analysis

C.2.1 Unless previously derived and confirmed by actual tests of a given design, critical speeds and their associated amplification factors shall be determined by means of a damped unbalanced rotor response analysis.

C.2.2 Unless known from previous tests of a given design, the location of all critical speeds below the trip speed shall be confirmed on the test stand during the mechanical running test (see C.3.1). The accuracy of the analytical model shall be demonstrated (see C.3).

C.2.3 Before carrying out the damped unbalanced response analysis, the vendor shall conduct an undamped analysis to identify the undamped critical speeds and determine their mode shapes. The analysis shall identify the first four undamped critical speeds, and cover as a minimum, the stiffness range from 0.1 times to 10 times the expected support stiffness (See Figure C.1).

NOTE For machinery with widely varying bearing loads or load direction such as overhung style machines, or both, the vendor sometimes substitute mode shape plots for the undamped critical speed map and list the undamped critical speed for each of the identified modes.

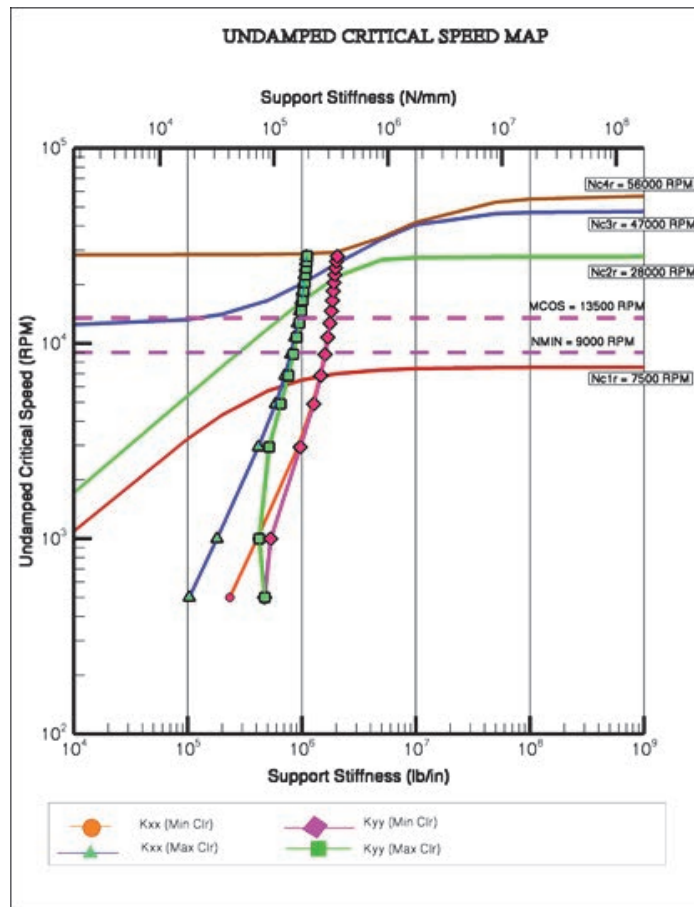


Figure C.1—Undamped Critical Speed Map

C.2.4 The rotor-dynamic analysis shall include, but shall not be limited to, the following:

NOTE The following list of items does not address the details and product of the analysis that is covered in C.2.7 and C.2.8.

- a) Rotor stiffness, mass and polar, and transverse moments of inertia, including coupling halves, and rotor stiffness changes due to shrink on components;
- b) Bearing lubricant film stiffness and damping values including changes due to speed, load, preload, range of oil temperatures, maximum to minimum clearances resulting from accumulated assembly tolerances, and the effect of asymmetrical loading which may be caused by gear forces (including the changes over range of maximum to minimum torque), eccentric clearances, volutes etc.;
- c) For tilt-pad bearings, the pad pivot stiffness;
- d) Structural support stiffness, mass, and damping characteristics, including effects of excitation frequency over the required analysis range. The term "support" includes the foundation or support structure, the base, the machine frame and the bearing housing as appropriate. For machines whose dynamic structural stiffness values are less than or equal to 3.5 times the bearing stiffness values in the range from 0 to 150 % of N_{mc} , the structure characteristics shall be incorporated as an adequate dynamic system model, calculated frequency dependent structure stiffness and damping values (impedances), or structure stiffness and damping values (impedances) derived from modal or other testing. The vendor shall state the structure characteristic values used in the analysis and the basis for these values (for example, modal tests of similar rotor structure systems or calculated structure stiffness values);

NOTE The support stiffness, in most cases is not more than 8.75×10^6 N/mm (5×10^6 lbs/in.).

- e) Rotational speed, including the various starting speed detents, operating speed and load ranges (including agreed upon test conditions if different from those specified), trip speed, and coast down conditions;
- f) The influence, over the operating range, of the hydrodynamic stiffness and damping generated by the casing shaft end seals. Minimum and maximum stiffness shall be considered taking into account the tolerance on the component clearance and the oil inlet temperature;
- g) The location and orientation of the radial vibration probes which shall be the same in the analysis as in the machine.

C.2.5 In addition to the damped unbalanced response analysis requirements of C.2.4, for machines equipped with rolling element bearings, the vendor shall state the bearing stiffness and damping values used for the analysis, and either the basis for these values or the assumptions made in calculating the values.

C.2.6 The effect of other equipment in the train is rarely necessary to be included in the damped unbalanced response analysis. A train lateral analysis shall only be performed if the drive train is rigidly coupled to the compressor.

NOTE This analysis is provided by the vendor with unit responsibility.

C.2.7 A separate damped unbalanced response analysis shall be conducted within the speed range of 0 to 150 % of N_{mc} . Unbalance shall analytically be placed at the locations defined in Figure C.2 and Figure C.3. For the translatory (symmetric) modes, the unbalance shall be based on the sum of the journal static loads. For conical (asymmetric) modes, these unbalances shall be 180° out of phase and of a magnitude based on the static load on the adjacent bearing. For overhung modes, the unbalances shall be based on the overhung mass. Figure C.2 and Figure C.3 show the typical mode shapes and indicate the location and definition of U_a for each of the shapes. The magnitude of the unbalances shall be 2 times the value of U_r as calculated by Equation (C.1) or Equation (C.2).

In SI units:

$$U_r = 6350 W / N_{mc} \quad (\text{for } N_{mc} < 25,000 \text{ rpm}) \quad (\text{C.1})$$

$$U_r = W / 3.937 \quad (\text{for } N_{mc} \geq 25,000 \text{ rpm}) \quad (\text{C.2})$$

In USC units:

$$U_r = 4 W / N_{mc} \quad (\text{for } N_{mc} < 25,000 \text{ rpm}) \quad (\text{C.3})$$

$$U_r = W / 6250 \quad (\text{for } N_{mc} \geq 25,000 \text{ rpm}) \quad (\text{C.4})$$

where

U_a is the input unbalance for the unbalance response analysis in g-mm (oz-in.)

$$= 2 * U_r;$$

U_r is the maximum allowable residual unbalance in g-mm (oz-in.);

N_{mc} is the maximum continuous operating speed, rpm;

W is the journal static load in kg (lbm), or for bending modes where the maximum deflection occurs at the shaft ends, the overhung mass (that is the mass of the rotor outboard of the bearing) in kg (lbm) (See Figures C.2 and C.3).

NOTE Above 25,000 rpm, the unbalance limit is based on 0.254 mm (10 in.) mass displacement, which is in general agreement with the capabilities of conventional balance machines, and is necessary to invoke for small rotors running at high speeds.

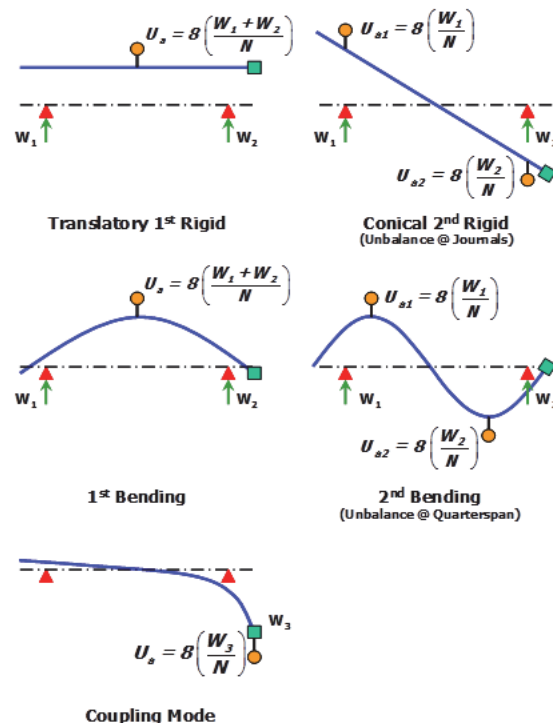


Figure C.2—Mode Shapes

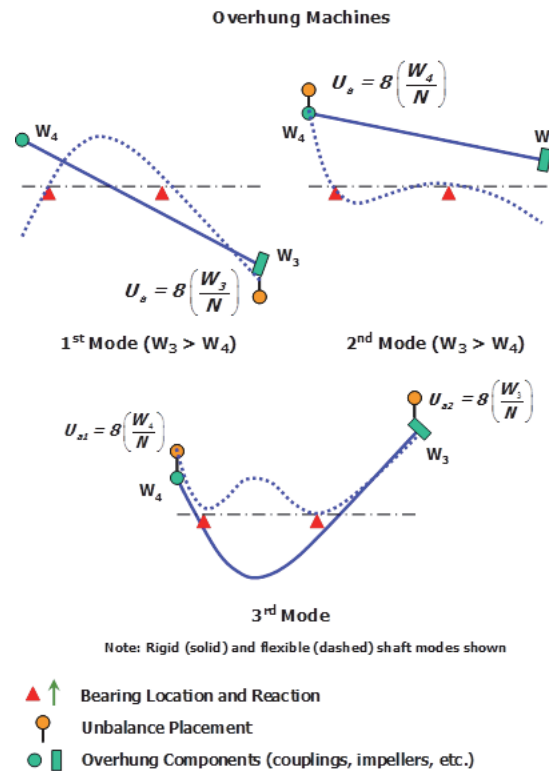


Figure C.3—Unbalance Placement and Mode Shapes for Overhung Machines

C.2.8 As a minimum, the unbalanced response analysis shall produce the following:

- a) Identification of the frequency of each critical speed in the range from 0 to 150 % of N_{mc} .
- b) Frequency, phase and response amplitude data (Bode plots) at the vibration probe locations through the range of each critical speed resulting from the unbalance specified in C.2.7.
- c) The plot of deflected rotor shape for each critical speed resulting from the unbalances specified in C.2.7, showing the major-axis amplitude at each coupling plane of flexure, the centerlines of each bearing, the locations of each radial probe, and at each seal throughout the machine as appropriate. The minimum design diametrical running clearance of the seals shall also be indicated.
- d) Additional Bode plots that compare absolute shaft motion with shaft motion relative to the bearing housing for machines where the support stiffness is less than 3.5 times the oil-film stiffness.

C.2.9 Additional analyses shall be made for use with the verification test described in C.3. The location of the unbalance shall be determined by the vendor. The unbalance shall not be less than two times or greater than eight times the value from Equation (C.1) or Equation (C.2), or as specified in C.2.7. Any test stand parameters that influence the results of the analysis shall be included.

C.2.9.1 For coupling unbalance placement (unbalance based on the coupling half weight), the unbalance shall be greater or equal to 16 times the value of Equation (C.1) or Equation (C.2).

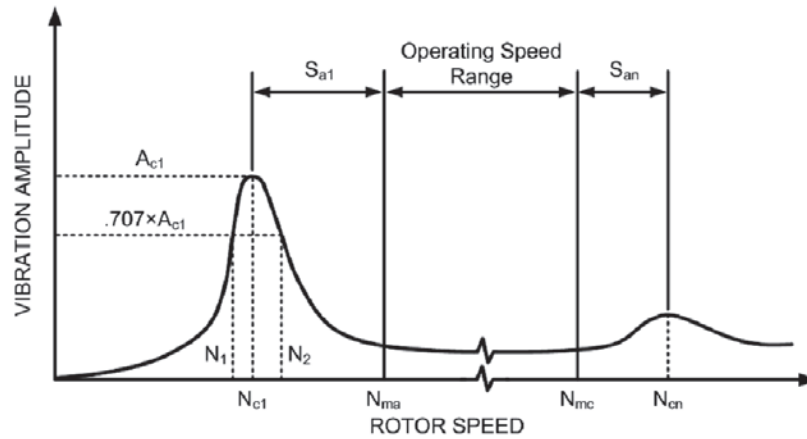
C.2.10 As shown in Equation (C.5), the damped unbalanced response analysis shall indicate that the machine will meet the following requirement (see Figure C.4):

$$SM_a \geq SM_r \quad (C.5)$$

where

SM_r is the required separation margin, %;

SM_a is defined in Figure C.4.



N_{c1} = Rotor first critical speed

N_{cn} = n^{th} critical speed

N_{ma} = Minimum allowable speed

N_{mc} = Maximum continuous speed

A_{c1} = Amplitude at N_{c1}

N_1 = Initial (lesser) speed at $0.707 \times A_{c1}$

N_2 = Final (greater) speed at $0.707 \times A_{c1}$

AF_1 = Amplification factor of the first critical speed

$$= N_{c1} / (N_2 - N_1)$$

S_{a1} = Actual separation between N_{c1} and the operating speed range

S_{an} = Actual separation between N_{cn} and the operating speed range

SM_{a1} = Actual separation margin of first critical speed (%)

$$= 100 \times S_{a1} / N_{ma}$$

SM_{an} = Actual separation margin of n^{th} critical speed (%)

$$= 100 \times S_{an} / N_{mc}$$

Note: The shape of the curve is for illustration only and does not necessarily represent any actual rotor response plot.

Figure C.4—Typical Rotor Response Plot

- If the amplification factor (AF) at a particular critical speed is less than 2.5, the response is considered critically damped and no separation margin is required ($SM_r = 0$).
- If the amplification factor at a particular critical speed is greater than or equal to 2.5 and that critical speed is below the minimum speed, the separation margin (SM) is given by Equation (C.6).

- c) If the amplification factor at a particular critical speed is equal to 2.5 or greater and that critical speed is above the maximum continuous speed (MCS), the separation margin (SM) is given by Equation (C.7).

$$SM_r = 17 \left(1 - \frac{1}{AF - 1.5} \right) \quad (C.6)$$

$$SM_r = 10 + 17 \left(1 - \frac{1}{AF - 1.5} \right) \quad (C.7)$$

C.2.11 The calculated unbalanced peak-to-peak response (see C.2.8, Item b) at each vibration probe, for each unbalance amount and case, shall not exceed the mechanical test vibration limit, A_{vl} , of $25.4 \mu\text{m}$ (1.0 mil) or Equation (C.8), whichever is less, over the range of N_{ma} to N_{mc} as shown in Figure C.5.

In SI units:

$$A_{vl} = 25.4 \sqrt{12000 / N_{mc}} \quad (C.8)$$

In USC units:

$$A_{vl} = \sqrt{12000 / N_{mc}}$$

where

A_{vl} is the mechanical test vibration limit, μm (mil);

N_{mc} is the maximum continuous speed (rpm).

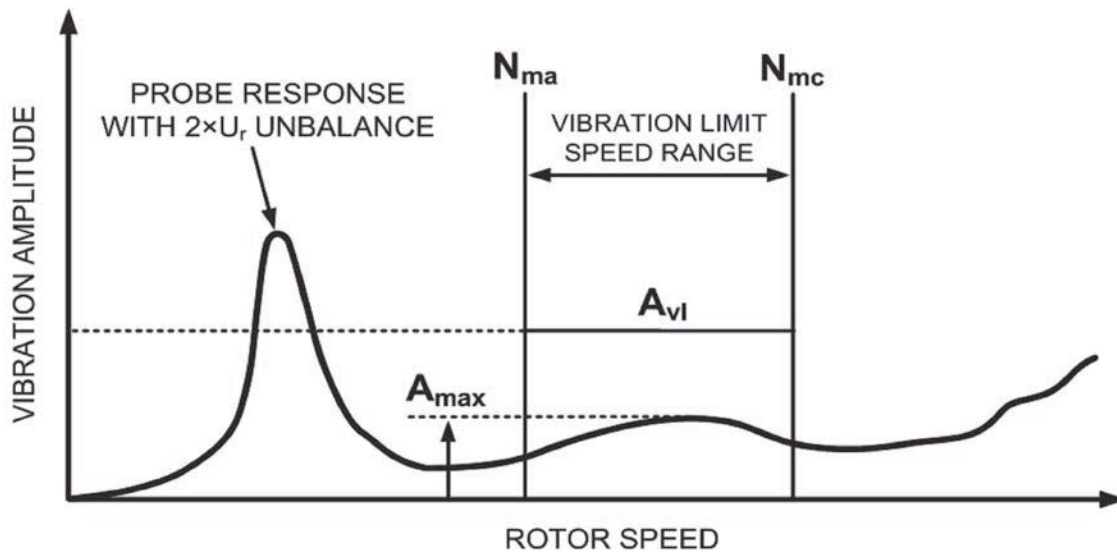


Figure C.5—Plot of Applicable Speed Range of Vibration Limit

C.2.12 For each unbalance amount and case as specified in C.2.7, the calculated major-axis, peak-to-peak response amplitudes at each close clearance location shall be multiplied by a scale factor defined by Equation (C.9).

$$S_{cc} = (A_{vl} / A_{max}) \text{ or } 6, \text{ whichever is less} \quad (C.9)$$

where

- S_{cc} is the scale factor for close clearance check;
- A_{vl} is the mechanical test vibration limit defined in C.2.11;
- A_{max} is the maximum probe response amplitude (p-p) considering all vibration probes, over the range of N_{ma} to N_{mc} , for the unbalance amount/case being considered.

NOTE To meet the requirements of C.2.10, the scale factor is normally greater than or equal to one.

C.2.12.1 For each close clearance location, the scaled response shall be less than 75 % of the minimum design diametral running clearance over the range from zero to trip speed.

Running clearances can be different than the assembled clearances with the machine shutdown.

Consideration shall be given to:

- a) centrifugal/thermal growth,
- b) bearing lift,
- c) nonconcentricity (of stator to the bearings).

C.2.13 If the analysis indicates that if either of the following requirements cannot be met:

- The required separation margins
- The requirements of C.2.11 and C.2.12

and the purchaser and vendor have agreed that all practical design efforts have been exhausted, then acceptable amplitudes, separation margins and amplification factors shall be agreed, subject to the requirements of C.4.

C.3 Unbalanced Rotor Response Verification Test

C.3.1 For previously untested designs, an unbalanced rotor response test shall be performed as part of the mechanical running test, and the results shall be used to verify the analytical model. The actual response of the rotor on the test stand to the same arrangement of unbalance as was used in the analysis specified in C.2.9 shall be the criterion for determining the validity of the damped unbalanced response analysis. To accomplish this, the requirements of C.3.1.1 through C.3.1.6 shall be followed.

C.3.1.1 During the mechanical running test, the amplitudes and phase angle of the shaft vibration from trip to slow roll speed shall be recorded at the end of the run test. The recording instrumentation resolution shall be at least 1.25 micron (0.05 mils).

NOTE This set of readings is normally taken during a coastdown, with convenient increments of speed such as 50 rpm. Since at this point the rotor is balanced, any vibration amplitude and phase detected is the result of residual unbalance, and mechanical and electrical runout.

C.3.1.2 The unbalance that was used in the analysis performed in C.2.9 shall be added to the rotor in the location used in the analysis.

C.3.1.3 The machine shall then be brought up to trip speed after being held at maximum continuous speed for at least 15 minutes and the indicated vibration amplitudes and phase shall be recorded during the coast down using the same procedure as C.3.1.1.

C.3.1.4 The location of critical speeds below the trip speed shall be established. If a clearly defined response peak is not observed during the test, then the critical speeds shall be identified as those in the lateral damped analysis report.

NOTE Slow roll runout is normally vectorially subtracted from the 1X Bode plots to accurately define the location of the critical speeds.

C.3.1.5 The corresponding indicated vibration data taken in accordance with C.3.1.1 and C.3.1.4 shall be vectorially subtracted.

NOTE If slow roll runout is checked prior to subtraction, the data is expected to be nearly identical for both runs.

C.3.1.6 The results of the mechanical run including the unbalance response verification test shall be compared with those from the analytical model specified in C.2.9.

NOTE It is necessary for probe orientation to be the same for the analysis and the machine for the comparison to be valid.

C.3.2 Using the unbalance response test results, the vendor shall correct the model if it fails to meet either of the following criteria:

The actual critical speed(s) determined on test shall not deviate from the corresponding critical speed ranges predicted by analysis by more than $\pm 5\%$.

The maximum probe response from the results of C.3.1.5 shall not exceed the predicted ranges.

C.3.3 The vendor shall determine whether the comparison made is for absolute or relative motion.

NOTE For absolute motion, bearing housing vibration will need to be vectorially added to relative probe readings. This is normal for machinery with soft supports.

C.3.4 Unless otherwise specified, the verification test of the rotor unbalance shall be performed only on the first rotor tested, if multiple identical rotors are produced.

C.3.5 After correcting the model, the response amplitudes shall be checked against the limits specified in C.2.11 and C.2.12.

C.4 Additional Testing

C.4.1 Additional testing is required if from the shop verification test data (see C.3) or from the damped, corrected unbalanced response analysis (see C.3.1.2), if either of the following conditions exists:

- a) Any critical speed fails to meet the SM_r requirements (see C.2.10).
- b) The requirements of C.2.11 and C.2.12 have not been met.

NOTE When the analysis or test data does not meet the requirements of the standard, additional more stringent testing is applied. The purpose of this additional testing is to determine on the test stand that the machine will operate successfully.

C.4.2 Unbalance weights shall be placed as described in C.2.7; this may require disassembly of the machine. Unbalance magnitudes shall be achieved by adjusting the indicated unbalance that exists in the rotor from the initial run to raise the displacement of the rotor at the probe locations to the vibration limit defined by Equation (C.8) (see

C.2.11) at the maximum continuous speed; however, the unbalance used shall be no less than twice nor greater than 8 times the unbalance limit specified in Equation C.1) or Equation (C.2). The measurements from this test, taken in accordance with C.3.1.1 through C.3.1.3, shall meet the following criteria:

- a) From zero to trip speed; the shaft deflections shall not exceed 90 % of the minimum design running clearances.
- b) Within the operating speed range, including the separation margins, shall the shaft deflections exceed 55 % of the minimum design running clearances or 150 % of the allowable vibration limit at the probes (see C.2.11).

C.4.3 The internal deflection limits specified in C.4.2 Items a. and b. shall be based on the calculated displacement ratios between the probe locations and the areas of concern (see C.2.12) based on a corrected model if required. Acceptance shall be based on these calculated displacements or inspection of the seals if the machine is opened.

NOTE Internal displacements for these tests are calculated by multiplying these ratios by the peak readings from the probes.

C.4.3.1 Damage to any portion of the machine as a result of this testing shall constitute failure of the test. Minor internal seal rubs that do not cause clearance changes outside the vendor's new part tolerance do not constitute damage.

C.5 Level 1 Stability Analysis

C.5.1 A stability analysis shall be performed on the initial design of all integrally geared centrifugal compressors rotors that meet the following:

- a) those rotors whose maximum continuous speed is greater than the first undamped critical speed in accordance with C.2.3;
- b) those rotors with fixed geometry bearings.

The stability analysis shall be carried out at the API defined maximum continuous speed.

NOTE Level 1 analysis was developed to fulfill two purposes: first, it provides an initial screening to identify rotors that do not require a more detailed study. The approach as developed is conservative and not intended as an indication of an unstable rotor. Second, the Level 1 analysis specifies a standardized procedure applied to all manufacturers similar to that found in C.2. (Refer to API 684 for a detailed explanation.)

C.5.2 The model used in the Level 1 analysis shall include the items listed in C.2.4.

C.5.3 All components shall be analyzed using the mean value of oil inlet temperature and the extremes of the operating limits for clearance.

C.5.4 When tilt pad journal bearings are used, the analysis shall be performed with synchronous tilt pad coefficients.

C.5.5 For integrally geared compressor rotors that have quantifiable external radial loading, the stability analysis shall also include the external loads associated with the operating conditions defined in C.5.1. For some rotors, the unloaded (or minimum load condition) may represent the worst stability case and shall be considered.

C.5.6 The anticipated cross coupling, Q_A , present in the rotor is defined by the following procedure:

The parameters in Equation (C.9) shall be determined based on the machine conditions in C.5.1 unless the vendor and purchaser agree upon another operating point.

$$q_A = \frac{HP * B_c * C}{D_c * H_c * N} * \frac{\rho_d}{\rho_s} \quad (C.9)$$

where

- HP is the rated power per impeller, Nm/s (HP);
- B_c is 3;
- C is 9.55 (63);
- ρ_d is the discharge gas density per impeller, kg/m³ (lbm/ft³);
- ρ_s is the suction gas density per impeller, kg/m³ (lbm/ft³);
- D_c is the impeller diameter, mm (in.);
- H_c is the minimum of diffuser or impeller discharge width per impeller, mm (in.);
- N_r is the normal operating speed for calculation of aerodynamic excitation (rpm);
- q_A is the cross coupling for each individual impeller, kN/mm (klbf/in.).

Equation (C.9) is calculated for each impeller of the rotor. Q_A is equal to the sum of q_A for all impellers.

C.5.7 An analysis shall be performed with a varying amount of cross coupling introduced at the center of gravity of the stage or impeller for single overhung rotors. For double overhung rotors, the cross coupling shall be placed at each stage or impeller concurrently and shall reflect the ratio of the anticipated cross coupling, q_A , calculated for each impeller or stage.

C.5.8 The applied cross coupling shall extend from zero to the minimum of:

A level equal to 10 times the anticipated cross coupling, Q_A .

The amount of the applied cross coupling required to produce a zero log decrement, Q_0 . This value can be reached by extrapolation or linear interpolation between two adjacent points on the curve shown in Figure C.6.

C.5.9 A plot of the calculated log decrement δ , for the first forward mode shall be prepared for the minimum and maximum component clearances. Each curve shall contain a minimum of five calculated stability points. The ordinate (y-axis) shall be the log decrement. The abscissa (x-axis) shall be the applied cross coupling with the range defined in C.5.8. For double overhung rotors, the applied cross coupling will be the sum of the cross coupling applied to each impeller or stage.

A typical plot is presented in Figure C6. Q_0 and δ_A are identified as the minimum values from either component clearance curves.

C.5.10 Level 1 Screening Criteria

If any of the following criteria apply, a Level 2 stability analysis shall be performed:

- 1) $Q_0/Q_A < 2.0$.
- 2) $\delta_A < 0.1$.
- 3) $Q_0/Q_A < 10$ and the point defined by CSR and the average density at the normal operating point is located in Region B of Figure C.7.

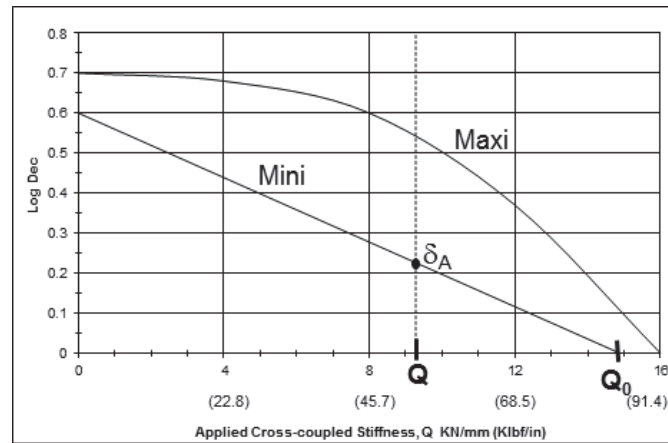


Figure C.6—Level 1 Stability Sensitivity Plot

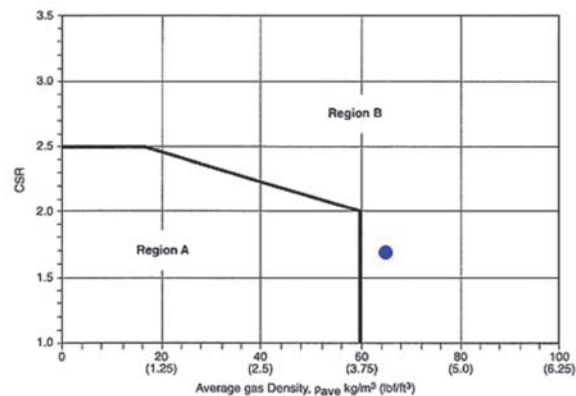


Figure C.7—Stability Experience Plot

Otherwise, the stability is acceptable and no further analyses are required.

C.6 Level 2 Stability Analysis

C.6.1 A Level 2 analysis, which reflects the actual dynamic forces (both stabilizing and destabilizing) of the rotor, shall be performed as required by C.5.10.

C.6.2 The Level 2 analysis shall include the dynamic effects from all sources that contribute to the overall stability of the rotating assembly as appropriate. These dynamic effects shall replace the anticipated cross coupling, Q_A . The following sources shall be considered:

- a) Labyrinth seals
- b) Impeller/blade flow aerodynamic effects
- c) Internal friction

The vendor shall state how the sources are handled in the analysis.

NOTE It is recognized that methods are not available at present to accurately model the destabilizing effects from all sources listed above.

C.6.3 The Level II analysis shall be carried out at N_{mc} ;

C.6.4 The operating conditions defined in C.5.1 extrapolated to the conditions that shall not fall outside the operating limits (the defined operating map) of the equipment train such as horsepower, discharge pressure, etc.

C.6.5 The modeling requirements of C.5.2, C.5.4 and C.5.5 shall also apply.

C.6.6 The dynamic coefficients of the labyrinth seals shall be calculated at minimum seal running clearance.

C.6.7 The frequency and log decrement of the first forward damped mode shall be calculated for the following conditions (except for double overhung machines where the first two forward modes must be considered):

- Rotor and support system only. (Basic log decrement, δ_b);
- Each source from C.6.2 included in the analysis;
- Complete model including all sources. (Final log decrement, δ_f).

C.6.8 Acceptance Criteria

The Level 2 stability analysis shall indicate that the machine, as calculated in C.6.1 through C.6.7, shall have a final log decrement, δ_f , greater than 0.1.

C.6.9 If after all practical design efforts have been exhausted to achieve the requirements of C.6.8, acceptable levels of the log decrement, δ_f , shall be mutually agreed upon by the purchaser and vendor.

NOTE It is recognized that other analysis methods and continuously updated acceptance criteria have been used successfully since the mid-1970s to evaluate stability. The historical data accumulated by machinery manufacturers for successfully operated machines can conflict with the acceptance criteria of this standard. If such a conflict exists and the suppliers can demonstrate that their stability analysis methods and acceptance criteria predict a stable rotor, then the suppliers' criteria can be the guiding principle in the determination of acceptability.

Symbols

B_c	= 3
B_t	= 1.5
C	= 9.55 (63)
D_c	= Impeller diameter, mm (in.)
D_t	= Blade pitch diameter, mm (in.)
H_c	= Minimum of diffuser or impeller discharge width per impeller, mm (in.)
H_t	= Effective blade height, mm (in.)
HP	= Rated power per stage or impeller, Nm/sec (HP)
CSR	= Critical speed ratio is defined as:
N	= Operating speed, rpm
Q_A	= Anticipated cross coupling for the rotor, KN/mm (Klbf/in.) defined as:

$$Q_A = \sum_{i=1}^S q_{Ai} \quad (C.10)$$

Q_0	= Minimum cross coupling needed to achieve a log decrement equal to zero for either minimum or maximum component clearance.
q_A	= Cross coupling defined in Equation (C.9) or Equation (C.10) for each stage or impeller, KN/mm (Klbf/in.)
S	= Number of stages or impellers
δ_A	= Minimum log decrement at the anticipated cross coupling for either minimum or maximum component clearance.
δ_b	= Basic log decrement of the rotor and support system only
δ_f	= Log decrement of the complete rotor support system from the Level 2 analysis
ρ_d	= Discharge gas density per stage or impeller
ρ_s	= Suction gas density per stage or impeller
ρ_{ave}	= Average gas density across the rotor, kg/m ³ (lbm/ft ³)

Definitions

- Stability analysis is the determination of the natural frequencies and the corresponding logarithmic decrements of the damped rotor/support system using a complex eigenvalue analysis.
- Synchronous tilt pad coefficients are derived from the complex frequency dependent coefficients with the frequency equal to the rotational speed of the shaft.
- Stage refers to an individual turbine or axial compressor blade row.
- Hysteresis or internal friction damping causes a phase difference between the stress and strain in any material under cyclic loading. This phase difference produces the characteristic hysteric loop on a stress-strain diagram and thus, a destabilizing damping force.
- Minimum clearance for a tilt pad bearing occurs at the maximum preload condition. These can be calculated using the following formulas:

$$Preload_{max} = 1 - \frac{Bearing\ Radius_{min} - Shaft\ Radius_{max}}{Pad\ Bore_{max} - Shaft\ Radius_{max}}$$

$$Bearing\ Clearance_{min} = Bearing\ Radius_{min} - Shaft\ Radius_{max}$$

For maximum clearance at minimum preload:

$$Preload_{min} = 1 - \frac{Bearing\ Radius_{max} - Shaft\ Radius_{min}}{Pad\ Bore_{min} - Shaft\ Radius_{min}}$$

$$Bearing\ Clearance_{max} = Bearing\ Radius_{max} - Shaft\ Radius_{min}$$

C.7 Torsional Analysis

C.7.1 For trains including synchronous motor-driven units or gears, and units comprising three or more coupled machines or when specified, the vendor having unit responsibility shall ensure that a torsional vibration analysis of the complete coupled train is carried out and shall be responsible for directing any modifications necessary to meet the requirements of C.7.3. through C.7.7.

C.7.2 If specified, for compressors directly driven by turbines, the supplier shall perform a torsional vibration analysis of the complete coupled train, and shall be responsible for directing any modifications necessary to meet the requirements of C.7.3 through C.7.7. For such trains, a simplified torsional model (lumped rotor inertia and stiffness) is sufficient.

NOTE The intent of the simplified analysis is to calculate the primary (coupling) modes of the system. Primary modes are those influenced primarily by the coupling torsional stiffness.

C.7.3 Excitation of torsional natural frequencies may come from many sources that may or may not be a function of running speed and should be considered in the analysis. These sources shall include but are not limited to the following:

- a) Gear characteristics such as unbalance, pitch line runout, and cumulative pitch error.
- b) Torsional pulsations due to gear radial vibrations.
- c) Cyclic process impulses.
- d) Torsional excitation resulting from electric motors and variable frequency drives..
- e) One and two times electrical line frequency.
- f) One and two times operating speed(s).

C.7.4 Primary (coupling) modes shall be at least 10 % above or 10 % below any 1X excitation frequency (mechanical or electrical) within the specified operating speed range.

C.7.5 All other torsional natural frequencies shall preferably be at least 10 % above or 10 % below any possible excitation frequency within the specified operating speed range (from minimum to maximum continuous speed).

C.7.6 Any interference resulting from C.7.5 shall be shown to have no adverse effect using C.7.7.

C.7.7 When torsional resonances are calculated to fall within the margin specified in C.7.5 (and the purchaser and the vendor have agreed that all efforts to remove the critical from within the limiting frequency range have been exhausted), a steady state stress analysis shall be performed to demonstrate that the resonances have no adverse effect on the complete train. The assumptions made in this analysis regarding the magnitude of excitation and the degree of damping shall be clearly stated. The analysis shall show that all shaft sections, couplings and gear mesh have infinite life using an agreed upon criteria.

C.7.8 In addition to the torsional analyses required in C.7.2 through C.7.7, the vendor shall perform a transient torsional vibration analysis for synchronous motor driven units, using a time-transient analysis. The following parameters shall be included in this analysis:

- a) Motor average torque, as well as pulsating torque (direct and quadrature axis) versus speed characteristics.
- b) Load torque versus speed characteristics.
- c) Electrical system characteristics affecting the motor terminal voltage or the assumptions made concerning the terminal voltage including the method of starting, such as across the line or some method of reduced voltage starting.

C.7.9 The analysis shall generate the maximum torque as well as a torque versus time history for each of the shafts in the compressor train.

The maximum torques shall be used to evaluate the peak torque capability of coupling components, gearing, and interference fits of components such as coupling hubs.

NOTE The torque vs time history is used to develop a cumulative damage fatigue analysis of shafting, keys, and coupling components.

C.7.10 Appropriate fatigue properties and stress concentrations shall be used.

C.7.11 An appropriate cumulative fatigue algorithm shall be used to develop a value for the safe number of starts. The safe number of starts shall be as agreed by the purchaser and vendor.

C.7.12 For VFD driven equipment trains, the vendor shall extend the analysis defined in C.7.2 through C.7.7 to include the following in C.7.13 through C.7.16.

C.7.13 In addition to the excitations of C.7.3, the following shall also be considered, but is not limited to:

- a) integer orders of the drive output frequency,
- b) sidebands of the pulse width modulation.

NOTE VFD produced broad band noise floor and feedback generated excitations can cause harmful torsional pulsations. Transient or mechanical/electrical coupled analyses, or both, can be applied to understand the effects of these excitations.

C.7.14 A steady state response analysis shall be performed from 0 to MCS to quantify the effects of the VFD excitation of C.7.13.

C.7.15 For interferences occurring below the minimum operating speed, an agreed criteria shall be used to establish acceptability of the train.

C.7.16 For interferences occurring within the operating speed range, the criteria set forth in C.7.7 shall be used.

C.7.17 If specified, for motor-driven equipment, a transient short circuit fault analysis shall be performed in accordance with C.7.18.

C.7.18 The following faults shall be considered, but are not limited to:

- a) short circuits:
 - 1) line-to-line,
 - 2) two phase,
 - 3) three phase,
 - 4) line-to-ground,
 - 5) line-to-line-to-ground.

C.7.19 For these fault conditions, generated stresses in the shafting and couplings shall not exceed the low-cycle fatigue limit.

NOTE The analysis for these fault conditions assumes a one-time event. It is possible that some components identified by the analysis will need to be replaced following the fault event.

Annex D

(informative)

Contract Documents and Engineering Design Data

D.1 General

D.1.1 If specified by the Purchaser (see 9.1.2), the contract documents and engineering design data shall be supplied by the Vendor, as listed in this annex.

D.1.2 The data shall be identified on transmittal (cover) letters, title pages, and in title blocks or other prominent position on drawings, with the following information:

- a) purchaser's/owner's corporate name,
- b) job/project number,
- c) equipment item number and service name,
- d) inquiry or purchase order number,
- e) any other identification specified in the inquiry or purchase order,
- f) vendor's identifying proposal number, shop order number, or serial number, or other reference required to completely identify return correspondence.

D.2 Proposals

D.2.1 General

D.2.1.1 The proposal shall include as a minimum, the data specified in D.2.2 through D.2.5 and a specific statement that the equipment and all its components and auxiliaries are in strict accordance with this standard.

D.2.1.2 If the equipment or any of its components or auxiliaries are not in strict accordance, the vendor shall include a list that details and explains each deviation to enable the purchaser to evaluate any proposed alternative designs.

D.2.1.3 The vendor shall provide sufficient detail to enable the purchaser to evaluate any proposed alternative designs.

D.2.1.4 All correspondence shall be clearly identified in accordance with D.1.2.

D.2.2 Drawings

D.2.2.1 The drawings indicated on the Vendor Drawing and Data Requirements or VDDR form (see Table D.1) shall be included in the proposal. All documentation supplied during proposal period is intended for reference only and it should be understood that some variation to this documentation will occur in order execution.

D.2.2.2 As a minimum, the following data shall be included:

- a) general arrangement or outline drawing for each major skid and remote mounted component, showing overall dimensions, maintenance clearance dimensions, overall weights, erection weights, maximum maintenance weights (indicated for each piece) the direction of rotation, and the size and location of major purchaser connections;

- b) cross-sectional drawings showing the details of the proposed equipment;
- c) schematics of all auxiliary systems, including the air, lube-oil, cooling water, seal air, control, and electrical systems, with bills of material identifying components by make, model, and materials of construction for each system.

D.2.3 Technical Data for Proposals

D.2.3.1 The following data shall be included in the proposal:

- a) purchaser's data sheets with complete vendor's information entered thereon and literature to fully describe details of the offering;
- b) predicted noise data;
- c) Vendor Drawing and Data Requirements form (see Annex D), indicating the schedule according to which the vendor agrees to transmit all the data specified;
- d) schedule for shipment of the equipment, in weeks after receipt of an order;
- e) list of major wearing components, showing any interchangeability with the owner's existing units;
- f) list of priced spare parts recommended for start-up and two or more years of normal operation;
- g) list of the special tools furnished for maintenance;
- h) description of any special weather protection and winterization required for start-up, operation, and periods of idleness under the site conditions specified and clearly indicating the protection to be furnished by the purchaser, as well as that included in the vendor's scope of supply;
- i) complete tabulation of utility requirements (clearly indicating approximate data where applicable), such as those for steam, water, electricity, air, and lube oil (including the quantity and supply pressure of the lube oil required, and the heat load to be removed by the oil), and the nameplate power rating and operating power requirements of auxiliary driver;
- j) description of any special requirements specified in the purchaser's inquiry and as outlined in 6.1.1.2, 6.1.1.4, 6.5.2.2, 6.10.1.2, and 7.6.8;
- k) listing of any components or maintenance requirements that would result in the need to shut down the equipment within the uninterrupted operational period noted in 6.1.1.5;
- l) allowable forces and moments on purchaser inlet, discharge and bypass air connections, as required by 6.4.1;
- m) description of the sealing system including air consumption as required by 6.6.3;
- n) description of the alarm and shutdown functions should refer to facilities requested in 7.4.5.3.3;
- o) vendor's recommended oil viscosity grade in accordance with ISO 3448 and the minimum allowable oil temperature as requested in API 614;
- p) description of standard tests including mechanical run and performance, control functionality, and oil system cleanliness;
- q) descriptive literature;

r) vendor's Quality Assurance Plan.

D.2.4 Curves

D.2.4.1 The vendor shall provide complete performance curves to encompass the map of operations, with any limitations indicated thereon.

D.2.4.2 Overall performance curves shall be submitted for rated, and any other specified conditions.

D.2.4.3 Curves shall include a plot of discharge pressure, and brake horsepower against delivered standard flow. Curves shall indicate theoretical surge, control line rated capacity, and any other specified operating points. Curves that show throttling effects at off-design inlet conditions shall also be provided.

D.2.4.4 Preliminary compressor and motor speed-torque curves shall be provided.

D.2.5 Additional Proposal Requirements for "Special Duty" Packages Only

D.2.5.1 If Special Duty has been specified, the following additional data shall be included in the proposal:

- a) list of similar machines installed and operating under conditions analogous to those specified in the proposal;
- b) any start-up, shutdown, or operating restrictions required to protect the integrity of the equipment;
- c) calculated values of gear-rated power, based on AGMA 6011.

D.3 Engineering Data

D.3.1 General

D.3.1.1 Engineering data shall be furnished by the vendor in accordance with the agreed VDDR form.

D.3.1.2 Each drawing shall have a title block in the lower right-hand corner with the date of certification, identification data specified in D.1.2, the revision number and date, and the title. Similar information shall be provided on all other documents including sub-vendor items.

D.3.1.3 The purchaser shall review the vendor's data upon receipt; however, this review shall not constitute permission to deviate from any requirements in the order unless specifically agreed upon in writing. After the data have been reviewed and accepted, the vendor shall furnish certified copies in the quantities specified.

D.3.1.4 A complete list of vendor data shall be included with the first issue of the major drawings. This list shall contain titles, drawing numbers, and a schedule for transmittal of each item listed. This list shall cross-reference data with respect to the VDDR form in Annex D.

D.3.2 Drawings and Technical Data

The drawings and data furnished by the vendor shall contain sufficient information so that together with the manuals specified in D5, the purchaser can properly install, operate, and maintain the equipment covered by the purchase order. All contract drawings and data shall be clearly legible (8-point minimum font size even if reduced from a larger size drawing), shall cover the scope of the agreed VDDR form, and shall satisfy the applicable detailed descriptions in Annex D as agreed upon between vendor and purchaser.

D.3.3 Progress Reports

The vendor shall submit progress reports to the purchaser at the intervals specified. Progress reports shall include not only vendor in-house engineering and manufacturing data, but also major sub-vendor parts status to ensure delivery of entire package on the contract schedule.

D.3.4 Parts Lists and Recommended Spares

D.3.4.1 The vendor shall submit complete parts lists for all equipment and accessories supplied. The lists shall include part names, manufacturers' unique part numbers, materials of construction where required to order the correct part, and delivery times. Each part shall be completely identified and shown on appropriate cross-sectional, assembly-type cutaway or exploded-view isometric drawings. Interchangeable parts shall be identified as such. Parts that have been modified from standard dimensions or finish to satisfy specific performance requirements shall be uniquely identified by part number. Standard purchased items shall be identified by the original manufacturer's name and part numbers.

D.3.4.2 The vendor shall indicate on these complete parts lists all those parts that are recommended as start-up or maintenance spares and the recommended stocking quantities of each. This should include spare parts recommendations of sub-vendors that were not available for inclusion in the vendor's original proposal (see D.2.3.1, item f).

D.3.5 Installation, Operation, Maintenance, and Technical Data Manuals

D.3.5.1 General

The vendor shall provide sufficient written instructions and all necessary drawings to enable the purchaser to install, operate, and maintain all of the equipment covered by the purchase order. This information shall be compiled in a manual or manuals with a cover sheet showing the information listed in D.1.2, an index sheet, and a complete list of the enclosed drawings by title and drawing number.

D.3.5.2 Installation Manual

D.3.5.2.1 Any special information required for proper installation design that is not on the drawings shall be compiled in a manual that is separate from the operating and maintenance instructions. This manual may be a preliminary section of the Operation and Maintenance Manual but is issued earlier than the Operation and Maintenance Manual to allow use for installation planning. This manual shall be forwarded at a time that is agreed upon in the order but not later than the issue of final certified drawings. A copy of this manual may be duplicated in the final Operating and Maintenance Manual.

D.3.5.2.2 The manual shall contain information for receiving the units and for preservation of the units prior to service. It will include information such as special alignment and grouting procedures, utility specifications (including quantities), and all other installation design data, including the drawings and data specified in D.2.2 and D.2.3. The manual shall also include sketches that show the location of the center of gravity and rigging provisions to permit the removal of the casing cover, rotors, and any sub-assemblies that weigh more than 136 kg (300 lbs.).

D.3.5.3 Operating and Maintenance Manual

A manual containing all required operating and maintenance instructions shall be supplied. In addition to covering operation at specified conditions, this manual shall also contain separate sections that provide special instructions for operation at specified extreme environmental conditions.

D.3.5.4 Technical Data Manual

The vendor shall provide the purchaser with a technical data manual within 30 days of completion of shop testing or with the machine at shipment, whichever comes first. (see Annex D for minimum requirements of this manual.)

D.3.5.5 Additional Design Data For “Special Duty” Packages Only

- If specified, the Installation, Operating and Maintenance Instructions (IOMI) Manual(s) shall be prepared for the equipment covered by the purchase order and generic or “pre-printed” material covering multiple models will not be allowed.

Table D.1—VDDR for Packaged, Integrally Geared Centrifugal Air Compressors ^a

Vendor drawing and data requirements for packaged, integrally geared centrifugal air compressors	Job number				Item number				
	Purchase order number				Date				
	Requisition number				Date				
	Inquiry number				Date				
	Revision by				Manufacturer				
	For				Unit				
	Site				Service				

Minimum requirements for basic systems indicated by "Required". ^e

Proposal ^b—Bidder shall furnish number of paper copies/number of electronic copies of data as indicated

Review—Vendor shall furnish number of paper copies /number of electronic copies of data as indicated

Final—Vendor shall furnish number of paper copies /number of electronic copies of data as indicated

				Description (see Table D.2)	Review due from vendor ^{c,d}	Review received from vendor	Review returned to vendor	Final due from vendor ^d	Final received from vendor
Required	/	/	/	1. Certified dimensional outline drawing and list of connections					
Required	/	/	/	2. Cross sectional drawings and bills of materials					
Required	/	/	/	3. Control, alarm, and trip settings (pressure and recommended temperature)					
Required	/	/	/	4. Sealing air system schematics and bill of materials					
Required	/	/	/	5. Electrical and instrumentation wiring diagram and bill of materials					
Required	/	/	/	6. Foundation loading diagram including dimensions of baseplates					
Required	/	/	/	7. Cooling or heating schematic and bill of materials					
Required	/	/	/	8. Lube oil schematic and bills of materials					
Required	/	/	/	9. Lube oil system assembly and arrangement drawings					
Required	/	/	/	10. Electrical, instrumentation and control schematics, wiring diagrams, and bill of materials					
Required	/	/	/	11. Electrical and instrumentation arrangement drawings					
Required	/	/	/	12. ISA data sheets for all instruments					
Required	/	/	/	13. Tabulation of utility requirements					
Required	/	/	/	14. Motor performance and electrical data and curves					
Required	/	/	/	15. Motor terminal box details and wiring instructions					
Required	/	/	/	16. Curves showing performance characteristics at rated condition					
Required	/	/	/	17. Curves showing performance characteristics at other specified inlet conditions					

Table D.1—VDDR for Packaged, Integrally Geared Centrifugal Air Compressors ^a

Vendor drawing and data requirements for packaged, integrally geared centrifugal air compressors	Job number				Item number				
	Purchase order number				Date				
	Requisition number				Date				
	Inquiry number				Date				
	Revision by				Manufacturer				
	For				Unit				
	Site				Service				

Minimum requirements for basic systems indicated by "Required". ^e

Proposal ^b—Bidder shall furnish number of paper copies/number of electronic copies of data as indicated

Review—Vendor shall furnish number of paper copies /number of electronic copies of data as indicated

Final—Vendor shall furnish number of paper copies /number of electronic copies of data as indicated

				Description (see Table D 2)	Distribution Record				
					Review due from vendor ^{c,d}	Review received from vendor	Review returned to vendor	Final due from vendor ^d	Final received from vendor
Required	/	/	/	18. Curve showing the effects of optional inlet guide vanes					
Required	/	/	/	19. Damped unbalanced response analysis (6.12.4)					
Required	/	/	/	20. Torsional vibration analysis (6.12.5)					
Required	/	/	/	21. Mechanical running test logs					
Required	/	/	/	22. Certified hydrostatic test logs					
Required	/	/	/	23. Material certifications					
Required	/	/	/	24. Dimensional drawings for all major auxiliary equipment or components					
Required	/	/	/	25. Data sheets applicable to proposals, purchase and as-built					
Required	/	/	/	26. Noise data sheets					
Required	/	/	/	27. Installation manual					
Required	/	/	/	28. Operating and maintenance manuals					
Required	/	/	/	29. Spare parts recommendations and price list					
Required	/	/	/	30. Preservation, packaging and shipping procedures					
Required	/	/	/	31. Material safety data sheets					
Required	/	/	/	32. Pressure vessel certification data					

^a See Table D.2 for details of the description.

^b Proposal drawings and data do not have to be certified. Typical data shall be clearly identified as such.

^c Purchase may indicate in the column the desired time frame for submission of data.

^d Bidder shall complete these two columns to reflect the actual distribution schedule and include this form with the proposal.

^e N/A signifies an item is generally not applicable to basic systems—Special Duty Only.

Table D.2—Description of VDDR

Item	Description
1	Certified dimensional outline drawings and list of connections, including the following: — Size, type, rating, location, and identification of all customer connections with respect to the datum (fixed) point — Thermal movement of nozzles — The weight of the package and approximate overall erection and maintenance handling weights of equipment and subassemblies that weigh more than 130 kilograms (300 lb). — Principal dimensions including overall package, withdrawal space/maintenance clearances, dismantling clearances, and those required for the piping design. — Shaft centerline height — Operating speeds of bull-gear, and pinions, direction of rotation for the bull-gear shaft — Location of the center of gravity in three planes and lifting points — Allowable piping loads — Vendor recommendations for piping, including requirements for straight length of air inlet piping or for straightening vanes where applicable. — Weight of rotating parts and inertia of bull gear, and pinion rotors in three planes — Purchaser's equipment tag numbers
2	Cross-sectional drawings and bill of materials, including the following: — Journal-bearing clearances and tolerances. — Axial rotor float for bull-gear and all pinions. — Axial position of impellers and tolerance allowed. — Air seal clearances and tolerances.
3	Control, alarm, and trip settings (pressure and recommended temperature).
4	Shaft-coupling assembly drawing and bill of materials, including the following: — The make, size, and type of the couplings — Mounting procedure — Shaft-end gap and tolerance — Coupling guards
5	Sealing air system schematics and bill of materials, including the following: — Air flows and control-valve (regulator) settings — Pipe and valve sizes — Instrumentation, safety devices, and control schemes — List of purchaser connections (if any)

Table D.2—Description of VDDR (Continued)

Item	Description
6	Foundation loading diagram including dimensions of baseplates, complete with the following: — Diameter, number, and locations of bolt holes; thickness of the metal through which the bolts must pass; and recommended clearance. — Weights and centers of gravity for major components.
7	Cooling or heating schematic and bill of materials, including cooling or heating media, fluid flows, pressure, pipe and valve sizes, instrumentation, and orifice sizes.
8	Lube oil schematic and bills of materials, including the following: — Oil flows, temperatures, and pressures at each use point. — Control, alarm, and trip settings (pressure and recommended temperatures). — Pipe, valve, and orifice sizes. — Instrumentation, safety devices, control schemes, and wiring diagrams.
9	Lube oil system assembly and arrangement drawing(s), including size, rating, and location of all customer connections.
10	Electrical, instrumentation and control schematics, wiring diagrams, and bill of materials for all systems. The schematics shall show all control settings, alarm, and shutdown limits (set points). Drawings shall include, but not be limited to the following: — Electrical one-line diagram. — Elementary (schematic) wiring diagram. — Interconnecting wiring/tubing diagrams. — Conduit/wiring installation plans and details.
11	Electrical and instrumentation arrangement drawings, including junction box location drawing and list of connections.
12	ISA data sheets for all instruments.
13	Tabulation of utility requirements (maybe on as built purchaser data sheets).
14	Motor performance and electrical data and curves.
15	Motor terminal box details and wiring instructions.
16	Curves showing performance characteristics at rated condition (discharge pressure and brake power plotted against delivered inlet flow at rated conditions. Performance curves shall indicate surge and rated capacity).
17	Curves showing performance characteristics at other specified inlet conditions.
18	Curve showing the effects of optional inlet guide vanes at off-design inlet conditions. — Damped unbalanced response analysis. — Complete description of the method used. — Graphic display of critical speeds versus operating speed. — Graphic display of bearing and support stiffness and its effect on critical speeds. — Graphic display of rotor response to unbalance (including damping). — Journal static loads.

Table D.2—Description of VDDR (Continued)

Item	Description
18	— Stiffness and damping coefficients. — Tilting-pad bearing geometry and configuration, including the following: a) Pad angle (arc) and number of pads; b) Pivot offset; c) Pad clearance (with journal radius, pad bore radius, and bearing-set bore radius); d) Preload.
19	Damped unbalanced response analysis. — Complete description of the method used. — Display of critical speeds versus operating speed. — Graphic display of bearing and support stiffness and its effect on critical speeds. — Graphic display of rotor response to unbalance (including damping). — Journal static loads. — Stiffness and damping coefficients. — Tilting-pad bearing geometry and configuration, including the following: a) Pad angle (arc) and number of pads; b) Pivot offset; c) Pad clearance (with journal radius, pad bore radius, and bearing-set bore radius); d) Preload.
20	Torsional vibration analysis report — Complete description of the method used. — Graphic display of the mass elastic system. — Tabulation identifying the mass moment and torsional stiffness of each component identified in the mass elastic diagram. — Graphic display of exciting forces versus speed and frequency. — Graphic display of torsional critical speeds and deflections (mode-shape diagram). — Effects of alternate coupling on the analysis.
21	Mechanical running test logs, including but not limited to the following:. — Oil pressures and temperatures. — Vibration, including (where applicable) an x-y plot of amplitude versus revolutions per minute during start-up and coast-down.
22	Certified hydrostatic test logs.

Table D.2—Description of VDDR (Continued)

Item	Description
23	Material certifications (Material certificates of compliance or mill test reports of items as agreed upon in the pre-commitment or pre-inspection meetings).
24	Dimensional drawings for all major auxiliary equipment or components.
25	Data sheets applicable to proposals, purchase and as-built.
26	Noise data sheets.
27	Installation manual describing, but not limited to the following: — Storage procedures; — Foundation plan; — Grouting details; — Setting equipment, rigging procedures, component weights, and lifting diagrams; — Coupling alignment diagram; — Allowable flange loads; — Dismantling clearances; — Written sequence for final tests and checks prior to initial start-up.
28	Operating and maintenance manuals describing, but not limited to the following: — Start-up; — Normal shutdown; — Emergency shutdown; — Operating limitations, other restrictions, and a list of undesirable speeds; — Lube-oil recommendations and specifications; — Routine operational procedures, including recommended inspection schedules and procedures. — Instructions for the following: a) Disassembly and reassembly of rotor in casing. b) Disassembly and reassembly of journal bearings. c) Disassembly and reassembly of thrust bearing. d) Disassembly and reassembly of seals (including maximum and minimum clearances). e) Torquing procedures and torque values. — Performance data per Item 16 above. — Vibration analysis data, per Item 20 above. — As-built data, including the following: a) As-built data sheets. b) Hydrostatic test logs, per Item 21 above. c) Mechanical running test logs, per Item 20 above. d) Rotor balancing logs. e) Rotor mechanical and electrical runout at each journal.

Table D.2—Description of VDDR (Continued)

Item	Description
28	f) Test logs of all specified optional tests. — Drawings and data, including the following: a) Certified dimensional outline drawing and list of connections. b) Cross-sectional drawing and bill of materials. c) Lube-oil schematics and bills of materials. d) Electrical and instrumentation schematics and bills of materials. e) Control- and trip-system data.
29	Spare parts recommendations and price list, complete with stocking level recommendations, including the following: — commissioning spares. — startup spares. — two-year's operating spares. — insurance capital spares.
30	Preservation, packaging and shipping procedures, including the following: — preservation specification. — painting specification. — export boxing details along with proper lifting procedures.
31	Material safety data sheets (OSHA Form 20).
32	Pressure vessel certification data: — design code forms; — nameplate rubbing; — test certifications.

Annex E (informative)

Compressor Control—Inlet Throttle Butterfly Valve versus Variable Inlet Guide Vanes

E.1 General

E.1.1 As shown in the typical compressor performance curves in Figure E.1, energy savings are possible by controlling compressor flow with variable inlet guide vanes rather than suction throttling with a butterfly valve. Inlet guide vane performance is shown as a solid line; butterfly valve throttling is shown as a dashed line.

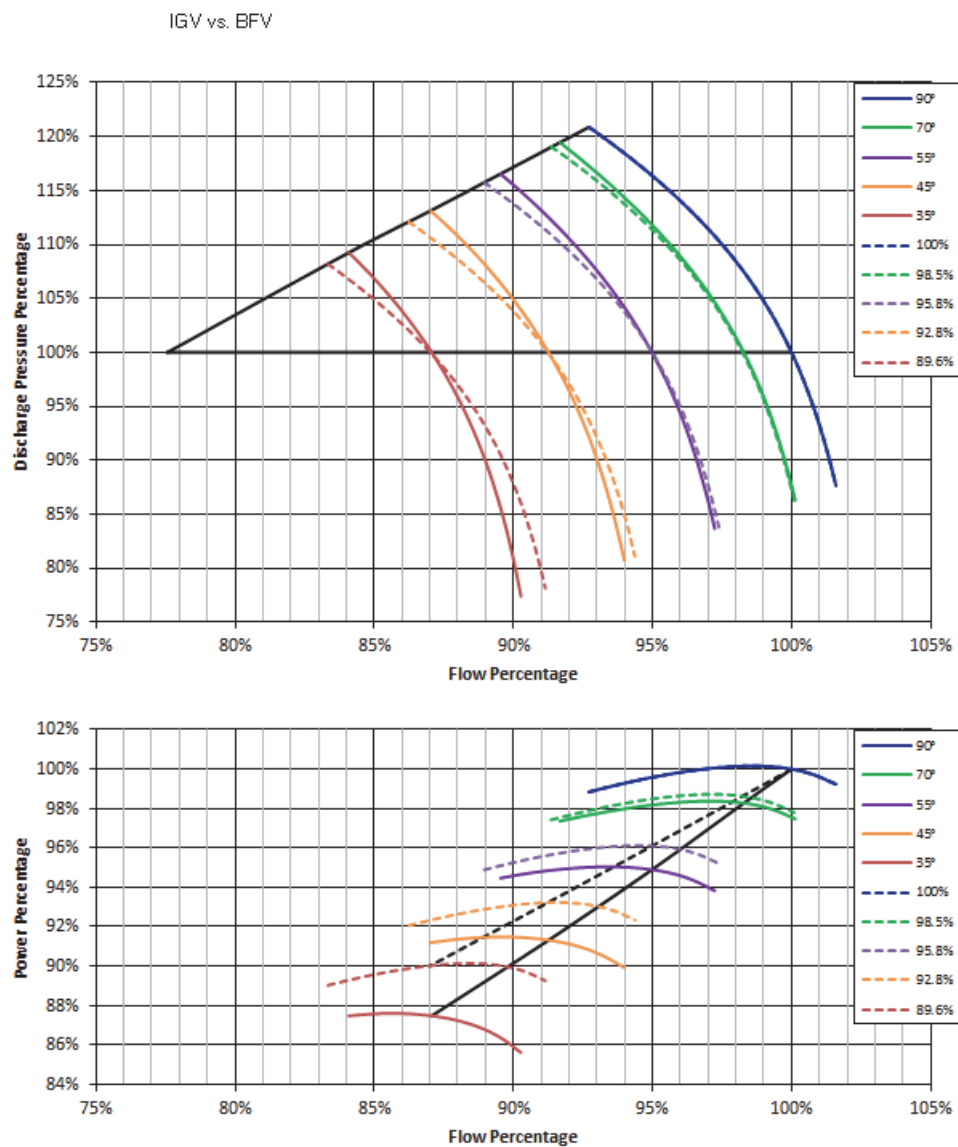


Figure E.1—Typical Performance Curve Showing BFV vs IGV Control

Annex F (normative)

Determination of Residual Imbalance

F.1 General

F.1.1 This annex describes the procedure to be used to determine residual unbalance in machine rotors. Although some balancing machines may be set up to read out the exact amount of unbalance, the calibration can be in error. The only sure method of determining is to test the rotor with a known amount of unbalance.

F.1.2 One way to prove calibration is with a calibrated rotor of known unbalance. Once machine calibration is verified, the read out can then be taken directly from a balancing machine with such readout capability.

F.1.3 The other method is as described in F.4. Some machines include software to store, transfer and compute the trial unbalance data electronically which reduces human errors associated with data entry, calculations and plotting.

F.1.4 Automated results either from calibrated machines or six radial position trial weight computations are considered equivalently acceptable to the manual calculation method described in F.4.

F.2 Residual Unbalance

Residual unbalance is the amount of unbalance remaining in a rotor after balancing. Unless otherwise specified, residual unbalance shall be expressed in g-mm (g-in.).

F.3 Maximum Allowable Residual Unbalance

F.3.1 The maximum allowable residual unbalance, per plane, shall be calculated according to the paragraph from the standard to which this annex is attached.

F.3.2 The static weight on each journal shall be determined by physical measurement. (Calculation methods may introduce errors). It should NOT simply be assumed that rotor weight is equally divided between the two journals. There can be great discrepancies in the journal weight to the point of being very low (even negative on over-hung rotors). In the example problem, the left plane has a journal weight of 3.377 kg (7.445 lb). The right plane has a journal weight of 5.155 kg (11.365 lb).

F.4 Residual Unbalance Check

F.4.1 General

F.4.1.1 When the balancing machine readings indicate that the rotor has been balanced within the specified tolerance, a residual unbalance check shall be performed before the rotor is removed from the balancing machine.

F.4.1.2 To check the residual unbalance, a known trial weight is attached to the rotor sequentially in six equally spaced radial positions (60 degrees apart), each at the same radius (i.e. same moment [g-in.]). The check is run at each balance machine readout plane, and the readings in each plane are tabulated and plotted on the polar graph using the procedure specified in F.4.2.

F.4.2 Procedure

F.4.2.1 Select a trial weight and radius that will be equivalent to between one and two times the maximum allowable residual unbalance [e.g. if U_{max} is 0.548 g-mm (0.0216 g-in.), the trial weight should cause 0.548 to 1.097 g-mm

(0.0216 to 0.0432 g-in.) of unbalance]. This trial weight and radius must be sufficient so that the resulting plot in F.4.2.5 encompasses the origin of the polar plot.

F.4.2.2 Starting at a convenient reference plane (i.e. last heavy spot), mark off the specified six radial positions (60-degree increments) around the rotor. Add the trial weight near the last known heavy spot for that plane. Verify that the balance machine is responding and is within the range and graph selected for taking the residual unbalance check.

F.4.2.3 Verify that the balancing machine is responding reasonably (i.e. no faulty sensors or displays). For example if the trial weight is added to the last known heavy spot, the first meter reading should be at least twice as much as the last reading taken before the trial weight was added. Little or no meter reading generally indicates that the rotor was not balanced to the correct tolerance, the balancing machine was not sensitive enough, or that a balancing machine fault exists (i.e. a faulty pickup). Proceed, if this check is satisfactory.

F.4.2.4 Remove the trial weight and rotate the trial weight to the next trial position (that is, 60, 120, 180, 240, 300 and 360 degrees from the initial trial weight position). Repeat the initial position as a check for repeatability on the Residual Unbalance Worksheet. All verification shall be performed using only one sensitivity range on the balance machine.

F.4.2.5 Plot the balancing machine amplitude readout versus angular location of trial weight (NOT balancing machine phase angle) on the Residual Unbalance Worksheet and calculate the amount of residual unbalance (refer to work sheets shown in Figure F.3 and Figure F.5).

NOTE The maximum reading occurs when the trial weight is placed at the rotor's remaining heavy spot; the minimum reading occurs when the trial weight is placed opposite the rotor's heavy spot (light spot). The plotted readings should form an approximate circle around the origin of the polar chart. The balance machine angular location readout should approximate the location of the trial weight. The maximum deviation (highest reading) is the heavy spot (represents the plane of the residual unbalance). Blank work sheets are shown in Figure F.1 and Figure F.2.

F.4.2.6 Repeat the steps described in F.4.2.1 through F.4.2.5 for each balance machine readout plane. If the specified maximum allowable residual unbalance has been exceeded in any balance machine readout plane, the rotor shall be balanced more precisely and checked again. If a balance correction is made in any balance machine readout plane, then the residual unbalance check shall be repeated in all balance machine readout planes.

F.4.2.7 For stacked component balanced rotors, a residual unbalance check shall be performed after the addition and balancing of the rotor after the addition of the first rotor component, and at the completion of balancing of the entire rotor, as a minimum.

NOTE 1 This ensures that time is not wasted and rotor components are not subjected to unnecessary material removal in attempting to balance a multiple component rotor with a faulty balancing machine.

NOTE 2 For large multi-stage rotors, the journal reactions can be considerably different from the case of a partially stacked to a completely stacked rotor.

Customer: _____
 Job / Project Number: _____
 OEM Equipment S / N: _____
 Rotor Identification Number: _____
 Repair Purchase Order Number: _____
 Vendor Job Number: _____
 Correction Plane (Left or Right) - use sketch _____ (plane)

Balancing Speed _____ (rpm)
 Maximum Rotor Operating Speed (N) _____ (rpm)
 Static Journal Weight Closest to This Correction Plane (W) _____ (kg) _____ (lbs)
 Trial Weight Radius (R) - the radius at which the trial weight will be placed _____ (mm) _____ (in)

Calculate Maximum Allowable Residual Unbalance (U_{max}):
 SI Units:
 $U_{max} = \frac{(6350) \times (W)}{(N)} = \frac{(6350) \times \underline{\hspace{1cm}}}{\underline{\hspace{1cm}}} = \underline{\hspace{1cm}} \text{ (g-mm)}$
 Customary Units:
 $U_{max} = \frac{(113.4) \times (W)}{(N)} = \frac{(113.4) \times \underline{\hspace{1cm}}}{\underline{\hspace{1cm}}} = \underline{\hspace{1cm}} \text{ (g-in)}$

Calculate the trial unbalance (TU):
 Trial Unbalance (TU) is between (1 X U_{max}) and (2 X U_{max})
 SI Units: _____ (1 X) to _____ (2 X) (Selected Multiplier is) _____
 Customary units: _____ to _____ = _____ (g-mm)
 _____ to _____ = _____ (g-in)

Calculate the trial weight (TW):
 $\text{Trial Weight (TW)} = \frac{U_{max}}{R} = \frac{\underline{\hspace{1cm}} \text{ g-mm}}{\underline{\hspace{1cm}} \text{ mm}} \text{ or } \frac{\underline{\hspace{1cm}} \text{ g-in}}{\underline{\hspace{1cm}} \text{ in}} = \underline{\hspace{1cm}} \text{ (g)}$

Conversion Information:
 1kg = 2.2046 lbs 1 ounce = 28.345 grams

Obtain the test data and complete the table:

Sketch the rotor configuration:

Test Data			
Position	Trial Weight Angular Location on Rotor (degrees)	Balancing Mach Readout	
		Amplitude (grams)	Phase Angle (degrees)
1	0		
2	60		
3	120		
4	180		
5	240		
6	300		
Repeat 1	0		

Rotor Sketch**PROCEDURE:**

- Step 1: Plot the balancing machine amplitude versus trial weight angular location on the polar chart (Figure F-2) such that the largest and smallest values will fit.
- Step 2: The points located on the Polar Chart should closely approximate a circle. If it does not, then it is probably that the recorded data is in error and the test should be repeated.
- Step 3: Determine the maximum and minimum balancing machine amplitude readings.
- Step 4: Using the worksheet, (Figure F-2), determine the Y and Z values required for the residual unbalance calculation.
- Step 5: Using the worksheet, (Figure F-2), calculate the residual unbalance remaining in the rotor.
- Step 6: Verify that the determined residual unbalance is equal to or less than the maximum allowable residual unbalance (U_{max}).

HALF KEYS USED FOR ROTOR BALANCING
(add sketch for clarification if necessary)

Location	Weight

NOTES:

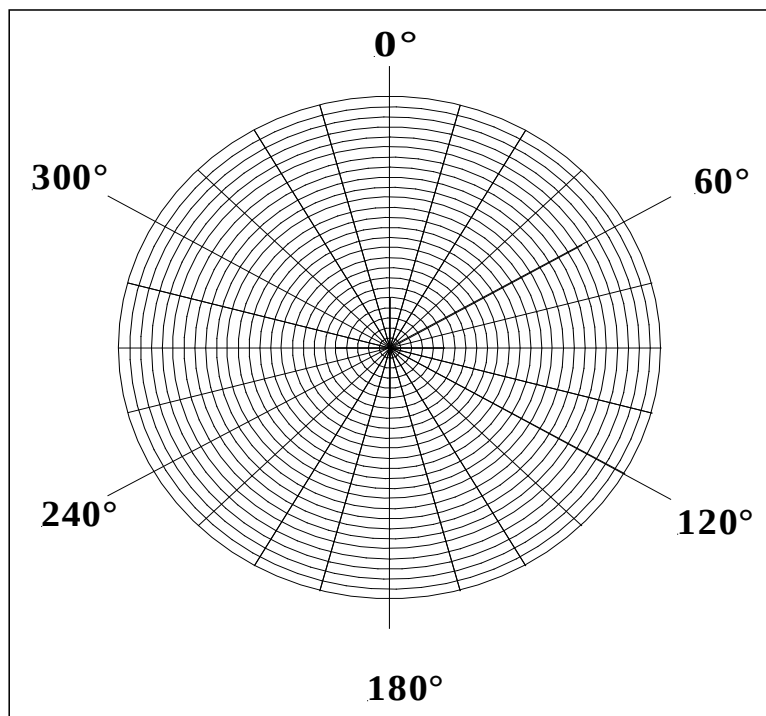
- The trial weight angular location should be referenced to a keyway or some other permanent marking on the rotor. The preferred location is the location of the once-per-revolution mark (for the phase reference transducer).
- The balancing machine amplitude readout for the 'Repeat of 1' should be the same as Position 1, indicating repeatability.
- A primary source for error is not maintaining the same radius for each trial weight location.

Balanced By: _____ Date: _____
 Approved By: _____ Date: _____

Figure F.1—(Blank) Residual Unbalance Work Sheet

Customer: _____
 Job / Project Number: _____
 OEM Equipment S / N: _____
 Rotor Identification Number: _____
 Repair Purchase Order Number: _____
 Vendor Job Number: _____
 Correction Plane (Left or Right) - use sketch _____

RESIDUAL UNBALANCE POLAR PLOT



Rotor Rotation: ☐ CCW ☐ CW Phase is layed out: ☐ CCW ☐ CW

Calculate Y and Z values:

Maximum amplitude value is: _____ grams Minimum amplitude value is: _____
 $Y = (\text{Maximum} - \text{Minimum}) / 2$ (_____ - _____) / 2 = _____
 $Z = (\text{Maximum} + \text{Minimum}) / 2$ (_____ + _____) / 2 = _____

Residual Unbalance

Left in Rotor = _____ (TU) X _____ (Y) / _____ (Z) = _____
 SI Units: _____ X _____ / _____ = _____
 Customary Units: _____ X _____ / _____ = _____

Allowable Unbalance Tolerance = Umax = _____ gm-mm _____ gm-in

RESULT: Residual unbalance left in the rotor is equal to or less than the allowable unbalance tolerance?

☐ As Received ☐ Final ☐ PASS ☐ FAIL
☐ Other: _____

Balanced By: _____ Date: _____
 Approved By: _____ Date: _____

Figure F.2—(Blank) Residual Unbalance Polar Plot Work Sheet

Customer:	ABC Refining Co.
Job / Project Number:	00 - 1234
OEM Equipment S / N:	C - 1234
Rotor Identification Number:	1234 - C - 4320
Repair Purchase Order Number:	PO 12345678
Vendor Job Number:	Shop - 00 - 1234
Correction Plane (Left or Right) - use sketch	Left (plane)

Balancing Speed	2250 (rpm)	
Maximum Rotor Operating Speed (N)	39096 (rpm)	
Static Journal Weight Closest to This Correction Plane (W)	3.377 (kg)	7.445 (lbs)
Trial Weight Radius (R) - the radius at which the trial weight will be placed	33.02 (mm)	1.3 (in)

Calculate Maximum Allowable Residual Unbalance (Umax):

SI Units:

$$U_{max} = \frac{(6350) \times (W)}{(N)} = \frac{(6350) \times 3.377}{39096} = 0.5485 \text{ (g-mm)}$$

Customary Units:

$$U_{max} = \frac{(113.4) \times (W)}{(N)} = \frac{(113.4) \times 7.445}{39096} = 0.0216 \text{ (g-in)}$$

Calculate the trial unbalance (TU):

Trial Unbalance (TU) is between (1 X Umax) and (2 X Umax)

SI Units:

Customary units:

Calculate the trial weight (TW):

$$\text{Trial Weight (TW)} = \frac{U_{max}}{R} = \frac{1.097 \text{ g-mm}}{33.02 \text{ mm}} = 0.033 \text{ (g)}$$

or

$$\frac{0.043 \text{ g-in}}{1.3 \text{ in}} = 0.033 \text{ (g)}$$

Conversion Information:

1kg = 2.2046 lbs 1 ounce = 28.345 grams

Obtain the test data and complete the table:

Test Data			
Position	Trial Weight Angular Location on Rotor (degrees)	Balancing Mach Readout Amplitude (grams)	Phase Angle (degrees)
1	0	0.613	358
2	60	0.624	59
3	120	0.633	123
4	180	0.605	182
5	240	0.618	241
6	300	0.636	301
Repeat 1	0	0.612	359

Sketch the rotor configuration:

Rotor Sketch

HALF KEYS USED FOR ROTOR BALANCING
(add sketch for clarification if necessary)

Location	Weight

PROCEDURE:

Step 1: Plot the balancing machine amplitude versus trial weight angular location on the polar chart (Figure F-2) such that the largest and smallest values will fit.

Step 2: The points located on the Polar Chart should closely approximate a circle. If it does not, then it is probably that the recorded data is in error and the test should be repeated.

Step 3: Determine the maximum and minimum balancing machine amplitude readings.

Step 4: Using the worksheet, (Figure F-2), determine the Y and Z values required for the residual unbalance calculation.

Step 5: Using the worksheet, (Figure F-2), calculate the residual unbalance remaining in the rotor.

Step 6: Verify that the determined residual unbalance is equal to or less than the maximum allowable residual unbalance (Umax).

NOTES:

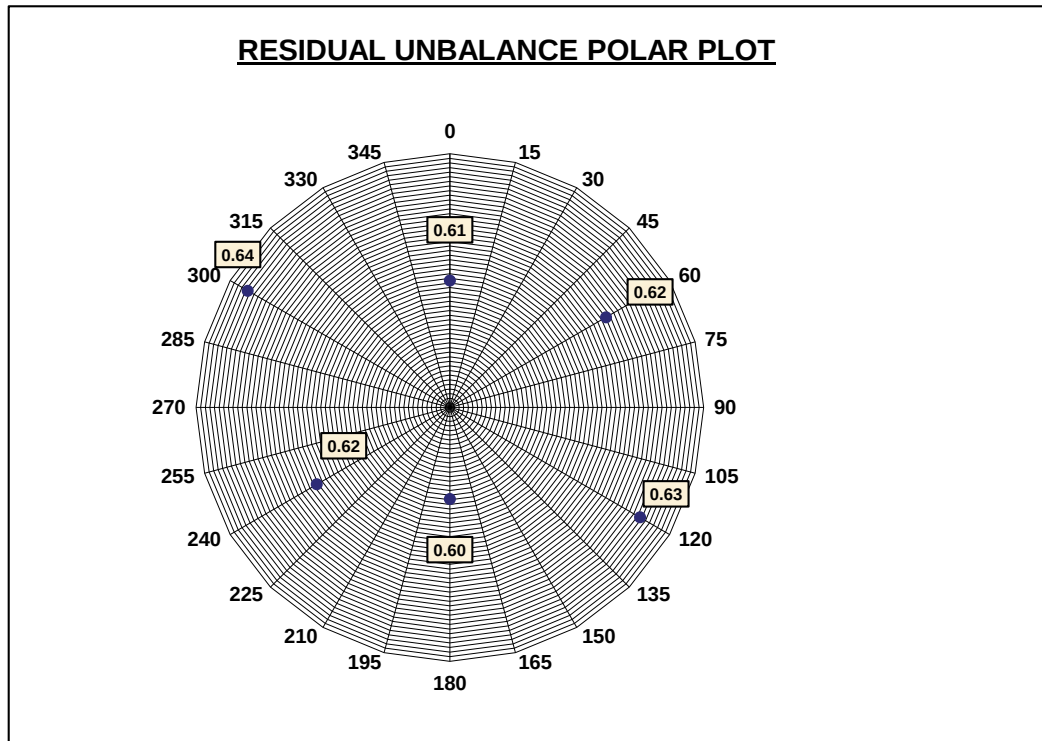
- The trial weight angular location should be referenced to a keyway or some other permanent marking on the rotor. The preferred location is the location of the once-per-revolution mark (for the phase reference transducer).
- The balancing machine amplitude readout for the 'Repeat of 1' should be the same as Position 1, indicating repeatability.
- A primary source for error is not maintaining the same radius for each trial weight location.

Balanced By:	AP, AP, & PP	Date:	7/7/2012
Approved By:	JB	Date:	7/7/2012

Figure F.3—Sample Residual Unbalance Work Sheet for Left Plane

Customer:
 Job / Project Number:
 OEM Equipment S / N:
 Rotor Identification Number:
 Repair Purchase Order Number:
 Vendor Job Number:
 Correction Plane (Left or Right) - use sketch

ABC Refining Co.
00 - 1234
C - 1234
1234 - C - 4320
PO 12345678
Shop - 00 - 1234
Left (plane)



Rotor Rotation: ☒ CCW ☐ CW Phase is layed out: ☐ CCW ☒ CW

Calculate Y and Z values:
 Maximum amplitude value is:

0.636

 grams Minimum amplitude value is:

0.605

 grams
 $Y = (\text{Maximum} - \text{Minimum}) / 2$ (

0.636

 -

0.605

) / 2 =

0.015

 $Z = (\text{Maximum} + \text{Minimum}) / 2$ (

0.636

 +

0.605

) / 2 =

0.620

Residual Unbalance
 Left in Rotor = (TU) X (Y) / (Z) =

1.097

 X

0.015

 /

0.620

 =

0.027

 g-mm
 SI Units:

1.097

 X

0.015

 /

0.620

 =

0.001

 g-in
 Customary Units:

0.043

 X

0.015

 /

0.620

 =

0.001

 g-in

Allowable Unbalance Tolerance = Umax =

0.548

 g-mm

0.022

 g-in

RESULT: Residual unbalance left in the rotor is equal to or less than the allowable unbalance tolerance?

☐ As Received ☐ Final ☒ Other: w/o trim hardware

PASS

Balanced By: AP, AP, & PP
 Approved By: JB

Date: 7/7/2012
 Date: 7/7/2012

Figure F.4—Sample Residual Unbalance Polar Plot Work Sheet for Left Plane

Customer:
 Job / Project Number:
 OEM Equipment S / N:
 Rotor Identification Number:
 Repair Purchase Order Number:
 Vendor Job Number:
 Correction Plane (Left or Right) - use sketch

ABC Refining Co.
 00 - 1234
 C - 1234
 1234 - C - 4320
 PO 12345678
 Shop - 00 - 1234
 Right (plane)

Balancing Speed
 Maximum Rotor Operating Speed (N)
 Static Journal Weight Closest to This Correction Plane (W)
 Trial Weight Radius (R) - the radius at which the trial weight will be placed

2250 (rpm)
 39096 (rpm)
 5.155 (kg) 11.365 (lbs)
 203.2 (mm) 8 (in)

Calculate Maximum Allowable Residual Unbalance (U_{max}):

SI Units:

$$U_{max} = \frac{(6350) \times (W)}{(N)} = \frac{(6350) \times 5.155}{39096} = 0.837 \text{ (g-mm)}$$

Customary Units:

$$U_{max} = \frac{(113.4) \times (W)}{(N)} = \frac{(113.4) \times 11.365}{39096} = 0.033 \text{ (g-in)}$$

Calculate the trial unbalance (TU):

Trial Unbalance (TU) is between (1 X U_{max}) and (2 X U_{max})

SI Units:

Customary units:

(1 X) to (2 X) (Selected Multiplier is) 2
 0.8 to 1.7 is 1.675 (g-mm)
 0.0 to 0.1 is 0.066 (g-in)

Calculate the trial weight (TW):

Trial Weight (TW) = $\frac{U_{max}}{R} = \frac{1.675 \text{ g-mm}}{203.2 \text{ mm}} = 0.008 \text{ (g)}$ or $\frac{0.066 \text{ g-in}}{8 \text{ in}} = 0.008 \text{ (g)}$

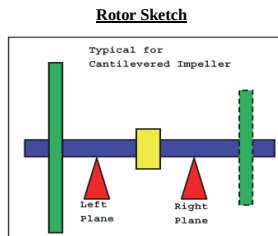
Conversion Information:

1kg = 2.2046 lbs 1 ounce = 28.345 grams

Obtain the test data and complete the table:

Test Data			
Position	Trial Weight Angular Location on Rotor (degrees)	Balancing Mach Readout Amplitude (grams)	Phase Angle (degrees)
1	0	0.590	3
2	60	0.622	58
3	120	0.603	121
4	180	0.582	180
5	240	0.631	235
6	300	0.617	301
Repeat 1	0	0.601	2

Sketch the rotor configuration:



PROCEDURE:

- Step 1: Plot the balancing machine amplitude versus trial weight angular location on the polar chart (Figure F-2) such that the largest and smallest values will fit.
- Step 2: The points located on the Polar Chart should closely approximate a circle. If it does not, then it is probably that the recorded data is in error and the test should be repeated.
- Step 3: Determine the maximum and minimum balancing machine amplitude readings.
- Step 4: Using the worksheet, (Figure F-2), determine the Y and Z values required for the residual unbalance calculation.
- Step 5: Using the worksheet, (Figure F-2), calculate the residual unbalance remaining in the rotor.
- Step 6: Verify that the determined residual unbalance is equal to or less than the maximum allowable residual unbalance (U_{max}).

HALF KEYS USED FOR ROTOR BALANCING
 (add sketch for clarification if necessary)

Location	Weight

NOTES:

- The trial weight angular location should be referenced to a keyway or some other permanent marking on the rotor. The preferred location is the location of the once-per-revolution mark (for the phase reference transducer).
- The balancing machine amplitude readout for the 'Repeat of 1' should be the same as Position 1, indicating repeatability.
- A primary source for error is not maintaining the same radius for each trial weight location.

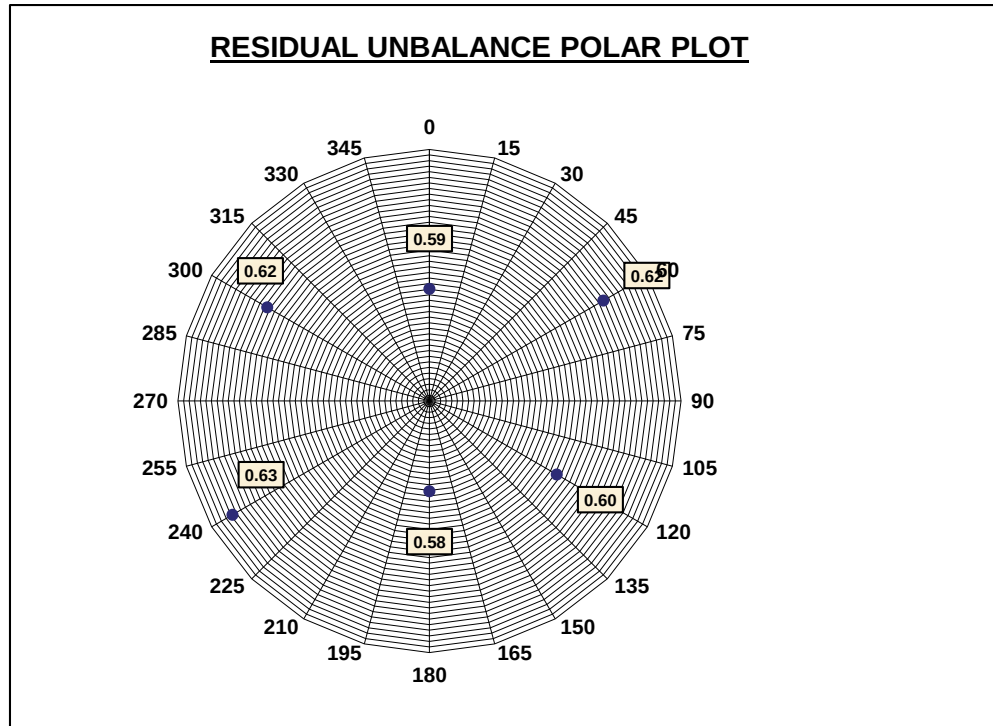
Balanced By: AP, AP, & PP
 Approved By: JB

Date: 7/7/2012
 Date: 7/7/2012

Figure F.5—Sample Residual Unbalance Work Sheet for Right Plane

Customer:
 Job / Project Number:
 OEM Equipment S / N:
 Rotor Identification Number:
 Repair Purchase Order Number:
 Vendor Job Number:
 Correction Plane (Left or Right) - use sketch

ABC Refining Co.
00 - 1234
C - 1234
1234 - C - 4320
PO 12345678
Shop - 00 - 1234
Right (plane)



Rotor Rotation: ☒ CCW ☐ CW Phase is layed out: ☐ CCW ☒ CW

Calculate Y and Z values:
 Maximum amplitude value is:

0.631
0.631
0.631

 grams Minimum amplitude value is:

0.582
0.582
0.582

 grams
 $Y = (\text{Maximum} - \text{Minimum}) / 2 = (0.631 - 0.582) / 2 = 0.025$
 $Z = (\text{Maximum} + \text{Minimum}) / 2 = (0.631 + 0.582) / 2 = 0.607$

Residual Unbalance
 Left in Rotor = (TU) X (Y) / (Z)
 SI Units:

1.675

 X

0.02455

 /

0.60685

 =

0.068

 g-mm
 Customary Units:

0.066

 X

0.02455

 /

0.60685

 =

0.003

 g-in

Allowable Unbalance Tolerance = Umax =

0.837

 g-mm

0.033

 g-in

RESULT: Residual unbalance left in the rotor is equal to or less than the allowable unbalance tolerance?

☐ As Received ☐ Final ☒ Other: w/o trim hardware

Balanced By: AP, AP, & PP Date: 7/7/2012
 Approved By: JB Date: 7/7/2012

PASS

Figure F.6—Sample Residual Unbalance Polar Plot Work Sheet for Right Plan

Annex G (normative)

Inspector's Checklist

G.1 General

The levels indicated in Table G.1 may be characterized as follows:

- Level 1 is typically used for packages in basic services;
- Level 2 comprises optional performance and material requirements and is more stringent than Level 1;
- Level 3 items should be considered for packages in special duty services.

The required inspection shall be indicated in the first column as:

- C—Certification only;
- O—Observed inspection;
- W—Witnessed inspection.

Table G.1—Inspector's Checklist

Inspection required C, O, or W	Item	API 672 section number	Date inspected	Inspected by	Status
	Level 1—Basic				
	Package scope	Contract; 6.1.7			
	Auxiliary systems per design	Contract; auxiliary system schematics			
	Overall dimensions and connection locations ^a	Outline drawing; 8.3.5			
	Anchor bolt layout and size ^a				
	Motors and electrical components area classification	6.1.11			
	Casing connections: nozzle size, rating and finish ^a	outline drawing; 6.1.14, 6.3			
	Bolting	6.1.15			
	Rotor balancing	6.7.4.1			
	Vibration within acceptance criteria	6.7.4.3			
	Lubrication system reservoir internal coating and cleanliness	6.9.4			
	Equipment nameplate data	6.11.4			
	Rotation arrows	6.11.2			
	Jackscrews on driver feet	7.1.1.6			
	Couplings proper type	7.2.1			

Table G.1—Inspector's Checklist (Continued)

Inspection required C, O, or W	Item	API 672 section number	Date inspected	Inspected by	Status
	Coupling guards with sufficient protection and sufficiently rigid	7.2.2			
	Baseplate with major components supported	7.3			
	Lifting lugs included and identified	7.3.4; 8.4.6			
	Mounting surfaces within tolerances	6.1.16; 7.3.6			
	Conduit routing, properly supported, properly shielded	7.4.1.5; 7.4.1.6; 7.4.6.5			
	Instrument control panel scope	7.4.3.1			
	Annunciator panel scope and function	7.4.5.2			
	Segregated instrument and control wiring from electrical power wiring	7.4.6.3			
	Piping fabrication and installation	7.5			
	Inlet air filter/silencer scope and construction materials	7.7			
	Pre-test static gear contact pattern	8.2.3.2			
	Hydrostatic tests	8.3.2			
	Impeller over-speed test	8.3.3			
	Combined mechanical performance test	8.3.4; 8.3.6			
	Preparation for shipment	8.4.1			
	Storage preservation instructions	8.4.2			
	Rust preventive	8.4.3			
	Painting	8.4.2			
	Shipping documents and tags	8.4.7			
	Level 2—Intermediate (Add to Level 1)				
	Material certification	8.2.1			
	Non-destructive examination (components)	7.3.4; 8.2.1c;			
	Hydrotest witnessed	8.3.2			
	Rotating elements balancing witnessed	6.7.4.1			
	Building records (runouts)	6.7.4.4; 8.3.4.6.4			
	Performance and Mechanical tests Witnessed	8.3.4			
	Inspection of cleanliness of internals	8.2.3.1			
	Level 3—Special (Add to Level 1 and 2)				
	Special devices used for maintenance	6.12.1			

Table G.1—Inspector's Checklist (Continued)

Inspection required C, O, or W	Item	API 672 section number	Date inspected	Inspected by	Status
	Confirm damped unbalanced response analysis	6.12.4			
	Dynamic, component balancing	6.12.6; 6.12.7			
	Residual unbalance check	6.12.8			
	Stainless steel oil reservoir	6.12.14			
	Proper preparation of grouted surfaces ^b	7.10.2			
	Provisions for phase reference ^b	7.10.11			
	Gear axial position probe provision ^b	7.10.8			
	Gear casing accelerometer mounting provisions ^b	7.10.9			
	Vibration and axial position probe transducers ^b	7.10.10			
	Vibration and axial position probe monitors ^b	7.10.12			
	Bearing temperature monitors ^b	7.10.13			
	Alarm and shutdown devices separate? ^b	7.10.15			
	Pilot lights on electrical circuits ^b	7.10.16			
	Stainless steel oil piping throughout ^b	7.10.18			
	Oil-actuated control valves vented back to reservoir ^b	7.10.19			
	All piping components of steel	7.10.20			
	Special cooler materials	7.10.21			
	Coolers TEMA C with removable channel covers ^b	7.10.22			
	Documentation for clearances	8.5.1			
	Impellers radiographed and inspected	8.5.2; 8.5.3			
	Non-synchronous vibration within tolerance ^b	8.5.10			
	Post test inspection ^b	8.5.12			
	Spare rotor mechanical test	8.5.14.2			

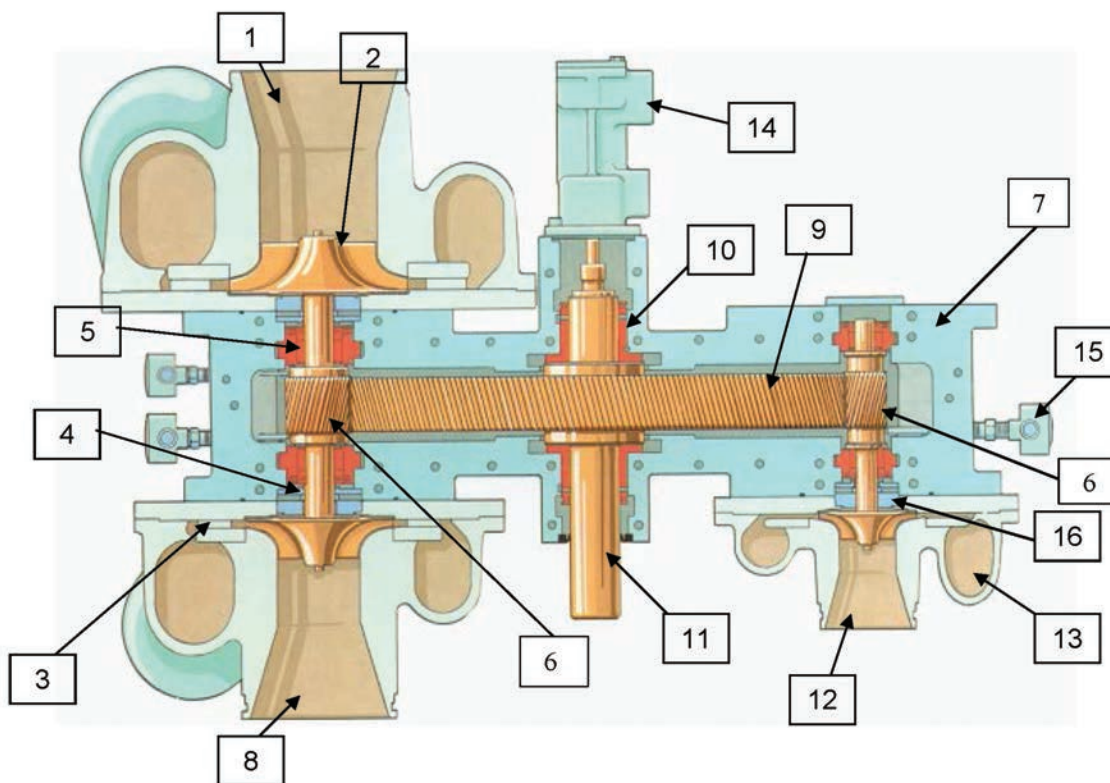
Annex H (informative)

Nomenclature for Integrally Geared Centrifugal Air Compressors

H.1 General

H.1.1 Figure H.1 and Figure H.2 shows sectional views with component identification for axially split compressors.

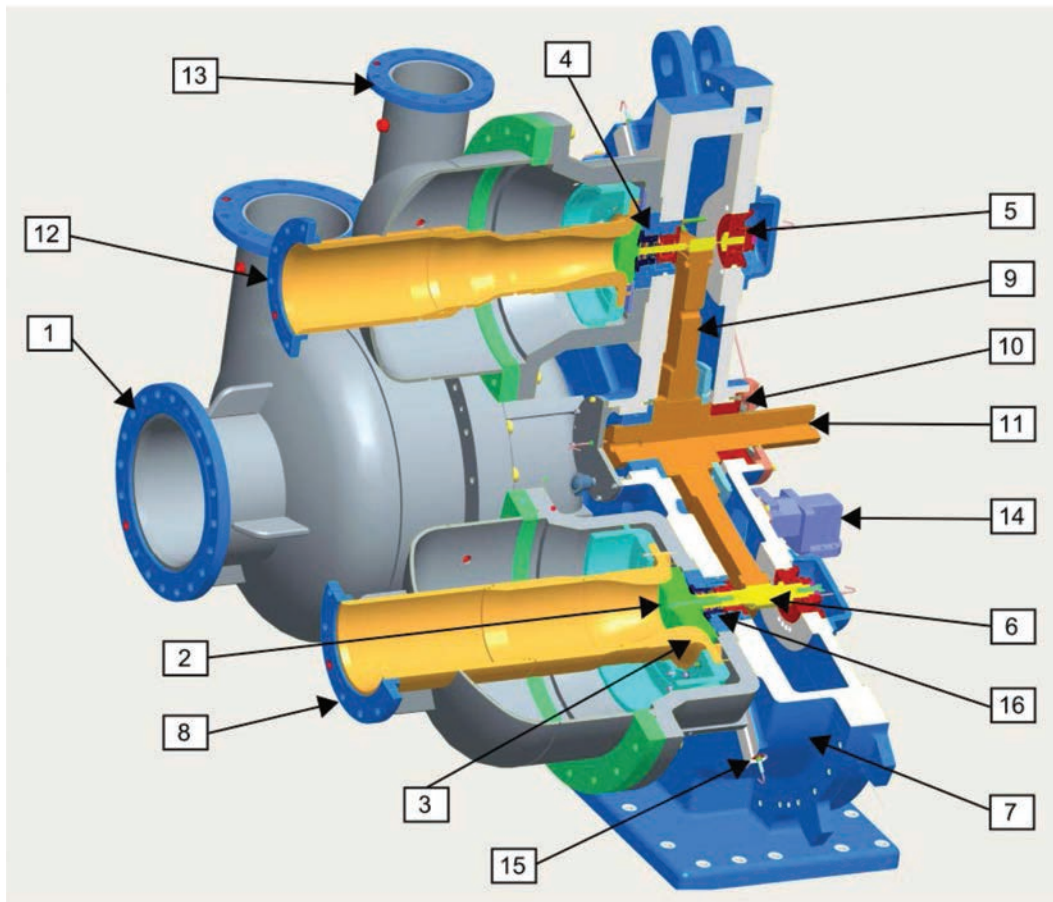
H.1.2 Figures H.3, H.4, and H.5 show typical nomenclature for package components.



Key

- | | |
|----------------------------------|---------------------------------------|
| 1) First Stage Inlet | 9) Gear Wheel (Bull Gear) |
| 2) Impeller | 10) Gear Wheel Journal/Thrust Bearing |
| 3) Diffuser | 11) Input (Drive) Shaft |
| 4) Oil seal | 12) Third Stage Inlet |
| 5) Pinion Journal/Thrust bearing | 13) Third Stage Discharge |
| 6) Pinion | 14) Shaft Driven Main Oil Pump |
| 7) Gear casing | 15) Vibration Instrument |
| 8) Second Stage Inlet | 16) Air Seal |

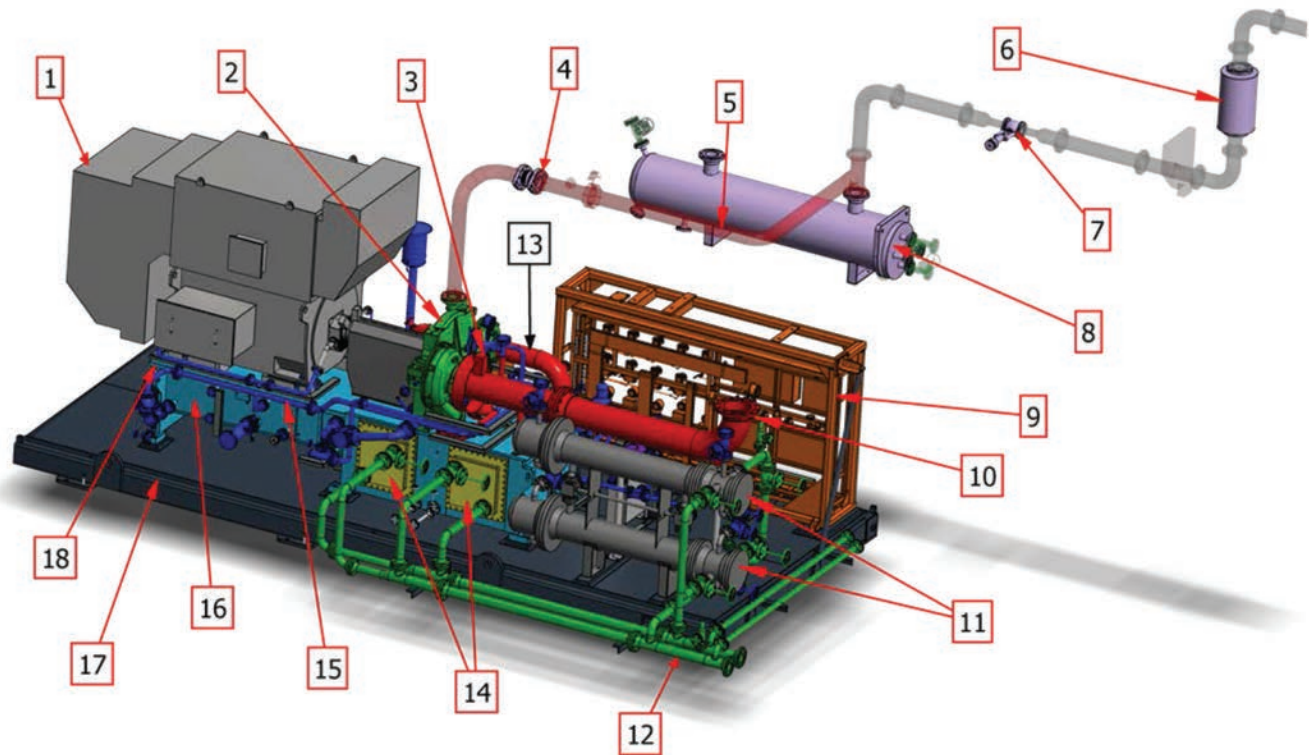
Figure H.1—Section of Axially (horizontally) Split Compressor



Key

- | | |
|----------------------------------|---------------------------------------|
| 1) First Stage Inlet | 9) Gear Wheel (Bull Gear) |
| 2) Impeller | 10) Gear Wheel Journal/Thrust Bearing |
| 3) Diffuser | 11) Input (Drive) Shaft |
| 4) Oil seal | 12) Third Stage Inlet |
| 5) Pinion Journal/Thrust bearing | 13) Third Stage Discharge |
| 6) Pinion | 14) Shaft Driven Main Oil Pump |
| 7) Gear casing | 15) Vibration Instrument |
| 8) Second Stage Inlet | 16) Air Seal |

Figure H.2—Section of Radially Split Compressor

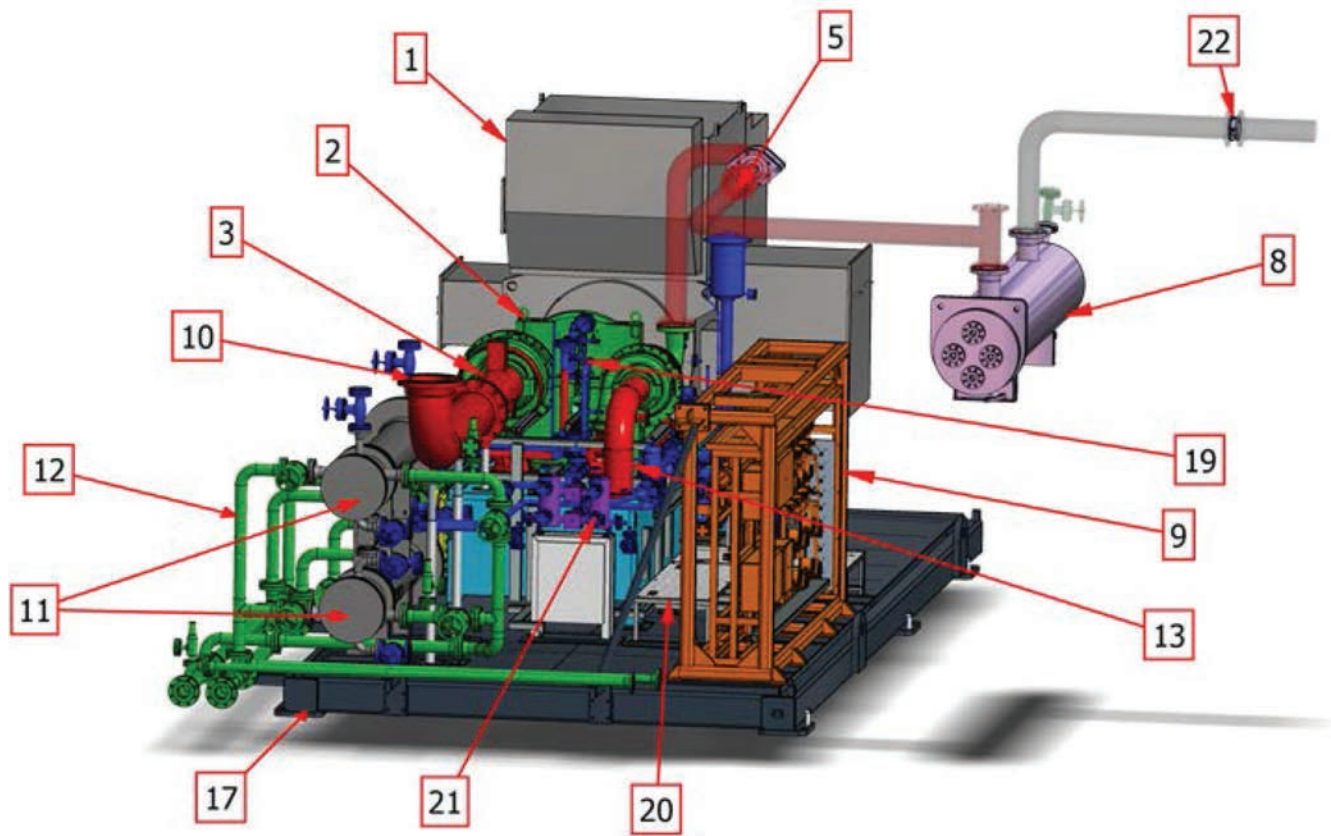


Key

- | | |
|--|--|
| 1) Driver | 14) Intercooler |
| 2) Air Compressor | 15) Oil System Piping |
| 3) Inlet Guide Vane | 16) Oil Reservoir |
| 4) Discharge Expansion Joint ^a | 17) Base Skid |
| 5) Discharge Piping ^a | 18) Baseplate |
| 6) Discharge Blowoff Silencer ^a | 19) Main Oil Pump |
| 7) Discharge Blowoff Valve | 20) Walkway |
| 8) Aftercooler ^a | 21) Oil Filter |
| 9) Instrument Rack | 22) Discharge Check Valve ^a |
| 10) Inlet | 23) Inlet Air Filter ^a |
| 11) Oil Cooler | 24) Oil Mist Eliminator |
| 12) Water Piping | 25) Auxiliary Oil Pump |
| 13) Air Piping | |

^a Components typically shipped loose.

Figure H.3—Nomenclature of Package Components, Part 1

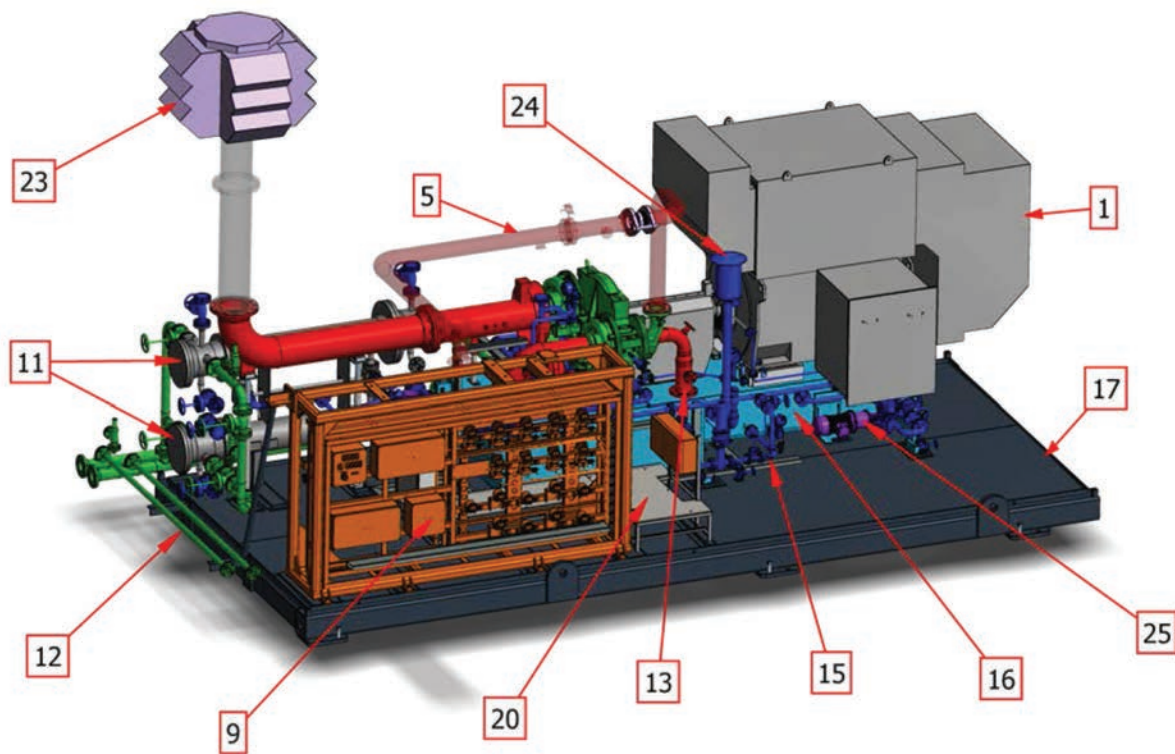


Key

- | | |
|--|--|
| 1) Driver | 14) Intercooler |
| 2) Air Compressor | 15) Oil System Piping |
| 3) Inlet Guide Vane | 16) Oil Reservoir |
| 4) Discharge Expansion Joint ^a | 17) Base Skid |
| 5) Discharge Piping ^a | 18) Baseplate |
| 6) Discharge Blowoff Silencer ^a | 19) Main Oil Pump |
| 7) Discharge Blowoff Valve | 20) Wa kway |
| 8) Aftercooler ^a | 21) Oil Filter |
| 9) Instrument Rack | 22) Discharge Check Valve ^a |
| 10) Inlet | 23) Inlet Air Filter ^a |
| 11) Oil Cooler | 24) Oil Mist Eliminator |
| 12) Water Piping | 25) Auxiliary Oil Pump |
| 13) Air Piping | |

^a Components typically shipped loose.

Figure H.4—Nomenclature of Package Components, Part 2



Key

- | | |
|--|--|
| 1) Driver | 14) Intercooler |
| 2) Air Compressor | 15) Oil System Piping |
| 3) Inlet Guide Vane | 16) Oil Reservoir |
| 4) Discharge Expansion Joint ^a | 17) Base Skid |
| 5) Discharge Piping ^a | 18) Baseplate |
| 6) Discharge Blowoff Silencer ^a | 19) Main Oil Pump |
| 7) Discharge Blowoff Valve | 20) Walkway |
| 8) Aftercooler ^a | 21) Oil Filter |
| 9) Instrument Rack | 22) Discharge Check Valve ^a |
| 10) Inlet | 23) Inlet Air Filter ^a |
| 11) Oil Cooler | 24) Oil Mist Eliminator |
| 12) Water Piping | 25) Auxiliary Oil Pump |
| 13) Air Piping | |

^a Components typically shipped loose.

Figure H.5—Nomenclature of Package Components, Part 3

Annex I (normative)

External Forces and Moments

I.1 General

For integrally geared compressors, it is not possible to give a formula to calculate the maximum allowable piping forces and moments on each casing flange. The limiting criteria are the gear contact pattern and the impeller/stator gap. The maximum value of the external forces and moments, which leads to acceptable deformations and therefore acceptable changes of the gear contact pattern and the impeller/stator gap, depends on various parameters.

These parameters include:

- Volute geometry, volute wall thickness, length of overhang, gear case geometry and gear case wall thickness. The possible combinations are nearly endless.
- Each manufacturer has limits based on their own experience for each volute size and gear case combination for a given specific machine. The values are available from the manufacturer with the quotation.
- It is a common practice on integrally geared compressors to supply expansion joints in order to minimize the piping loads on the machine flanges and to insure that piping loads are within the allowable limits for the particular unit.

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